Wonderware

Alarm Adviser

Installation and Configuration Guide

September 2017

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Contents

Contents **3**

Safety Information **7**

Chapter 1: Installing and Configuring Alarm Adviser **9**

Related Documents 9

Alarm Adviser Architecture 10

Introduction 10

Collectors 12

Alarm Adviser Database 13

Service 13

Web Server 13

Clients 14

Chapter 2: Deploying Alarm Adviser **15**

Introduction 15

Deployment Scenarios 15

Chapter 3: Alarm Adviser System Requirements **19**

Server Requirements 19

Required Software 19

Recommended Minimum Hardware* 19

Client PC Requirements 20

Chapter 4: Pre-Installation 21

Installation Prerequisites 21

Supported SCADA Systems 22

Before You Install the Vijeo Citect Collector 22

Data Collection 23

Upgrading Vijeo Citect 23

Before You Install the ClearSCADA Collector 24

Data Collection 24

Upgrading ClearSCADA 27

Chapter 5: Installing Alarm Adviser 31

Starting the Installation 31

Installing Alarm Adviser Components 32

Installing the Wonderware Collector 36

Installing the Vijeo Citect Collector 39

Installing the ClearSCADA Collector 42

Completing the Installation 46

Chapter 6: Configuring Alarm Adviser 49

Configurator Interface 50

Configuring the Alarm Adviser Service 52

Configuring the Alarm Adviser Web Server 56

Localizing Alarm Adviser 60

Prerequisites for Localization 60

Localizing the Web Client 61

Localizing the Online Help 62

Configuring Wonderware Collector 62

Configuring the Vijeo Citect Collector 67

Configuring the ClearSCADA Collector 74

Add a Connection to the ClearSCADA System 80

Client Applet Buttons 83

Configure the TCP/IP Tuning Settings 85

Change the Range of Ports for the Collector's Incoming Data 87

Configure the Node Settings 89

Node Cost 93

Configure the Connection Port and Poll Timeouts and Intervals 94

Changes to Collector Configuration 96

Creating Alarm Adviser Users 97

Adding Users to an Alarm Adviser Group 99

Chapter 7: Alarm Adviser Licensing 103

Alarm Adviser Licensing for InTouch 105

Chapter 8: Maintaining an Alarm Adviser Installation 107

Maintaining the Alarm Adviser Database 107

Backing Up the Alarm Adviser Database 108

Re-deploying Installed Components 108

Moving Web Server, Service or Collector 108

Moving the Alarm Adviser Database 109

Modifying an Installation 109

Repairing an Installation 112

Uninstalling Alarm Adviser 115

Chapter 9: Running System Diagnostics 119

Web Page Error Notifications 119

Alarm Adviser Component Logging 119

Viewing Alarm Adviser Events 120

Configuring Events for Display 122

Unhandled Exceptions 123

Appendix A: Supported Language Codes 124

Index 128

Safety Information

Read these instructions carefully, look at the equipment and familiarize yourself with Alarm Adviser before trying to install, operate, service or maintain your system. The following special messages may appear throughout this documentation or on the Alarm Adviser application to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a Danger or Warning safety label indicates that an electrical hazard exists, which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

 DANGER
DANGER indicates a hazardous situation which, if not avoided**, will result in** death or serious injury**.**

 WARNING
WARNING indicates a hazardous situation which, if not avoided**, could result in** death or serious injury.

 CAUTION
CAUTION indicates a hazardous situation which, if not avoided, **could result in** minor or moderate injury.

NOTICE
NOTICE is used to address practices not related to physical injury. The safety alert symbol shall not be used with this signal word.

Please Note

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction, installation, and operation of electrical equipment, and has received safety training to recognize and avoid the hazards involved.

Chapter 1: Installing and Configuring Alarm Adviser

Welcome to the Alarm Adviser Installation and Configuration Guide.

This guide provides information about:

- System requirements for running Alarm Adviser
- Prerequisites for installing Alarm Adviser
- Common deployment scenarios
- Instructions for:
    - installing Alarm Adviser
    - modifying an installation
    - repairing an installation
    - uninstalling Alarm Adviser
    - configuring Alarm Adviser
- Maintaining your Alarm Adviser installation

Related Documents

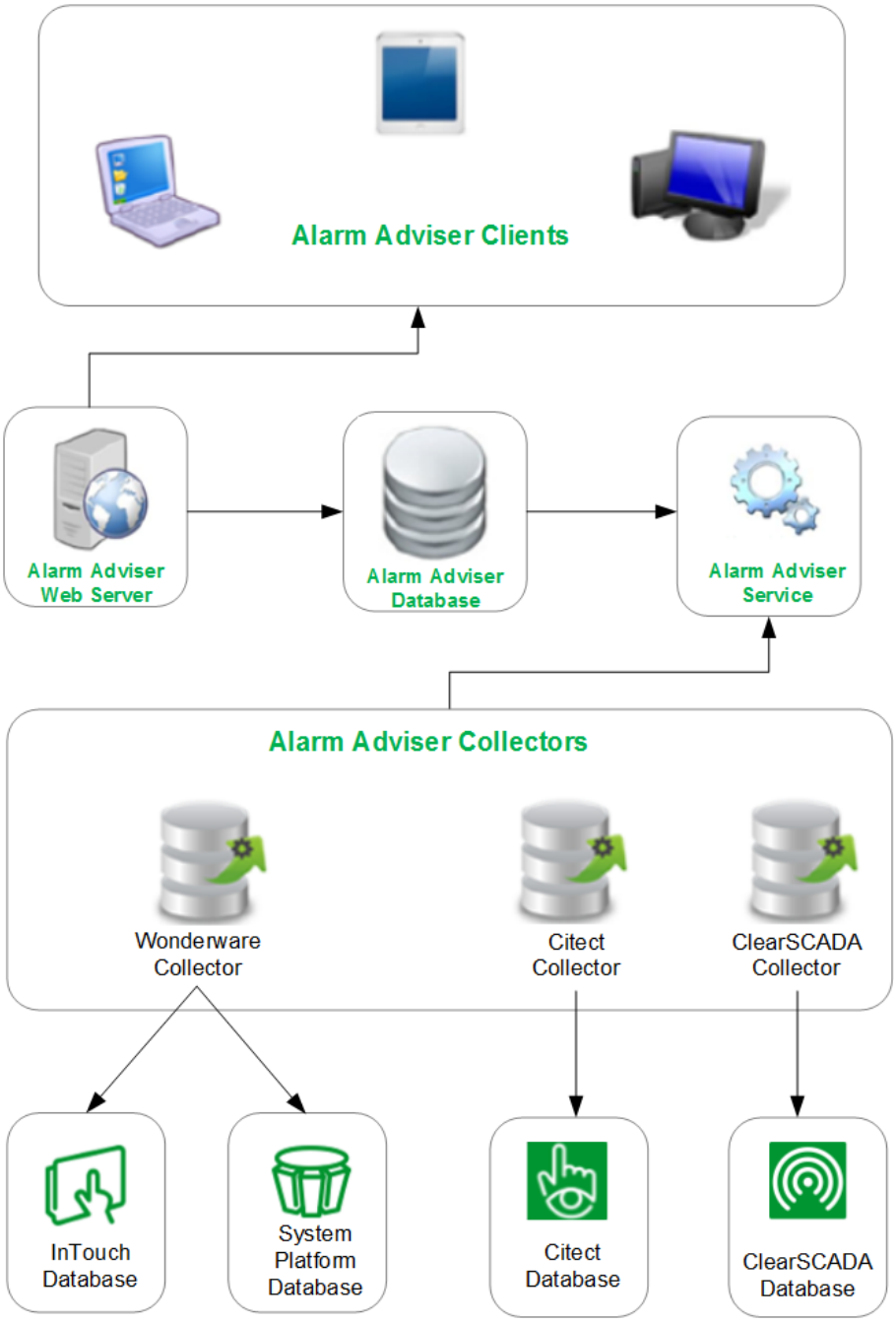
- Alarm Adviser Online Help - Provides information about using Alarm Adviser to analyze alarm data from your SCADA system. Online help is accessed via a web client.
- Schneider Electric Licensing Guide - Provides information about using the Schneider Electric License Manager.

Alarm Adviser Architecture

Alarm Adviser is an advanced, web-based application designed to provide functionality for interactive analysis of alarms within SCADA systems such as Wonderware System Platform™, Wonderware InTouch™, Vijeo Citect™ and ClearSCADA™. This section provides an overview of the underlying architecture that facilitates this functionality. For more information, see installing, deploying and configuring Alarm Adviser.

Introduction

The illustration below provides an overview of the Alarm Adviser architecture.



As illustrated above, Alarm Adviser comprises the following main components:

- Collectors
- Alarm Adviser Database

- Alarm Adviser Service
- Alarm Adviser Web Server
- Alarm Adviser Clients

These components interact with each other to retrieve, process and present alarm information. To understand the Alarm Adviser architecture, let us first trace the process flow, in a simplified manner, from gathering alarm information from a SCADA system to displaying the information to the end user. The table below lists the process flow tasks and the component responsible for performing each task.

STEP	TASK	COMPONENT RESPONSIBLE
1	Collect information from SCADA system.	Collector
2	Store the information collected in the Alarm Adviser database.	Service
3	Retrieve information from the Alarm Adviser Database.	Web Server
4	Perform the required calculations and transform the information for presentation.	
5	Display information to the end user.	Client

The role of each component and the tasks listed above are described in detail in the following sections.

The components listed in this section may be deployed in different ways. For detailed information about deployment, refer to Deploying Alarm Adviser.

Collectors

Alarm Adviser uses one or more Collectors to gather data for the Alarm Adviser Database.

The Collector runs as a Windows service. It retrieves alarm data from the SCADA system and uses a local cache to pre-process the data before transferring it to the Alarm Adviser Database.

Note: A Collector can collect data from a single SCADA system only.

Alarm Adviser Database

The Alarm Adviser Database is a centralized SQL Server analytics repository for alarm data retrieved from one or more SCADA systems being analyzed.

The database has been designed to accommodate alarm data from different sources, and optimized for quick alarm data retrieval. Both the Service and Web Server access this database.

Service

The Adviser Service handles connections to the Collectors and communicates with the Alarm Adviser Database.

The service runs as a Microsoft Windows service. It communicates with the Collector at a regular interval, determined by the Poll Rate, to retrieve alarm data. When the collector cache is reaching capacity due to a larger amount of alarm data to transfer, the Alarm Adviser service switches to use the faster Load Poll Rate. This helps maintain a balance between bandwidth utilization and alarm data updates.

The Poll Rate, Load Poll Rate and the number of records retrieved in each poll are settings that can be tuned as required.

Refer to the Online help for more information on configuring a system.

Web Server

The Web Server processes alarm data and shows alarm analytics. To do this, the Web Server accesses alarm data stored in the Alarm Adviser Database.

The Web Server is a standalone Windows service. It analyzes the gathered alarm data, performs the required calculations and serves data to web clients. This allows users to log on to the clients to analyze alarm data.

Clients

The Alarm Adviser clients provide access to the application interface through a web browser.

Alarm Adviser clients allow:

- users to access alarm data for analysis
- administrators to configure Alarm Adviser settings

For more information, see installing, deploying and configuring Alarm Adviser.

Chapter 2: Deploying Alarm Adviser

Introduction

You can deploy Alarm Adviser in several ways depending upon the size of your system and operational infrastructure. Its components may be installed on a single computer or be fully distributed across multiple computers, allowing you to conform to the performance, infrastructure and security requirements of your organization.

A complete Alarm Adviser installation comprises the following components:

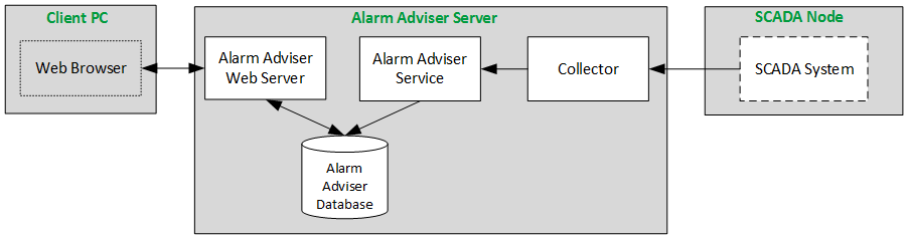
- Web Server
- Adviser Service
- SCADA Collector
- Alarm Adviser Database

Before you proceed further, it is recommended that you review the previous section to familiarize yourself with the function of each of the above components.

Deployment Scenarios

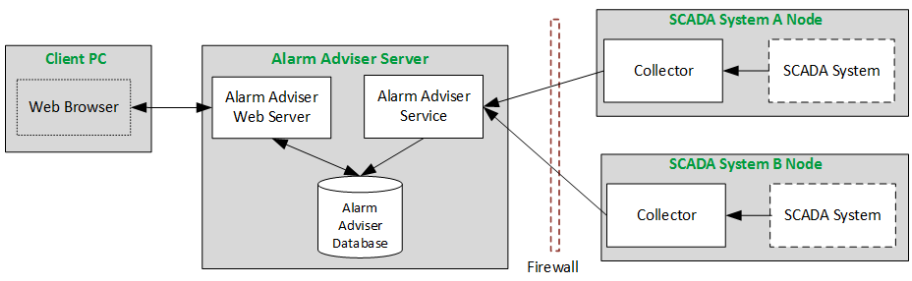
A typical deployment for Alarm Adviser would have the Alarm Adviser Service, the Web Server and database instance located together.

The Alarm Adviser Collector is typically located with the Alarm Adviser Service or with the remote data source system, that is, a SCADA system. It is recommended that the Adviser Service be installed on a computer other than the one on which the SCADA system is installed.



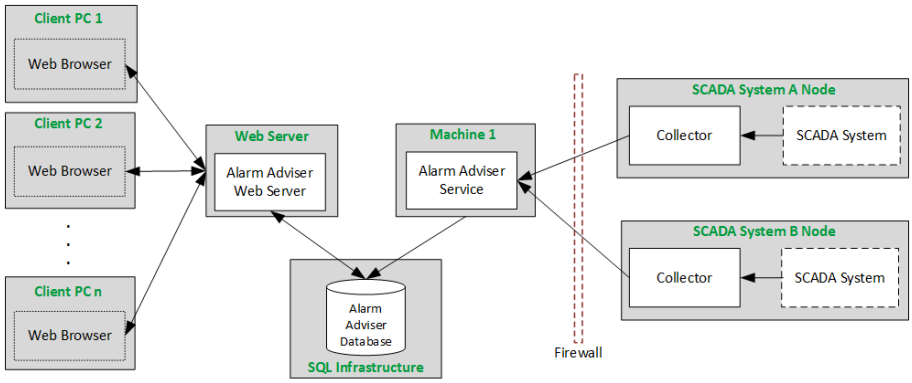
Consider the following when choosing the preferred location for the Collector:

- When Alarm Adviser is to be installed on the business network and the SCADA network is separate and could also have restricted access, the Collector is typically located on the SCADA network. The Collector is designed to only require an outbound connection to the Adviser Service to assist with network topologies using a firewall between the networks.
- When installing the Collector on the same computer as the SCADA system, take into account the amount of available capacity (CPU, memory, disk space) so as not to impact the SCADA system operation.
- When the Collector is separate from the SCADA system, the network load profile will consist of multiple smaller network packets.
- When the Collector is separate from the Adviser Service, the network load profile will consist of fewer larger network packets.
- When Alarm Adviser is connected to multiple systems, each Collector would be typically located with each SCADA system.



The Alarm Adviser Database should be located on a separate computer if your infrastructure requires databases to be on dedicated hardware, or if you are planning on running a large database and need additional hardware resources.

The Web Server should be located on a separate computer if your infrastructure requires specific deployments for a web server, or your system needs dedicated resources to support a large number of clients.



Chapter 3: Alarm Adviser System Requirements

Your Alarm Adviser DVD contains the files required to install Alarm Adviser. However, you should only install Alarm Adviser on computers that meet the required software and hardware specifications.

Server Requirements

This section describes the recommended minimum specification for servers on which Alarm Adviser components will be installed.

Required Software

SOFTWARE	SUPPORTED VERSION
Microsoft Windows® (32-bit or 64-bit)	7 SP1, 8.1, 10
Microsoft Windows® Server	2008 R2 SP1, 2012 R2, 2016
Microsoft SQL Server® (32-bit or 64-bit)	2012, 2016 SP1
Note: *SQL Server Express editions are supported for small systems.*	
Microsoft .NET Framework	4.5.1

Recommended Minimum Hardware*

CPU	Intel® Core™ 2 Duo 2.93GHz x 2
MEMORY	4GB
HARD DISK	8GB
NETWORK SPEED	Minimum 100Mbps

* Hardware recommendations are based on the Collector being installed on a computer different from the one on which the Adviser Service and Web Server are installed.

Note: If you are running Alarm Adviser on an Unlimited license, it is recommended that you evaluate the hardware requirements based on the amount of data you expect to collect.

Client PC Requirements

Confirm that the client PCs meet the minimum requirements to support one of the following web browsers required to run the Alarm Adviser web interface:

- Microsoft Internet Explorer 10
- Microsoft Internet Explorer 11
- Google Chrome 33 and higher
- Microsoft Edge

Chapter 4: Pre-Installation

Installation Prerequisites

The installation program checks for the following prerequisite software, and allows you to install them through the Alarm Adviser installer.

Note: Microsoft SQL Server needs to be pre-installed/available to complete Alarm Adviser configuration. SQL Server is not required if the computer has only a collector installed. Refer to System Requirements for more information.

Note: It is recommended that Microsoft .NET Framework be installed prior to running the Alarm Adviser installer. If it is not installed, the Alarm Adviser installer will automatically start the .NET Framework installer first.
The other prerequisites can be installed during the Alarm Adviser installation by clicking the **Install Prerequisites** button.

Prerequisite Software	Required by	Download Location
Microsoft .NET Framework™ 4.5.1	All Alarm Adviser components and collectors	http://www.microsoft.com/en-us/download/details.aspx?id=40779
Microsoft Visual C++® 10.0 SP1 x64 Running Redistributable **- OR -** Microsoft Visual C++® 10.0 SP1 x86 Running Redistributable	All Alarm Adviser components and ClearSCADA Collector	http://www.microsoft.com/en-us/download/details.aspx?id=13523 http://www.microsoft.com/en-us/download/details.aspx?id=8328

Prerequisite Software	Required by	Download Location
Microsoft SQL Server® Native Client 2012	Wonderware Collector	http://www.microsoft.com/en-us/download/details.aspx?id=35580
Microsoft SQL Server® 2012 SP1 Shared Management Objects (SMO)	All Alarm Adviser components and Wonderware Collector	
Microsoft SQL Server® 2012 SP1 System CLR Types		
Microsoft Visual C++® 2012 Update 4	Vijeo Citect Collector and ClearSCADA Collector	https://www.microsoft.com/en-us/download/details.aspx?id=30679
Microsoft Visual Studio C++® 2015 Update 3	ClearSCADA Collector	https://www.microsoft.com/en-us/download/details.aspx?id=53587

Supported SCADA Systems

Alarm Adviser currently supports the following SCADA systems:

SCADA System	Supported Versions
ClearSCADA	2012 R1 (v73) onwards
InTouch	2012 (v10.5) and higher (WWALMDB)
Vijeo Citect	7.20 SP5A and SP4, v7.40 SP1, 2015 (v7.50) SP1 with Patch 18 and 2016 (v8.00) July 11 2017 update

Before You Install the Vijeo Citect Collector

Different versions of Vijeo Citect exhibit different behavior when the Collector interacts with the SCADA system for collecting alarm data to be written to the Alarm Adviser Database. Version-specific differences in the behavior are described below. It is recommended that you review this section before you install the Vijeo Citect Collector.

Data Collection

Data collection occurs in polls for a particular period; this period is based on a last known sample time, and is limited to 2 weeks. The last known sample time is updated to the most recent alarm event time during the poll period, or the end of the poll period if no events have occurred.

VERSION	BEHAVIOR
v7.20 SP5A v7.20 SP4	Data collection for Vijeo Citect v7.20 is based on the time stamp of the alarm, that is, the time that the Vijeo Citect alarm server records as the time of occurrence. Samples arriving in subsequent polls with alarm event times before this time will not be collected. For timestamped analog, timestamped digital and timestamped alarms, the first alarm event cannot be collected. For time stamped alarms, the poll rate of the Collector needs to be set to double the slowest rate at which time stamped alarms arrive. The Vijeo Citect alarm server time and the time on the computer where the Collector is installed need to be synchronized. For more information, see the section Configuring the Vijeo Citect Collector.
v7.30 and higher	Data collection for Vijeo Citect v7.30 and higher versions is based on the time stamp of an event occurring, that is, the time that Citect records an alarm event (called the receipt time). Time stamped alarms that are reported by Vijeo Citect 7 days after they occur will not be collected for analysis in Alarm Adviser.

Upgrading Vijeo Citect

If you wish to upgrade your current Vijeo Citect version, you should review this section.

FROM VERSION	TO VERSION	BEHAVIOR
v7.20 SP5A	v7.40 SP1, v7.50	The Collector will process all active alarms stored by the SCADA system as it attempts to backfill 6 months of alarm data from the time of upgrade.
v7.20 SP4		All alarm data will be upgraded. The Collector will process active alarms stored by the SCADA system and backfill 6 months of historical alarm data.

Before You Install the ClearSCADA Collector

ClearSCADA Collectors can communicate with the following versions of ClearSCADA to collect alarm data:

- 2012 R1 (v73) onwards

Different versions of ClearSCADA exhibit different behavior when the Collector interacts with the SCADA system for collecting alarm data to be written to the Alarm Adviser Database. Version-specific differences in the behavior are described below. It is recommended that you review this section before you install the ClearSCADA Collector.

Data Collection

Data collection occurs in polls, at the **Read Interval** frequency specified on the ClearSCADA Collector's configuration form (*see* Configuring the ClearSCADA Collector). Regardless of the means of data collection (OPC AE, or Alarm Summary), the raised, cleared, and acknowledged times of the alarms are derived from the timestamps that are assigned to those alarms in the ClearSCADA system to which the Collector is connected.

VERSION	BEHAVIOR
2012 R1 (v73) to 2014 R1.1 (v75)	The ClearSCADA Collector uses OPC AE to retrieve alarm data from the ClearSCADA system to which it is connected. With this means of data retrieval, there is no back filling of historic alarm data. When the ClearSCADA Collector connects to a ClearSCADA system, the Collector receives information about alarms that are active, or inactive (cleared) but unacknowledged. The Collector can only generate event information for alarms that are in an active state; as such, it ignores any cleared but unacknowledged alarms as they contain insufficient information to generate valid data for Alarm Adviser. Once a connection is established with ClearSCADA, the ClearSCADA server will send the Collector updates on any new alarms and alarms for which the alarm condition has changed. The Collector will poll ClearSCADA at every **Read Interval** to check that the ClearSCADA server is still available. If the connection between ClearSCADA and the ClearSCADA Collector is lost, any alarms that become active but also clear during the period of communications loss will not be reported to Alarm Adviser. Alarms that were active when the connection was healthy, but then cleared while the connection was down, will be logged in Alarm Adviser as having cleared at the time that the connection to the ClearSCADA server is re-established.
2015 R1 (v76) onwards with the Alarm Summary feature disabled	The ClearSCADACollector uses OPC AE to retrieve alarm data from the ClearSCADA system to which it is connected. For more information, see above. **Note:** If the Collector is configured to collect alarms from a ClearSCADA system on which the Alarm Summary feature is supported but disabled, then later enabling the Alarm Summary feature will not automatically change the Collector's retrieval method. Instead, it will continue to use OPC AE for collecting alarm data. For information about changing the means by which a ClearSCADA Collector retrieves alarm data, see 'Upgrading ClearSCADA'.

<table>
<tr><th>VERSION</th><th>BEHAVIOR</th></tr>
<tr><td>2015 R1 (v76) onwards with the Alarm Summary feature enabled</td><td>
When the ClearSCADA Collector first connects to a ClearSCADA system on which the Alarm Summary feature is enabled, the Collector will retrieve the available alarm summary data for the last 180 days from that system. (If alarm summary data is only retained online in ClearSCADA for a shorter period than 180 days, the Collector will retrieve the available alarm summary data for that shorter period.)

Once a connection is established with ClearSCADA, the Collector will read the alarm summary data files for the last 56 days in ClearSCADA. The Collector will do this at the configured Read Interval. From those files, the Collector will retrieve data or new alarms and alarms for which the alarm condition has changed within the last 56-days. This will include data for alarms that no longer active in ClearSCADA (alarms for which the abnormal condition that triggered the alarm is no longer in effect, and the alarm entry has been acknowledged within the last 56 days).

If the connection between the Collector and ClearSCADA is lost, once the connection is re-established, the Collector will read the alarm summary records for the last 56-days. The Collector will then send the data relating to any new alarms and alarm changes that have occurred in that period to Alarm Adviser. (If the connection remains down for longer than 56 days, any alarms that became active but were also cleared and acknowledged more than 56 days ago will be omitted from the data collection once the connection is re-established.)

<table>
<tr><td>NOTICE</td></tr>
<tr><td>UNEXPECTED CHANGE IN DATA RETRIEVAL MECHANISM</td></tr>
<tr><td>Take care when considering whether to disable the Alarm Summary feature on a ClearSCADA system on which the feature was previously enabled. Be aware that once the Alarm Summary feature is disabled, the ClearSCADA collector will discard its stored alarm summary data, and will then start using OPC AE to collect alarm data.</td></tr>
<tr><td>Failure to follow these instructions can result in unexpected data retrieval changes.</td></tr>
</table>
</td></tr>
</table>

Upgrading ClearSCADA

If you wish to upgrade your current ClearSCADA version, you should review this section.

Unless stated otherwise in the table below, you do not need to stop the connection between the Collector and ClearSCADA during a ClearSCADA upgrade. Instead, the link between the Collector and ClearSCADA system will fail and then automatically re-establish, either by connecting to another available server (if multiple servers exist), or when the ClearSCADA upgrade completes.

From Version	To Version	Behavior
2012 R1 (v73) to 2014 R1.1 (v75)	2015 R1 (v76) onwards with the Alarm Summary feature disabled	When the connection re-establishes between the ClearSCADA Collector and ClearSCADA following the ClearSCADA upgrade, the Collector will resume using OPC AE to retrieve alarm data from ClearSCADA. The Collector will behave as described in the Data Collection section for 2012 R1 (v73) to 2014 R1.1 (v75). Any alarms that were already active before the upgrade and have not changed state during the upgrade will not be resent to Alarm Adviser.

From Version	To Version	Behavior
2012 R1 (v73) to 2014 R1.1 (v75)	2015 R1 (v76) onwards with the Alarm Summary feature enabled	With this type of upgrade, the Collector will continue to use OPC AE to retrieve alarm data from ClearSCADA, unless you perform the following procedure to change the means of data retrieval: 1. Use the Windows Service Manager (or equivalent) to stop the 'Alarm Adviser Collector ClearSCADA' service. 2. To retain the existing data for archival purposes, take a backup of the files at: AppData/Local/AlarmAdviser/ClearSCADACollector 3. Manually delete the directory and files at: AppData/Local/AlarmAdviser/ClearSCADACollector 4. Upgrade the ClearSCADA system, and ensure that the Alarm Summary feature is enabled (see the ClearSCADA online help). 5. Use the Windows Service Manager (or equivalent) to start the 'Alarm Adviser Collector ClearSCADA' service. When the connection is re-established between Alarm Adviser and ClearSCADA, the Collector will start to use the Alarm Summary means of retrieving alarm data from ClearSCADA. The Collector will behave as described in the Data Collection section for 2015 R1 (v76) onwards with the Alarm Summary feature enabled.
2015 R1 (v76) onwards with the Alarm Summary feature disabled	2015 R1 (v76) onwards with the Alarm Summary feature enabled	With this type of upgrade, the Collector will continue to use OPC AE to retrieve alarm data from ClearSCADA unless you perform the procedure shown above.

<table>
<tr><th>FROM VERSION</th><th>TO VERSION</th><th>BEHAVIOR</th></tr>
<tr><td>2015 R1 (v76) onwards with the Alarm Summary feature enabled</td><td>2017 (v8) onwards with the Alarm Summary feature enabled</td><td>Following the ClearSCADA upgrade, the Collector will continue to use the Alarm Summary means of data retrieval.

When the connection re-establishes between the ClearSCADA Collector and ClearSCADA following the ClearSCADA upgrade, the Collector will back fill up to 56-days' worth of data that occurred while the connection was down. (If the connection remains down for longer than 56 days, any alarms that became active but were also cleared and acknowledged more than 56 days ago will be omitted from the data collection once the connection is re-established.)

Data collection will then continue as usual, at the configured poll interval.</td></tr>
<tr><td>2015 R1 (v76) onwards with the Alarm Summary feature enabled</td><td>2015 R1 (v76) onwards with the Alarm Summary feature disabled</td><td>With this type of upgrade, the Collector will discard the stored alarm summary data and start using OPC AE to collect alarm data.

<table><tr><td>NOTICE</td></tr><tr><td>UNEXPECTED CHANGE IN DATA RETRIEVAL MECHANISM</td></tr><tr><td>Take care when considering whether to disable the Alarm Summary feature on a ClearSCADA system on which the feature was previously enabled. Be aware that once the Alarm Summary feature is disabled, the ClearSCADA collector will discard its stored alarm summary data, and will then start using OPC AE to collect alarm data.</td></tr><tr><td>Failure to follow these instructions can result in unexpected data retrieval changes.</td></tr></table></td></tr>
</table>

Chapter 5: Installing Alarm Adviser

The Alarm Adviser installer installs the Service, Web Server and Collectors. All components may be installed together or separately, depending on the desired deployment configuration.

Note: The licensing components need to be installed on the same computer as the Adviser Service.

To install Alarm Adviser:

1. Start the installation.
2. Depending upon your installation, refer to the relevant section:

- Installing Alarm Adviser Components
- Installing the Wonderware Collector
- Installing the Vijeo Citect Collector
- Installing the ClearSCADA Collector

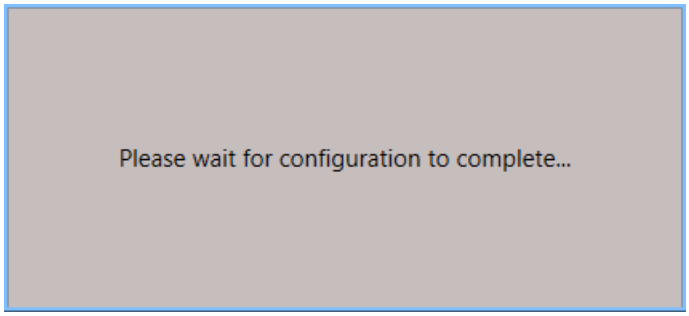
3. Complete the installation.

Starting the Installation

1. Insert the CD into your CD-ROM drive, and run **Setup.exe**. The Alarm Adviser splash screen appears.



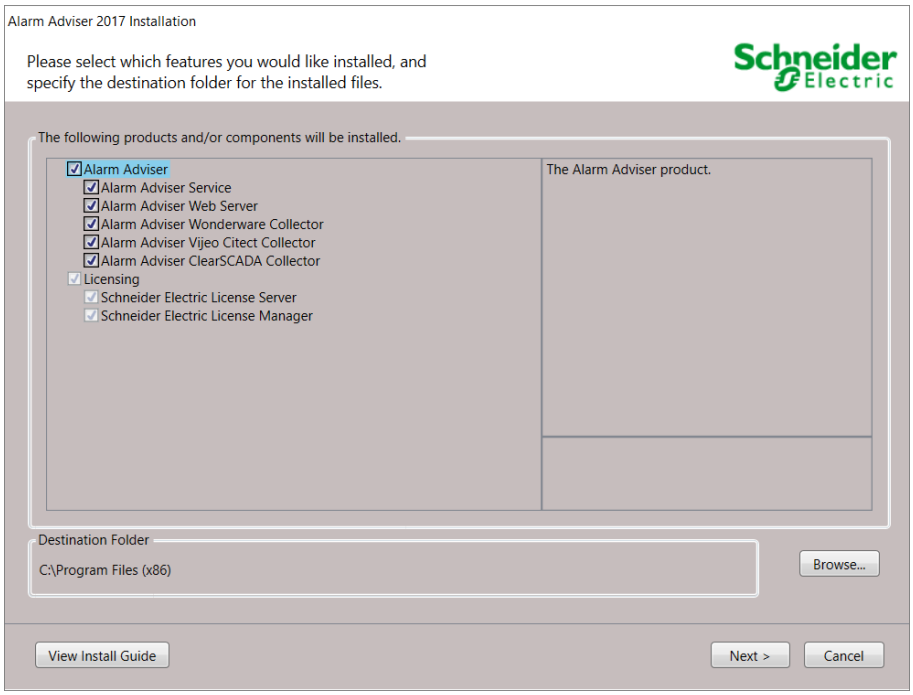
The following message appears as the installer prepares to start the installation.



2. The installer shows a screen that allows you to select the components you wish to install.

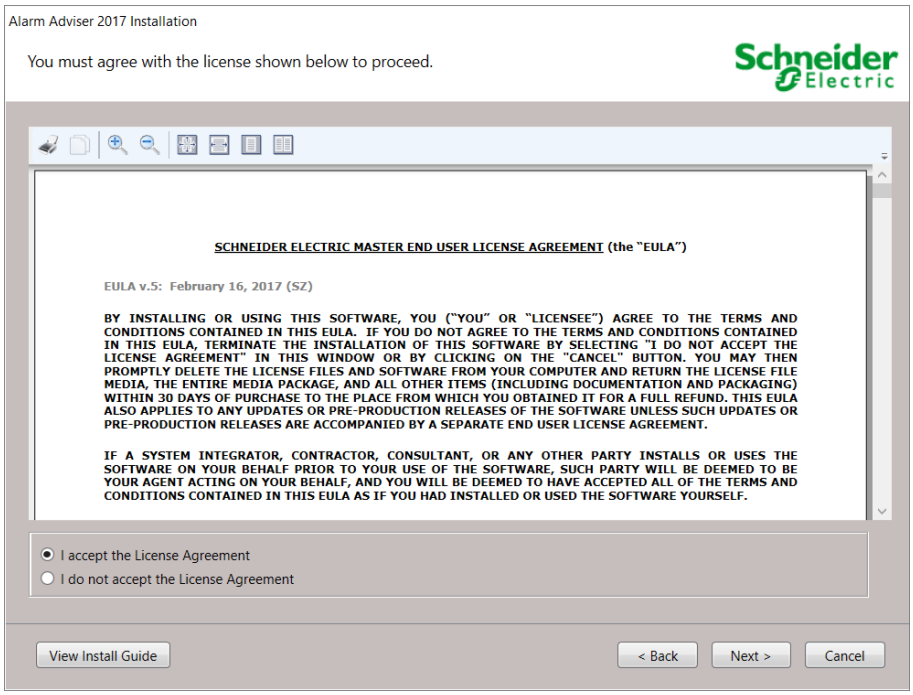
Installing Alarm Adviser Components

1. Start the installation.
2. If you wish to install all Alarm Adviser, Licensing components and the Collector(s) on a single computer, select the **Alarm Adviser** and **Licensing** check boxes.



By default, Alarm Adviser is installed into the program files location such as *c:\Program Files (x86)*. To select a different folder for installation, click **Browse**.

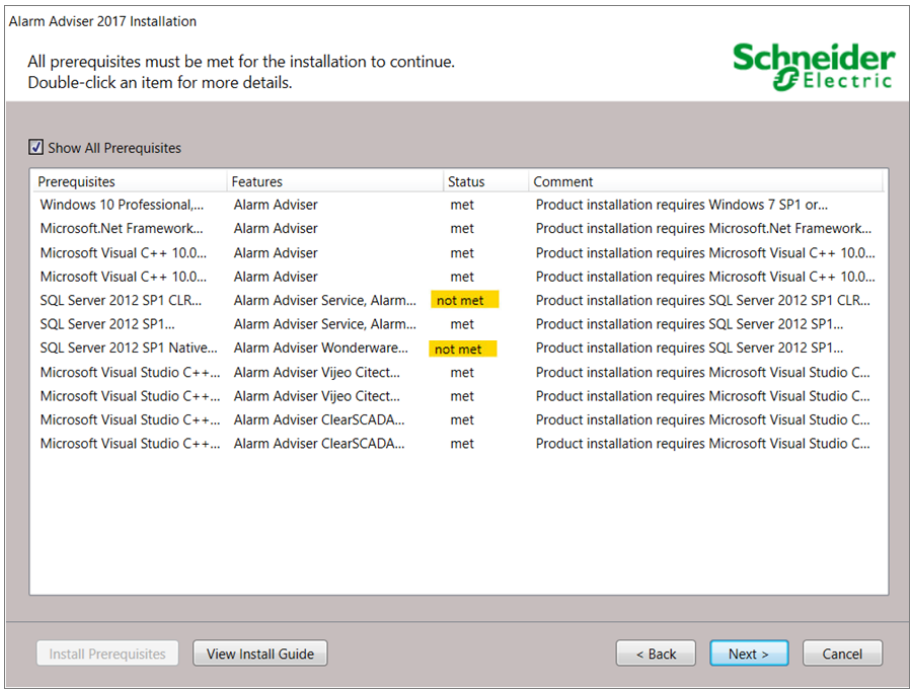
3. Click **Next**. The License Agreement screen appears. Read the license agreement carefully, and select the **I accept the License Agreement** option to continue. Selecting **I do not accept the License Agreement** will prevent you from proceeding with the installation.



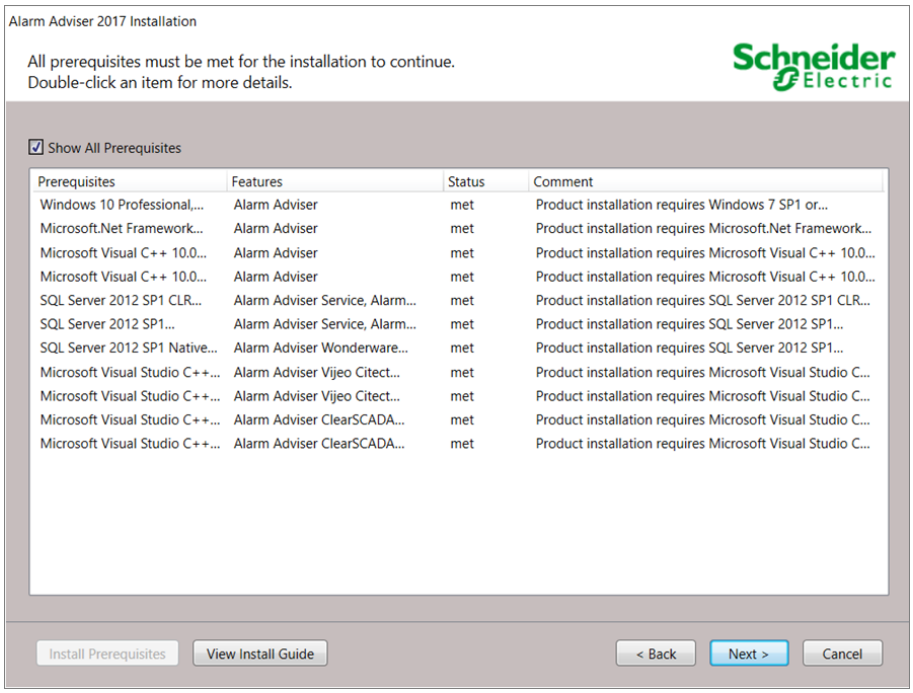
4. Click **Next**. The Installation Prerequisites screen appears.
   The installation program checks whether or not the installation prerequisites are met, and shows the missing prerequisites.

Note: The **Next** button will be disabled until all prerequisites are met.

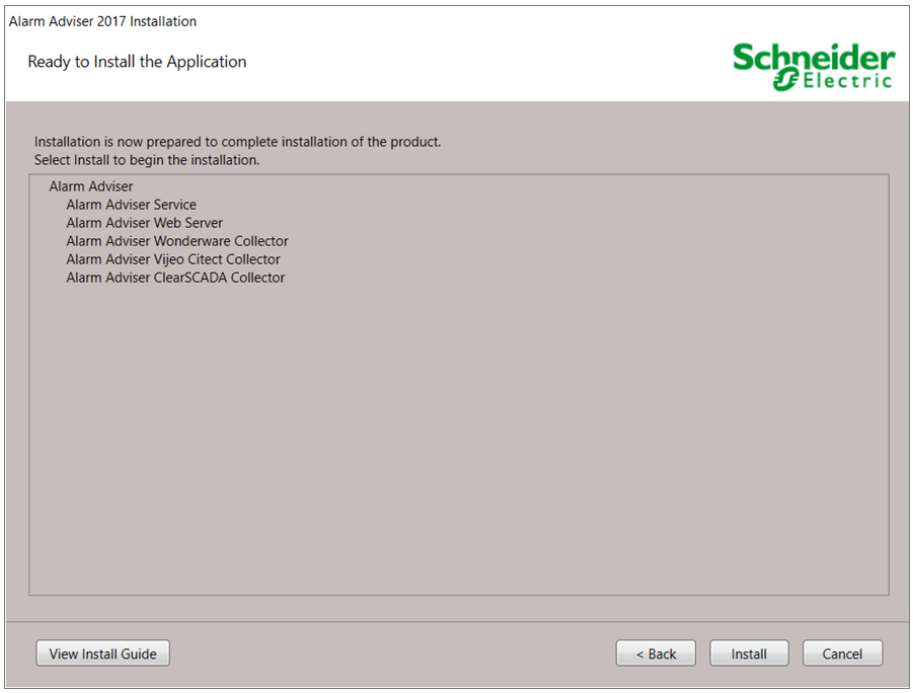
5. To view a list of all prerequisites, select the **Show All Prerequisites** check box.



6. Click **Install Prerequisites** to install the missing prerequisite(s).



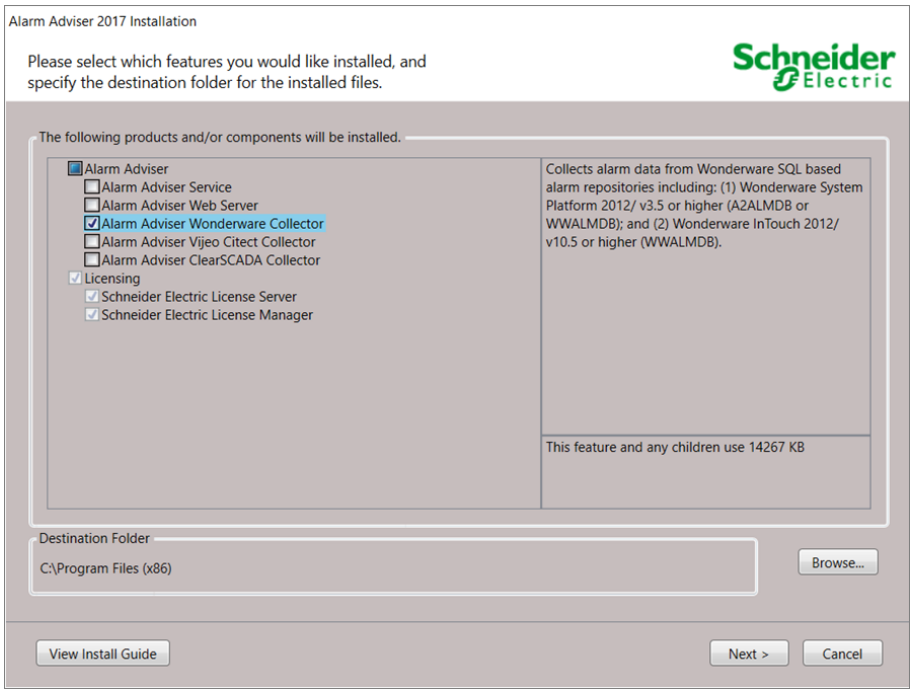
7. Click **Next**. The installer shows the list of components selected for installation.



Installing the Wonderware Collector

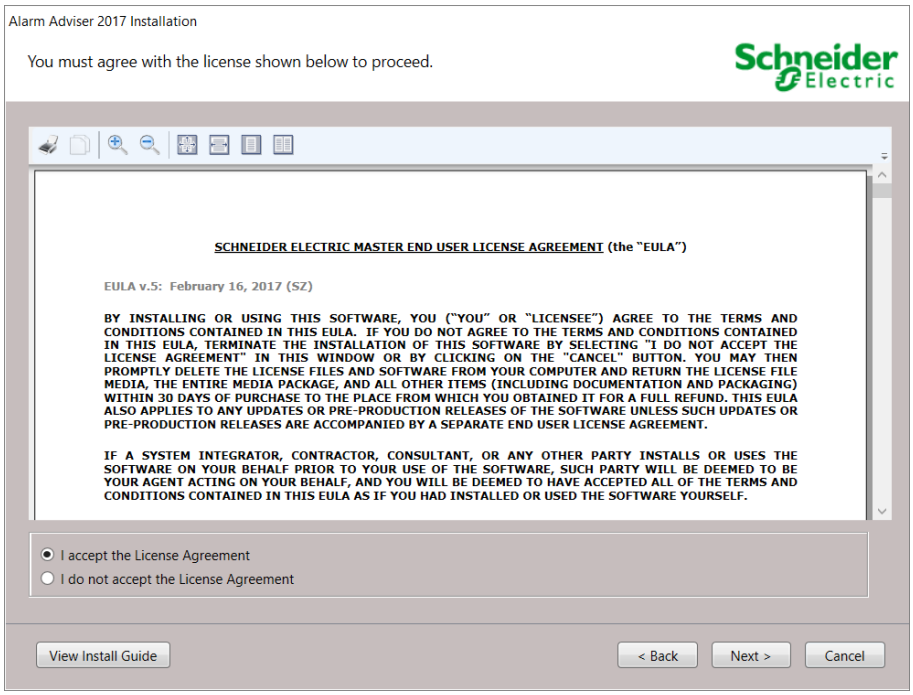
To install the Wonderware Collector:

1. Start the installation.
2. Select the Alarm Adviser Wonderware Collector check box.

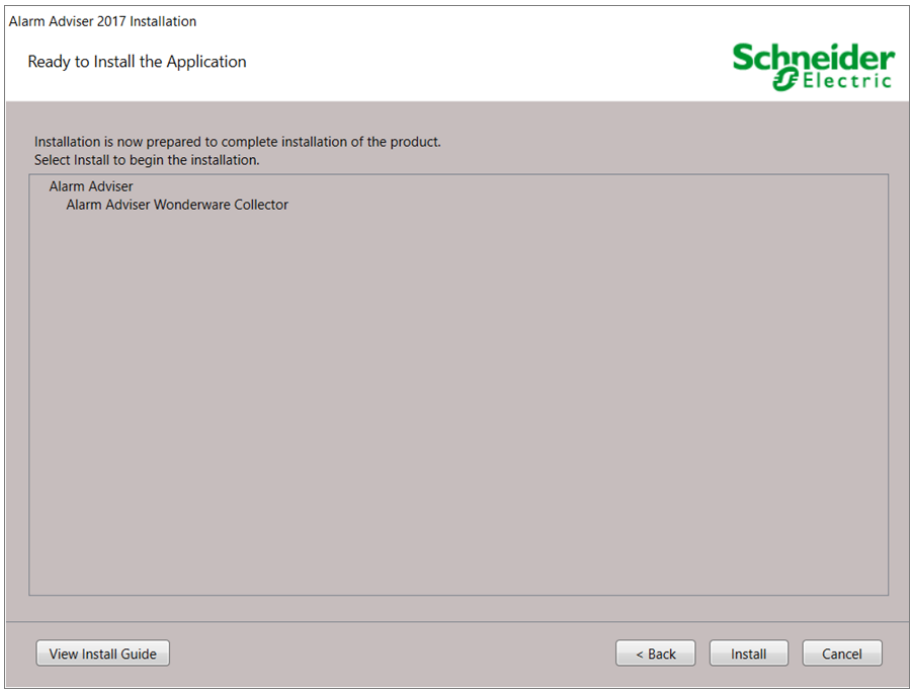


By default, the Collector is installed into the program files location such as *c:\Program Files (x86)*. To select a different folder for installation, click **Browse**.

3. Click **Next**. The License Agreement screen appears. Read the license agreement carefully, and select the **I accept the License Agreement** option to continue. Selecting **I do not accept the License Agreement** will prevent you from proceeding with the installation.



4. Click **Next**. The Installation Prerequisites screen appears.
   The installation program checks whether or not the installation prerequisites are met, and shows the missing prerequisites.
5. To view a list of all prerequisites, select the **Show All Prerequisites** check box.
6. Click **Install Prerequisites** to install the missing prerequisite(s).
7. Click **Next**. The installer shows the list of components selected for installation.

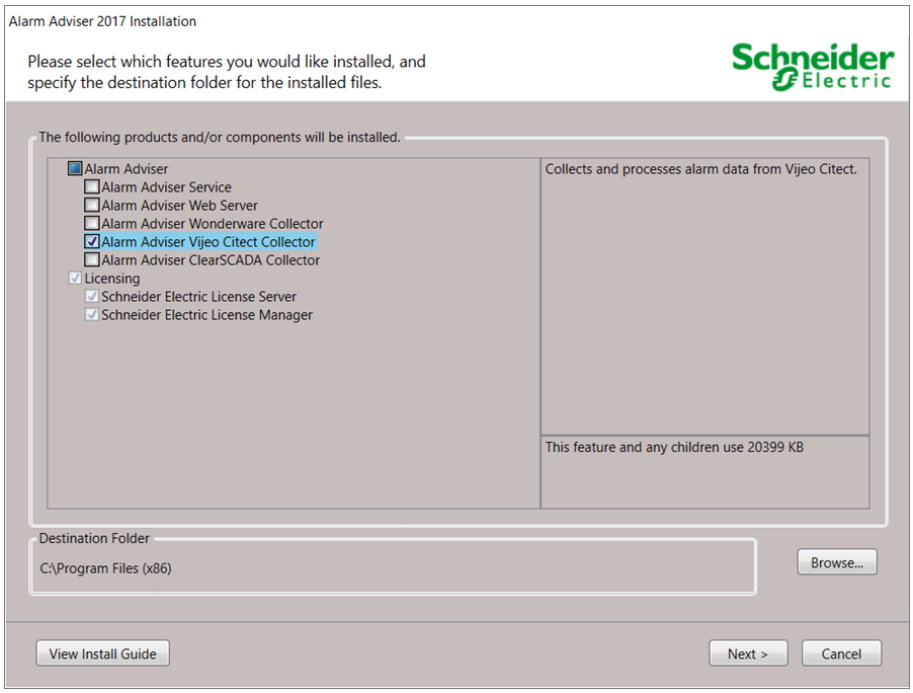


8. Complete the installation.

Installing the Vijeo Citect Collector

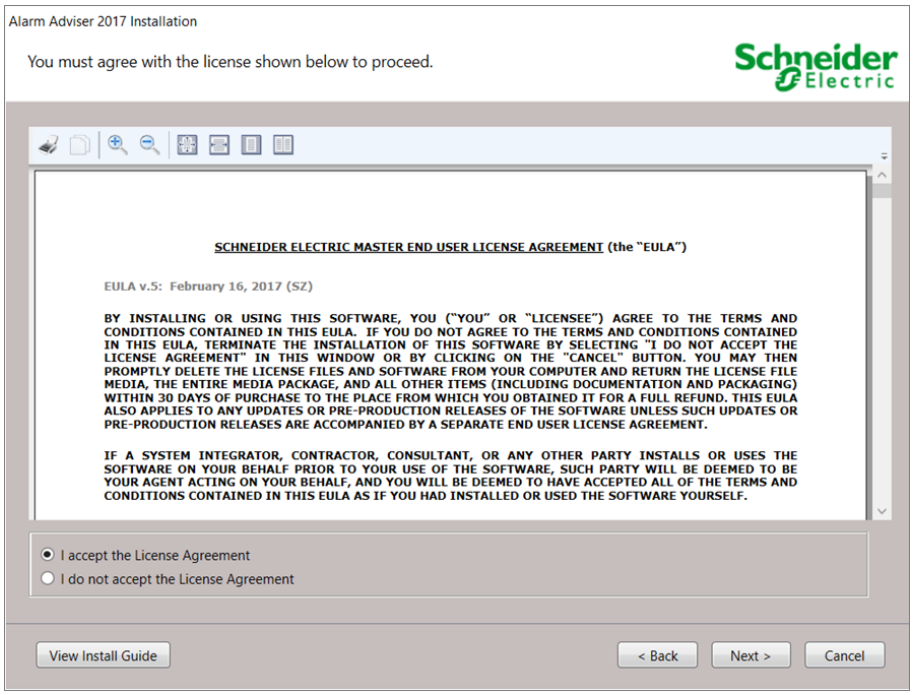
To install the Vijeo Citect Collector:

1. Start the installation.
2. Select the **Alarm Adviser Vijeo Citect** Collector check box.

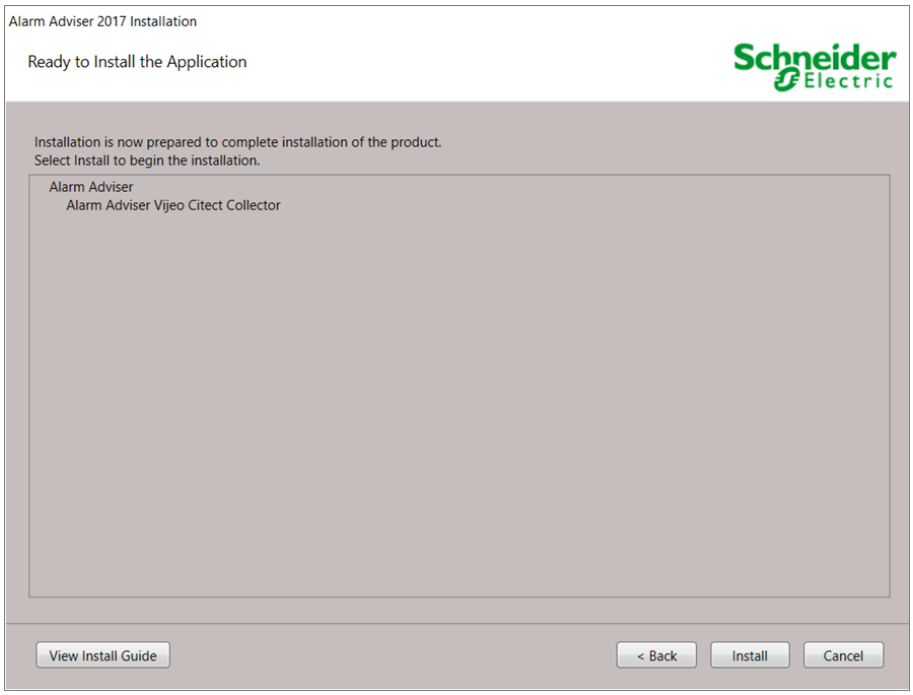


By default, the Collector is installed into the program files location such as *c:\Program Files (x86)*. To select a different folder for installation, click **Browse**.

3. Click **Next**. The License Agreement screen appears. Read the license agreement carefully, and select the **I accept the License Agreement** option to continue. Selecting **I do not accept the License Agreement** will prevent you from proceeding with the installation.



4. Click **Next**. The Installation Prerequisites screen appears. To view a list of all prerequisites, select the **Show All Prerequisites** check box.
   The installer checks for the prerequisite software, and installs the required software before the Vijeo Citect Collector installation starts.
5. Click **Next**. The installer shows the list of components selected for installation.

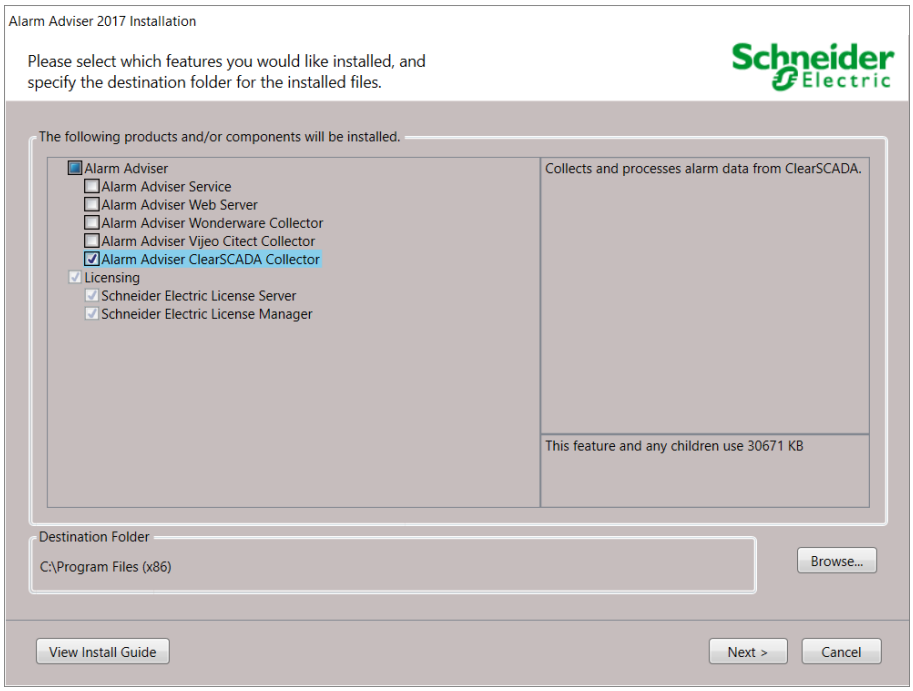


6. Complete the installation.

Installing the ClearSCADA Collector

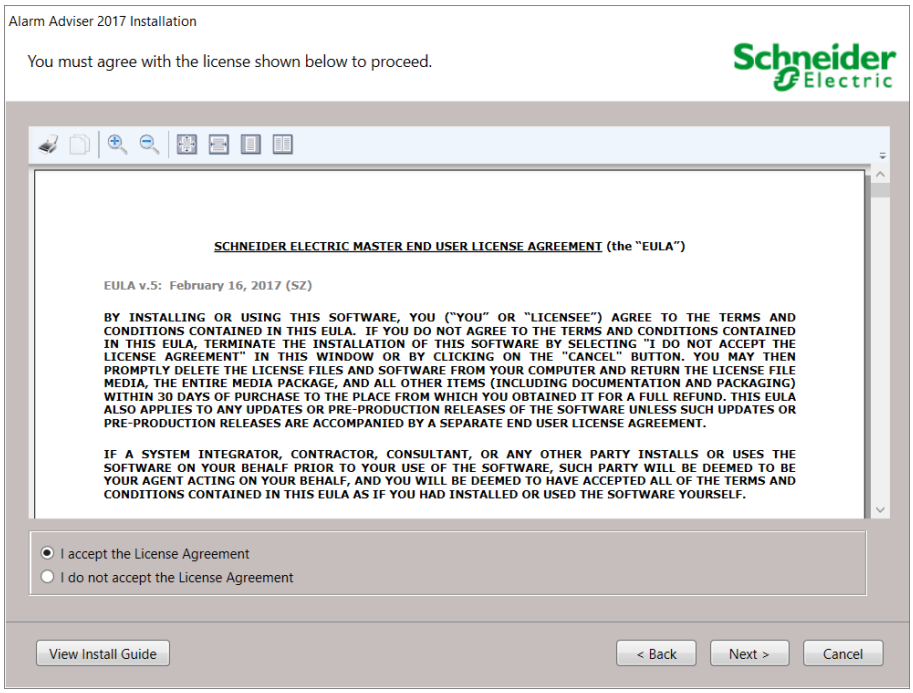
To install the ClearSCADA Collector:

1. Start the installation.
2. Select the **Alarm Adviser ClearSCADA** Collector check box.

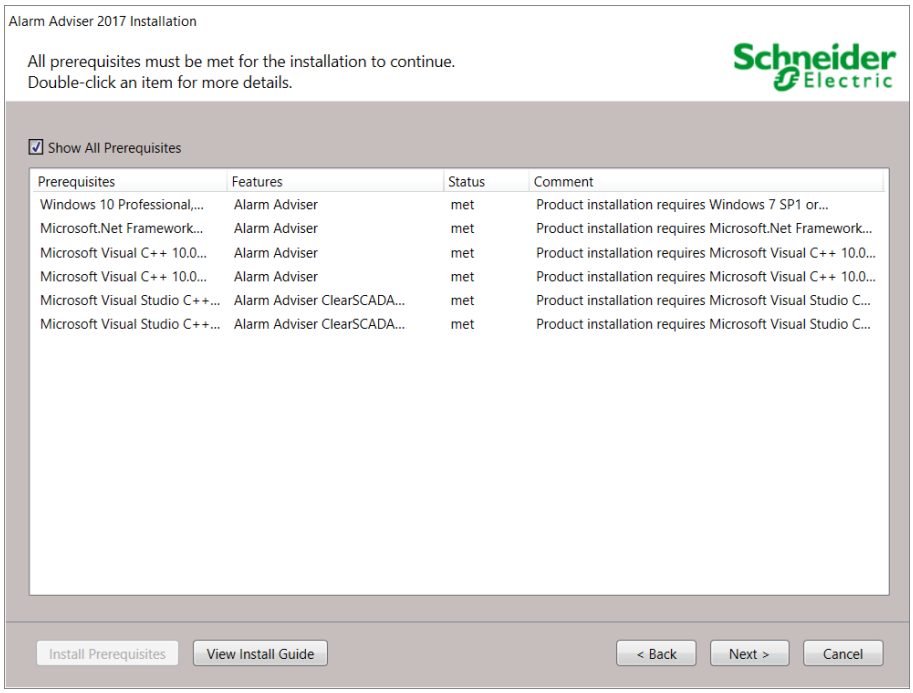


By default, the Collector is installed into the program files location such as *c:\Program Files (x86)\Schneider Electric*. To select a different folder for installation, click **Browse**.

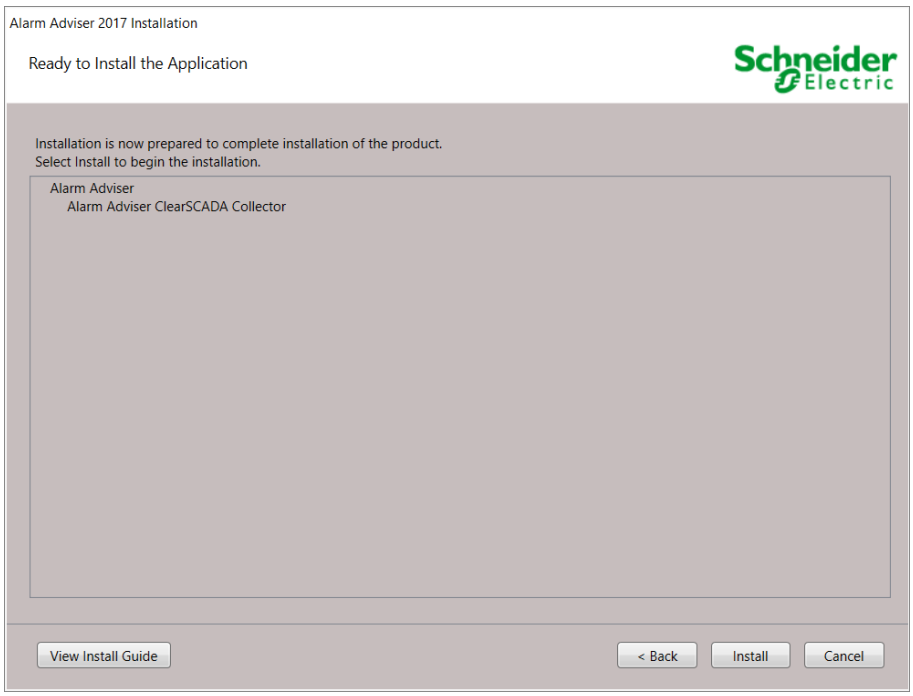
3. Click **Next**. The License Agreement screen appears. Read the license agreement carefully, and select the **I accept the License Agreement** option to continue. Selecting **I do not accept the License Agreement** will prevent you from proceeding with the installation.



4. Click **Next**. The Installation Prerequisites screen appears. To view a list of all prerequisites, select the **Show All Prerequisites** check box.
   The installer checks for the prerequisite software, and installs the required software before the ClearSCADA Collector installation starts.



5. Click **Next**. The installer shows the list of components selected for installation.

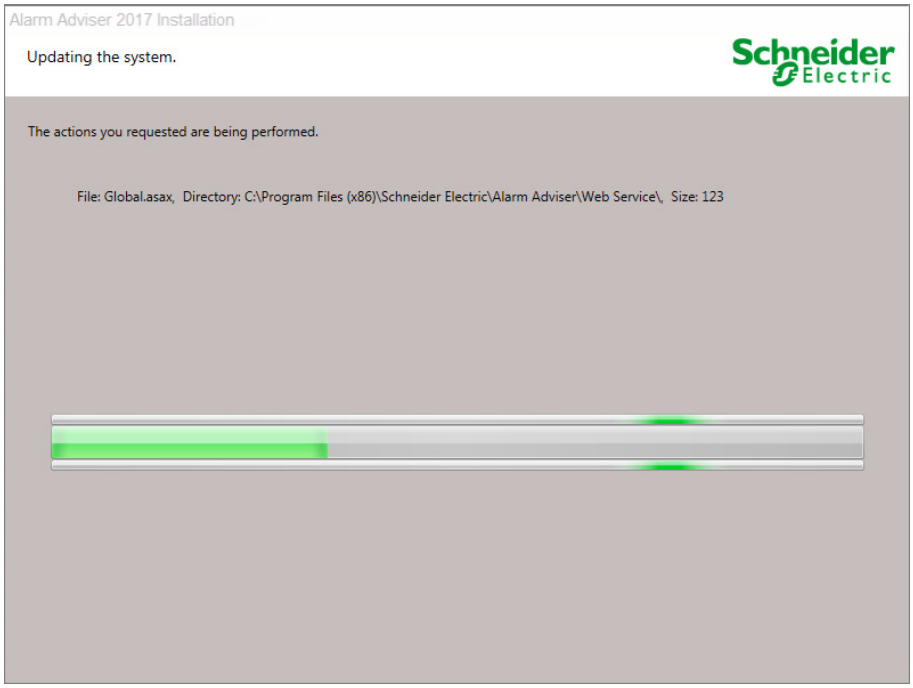


6. Complete the installation.

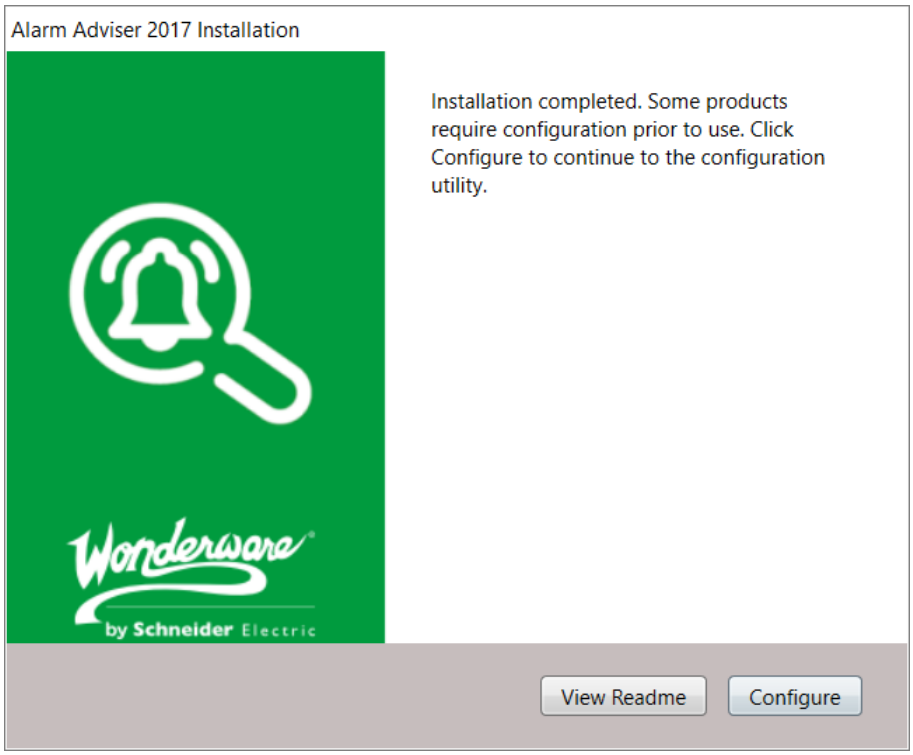
Completing the Installation

To complete your installation:

1. Click **Install**. The progress bar appears.



2. After the installation is over, the Installation Completed dialog box appears. Click **View Readme** to open the Readme.html file, which contains details about Alarm Adviser features, or click **Configure**. For more information on configuring Alarm Adviser, refer to the chapter Configuring Alarm Adviser.


Alarm Adviser 2017 Installation
Installation completed. Some products require configuration prior to use. Click Configure to continue to the configuration utility.
Wonderware
by Schneider Electric
View Readme
Configure

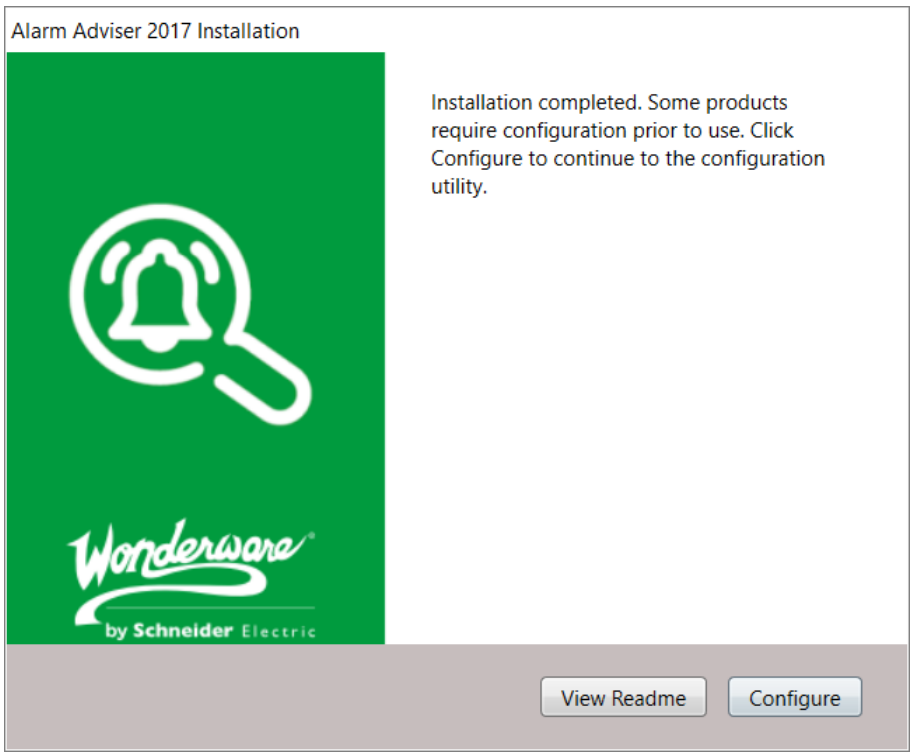
Chapter 6: Configuring Alarm Adviser

After you install Alarm Adviser components, use the Configurator utility to configure the components to communicate with each other, and access the required databases to retrieve and process alarm data.

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To start the Configurator:

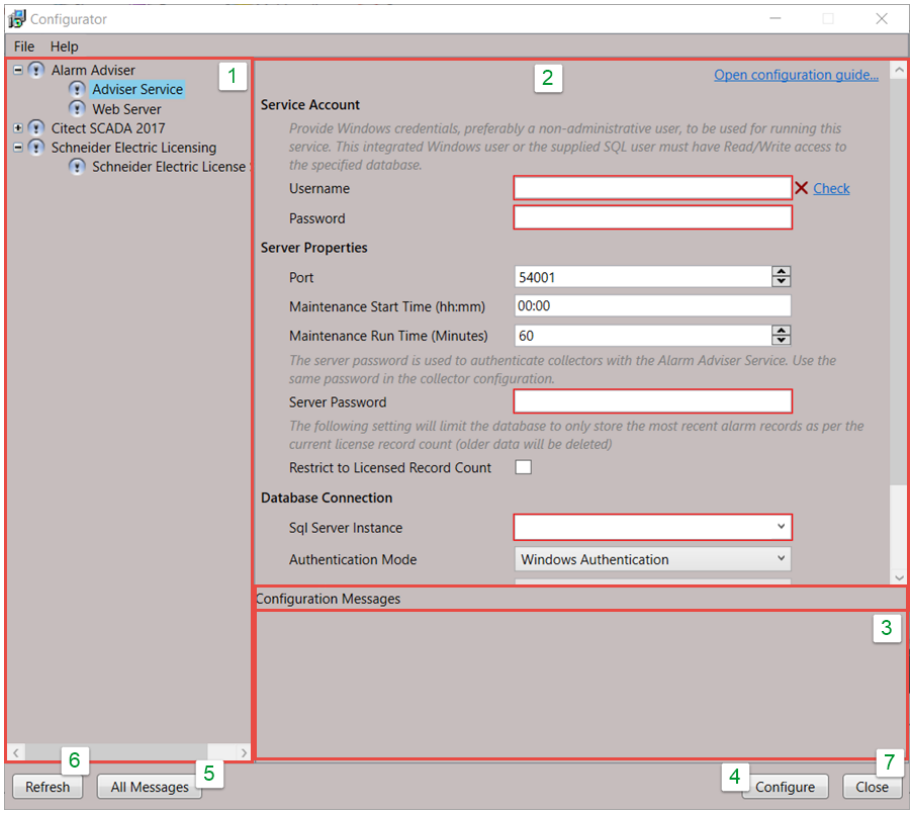
1. Click **Configure** after Alarm Adviser is installed.



The Configurator dialog box lists all product components you have installed.

Alternatively, start the Configurator from the Windows **Start** menu by selecting **Wonderware > Common > Configurator**.

Configurator Interface



The Configurator comprises the following sections:

REF	SECTION	DESCRIPTION
1	Application Components	Shows the Alarm Adviser and licensing components that need to be configured. Click on a component to view and edit the settings for the component.

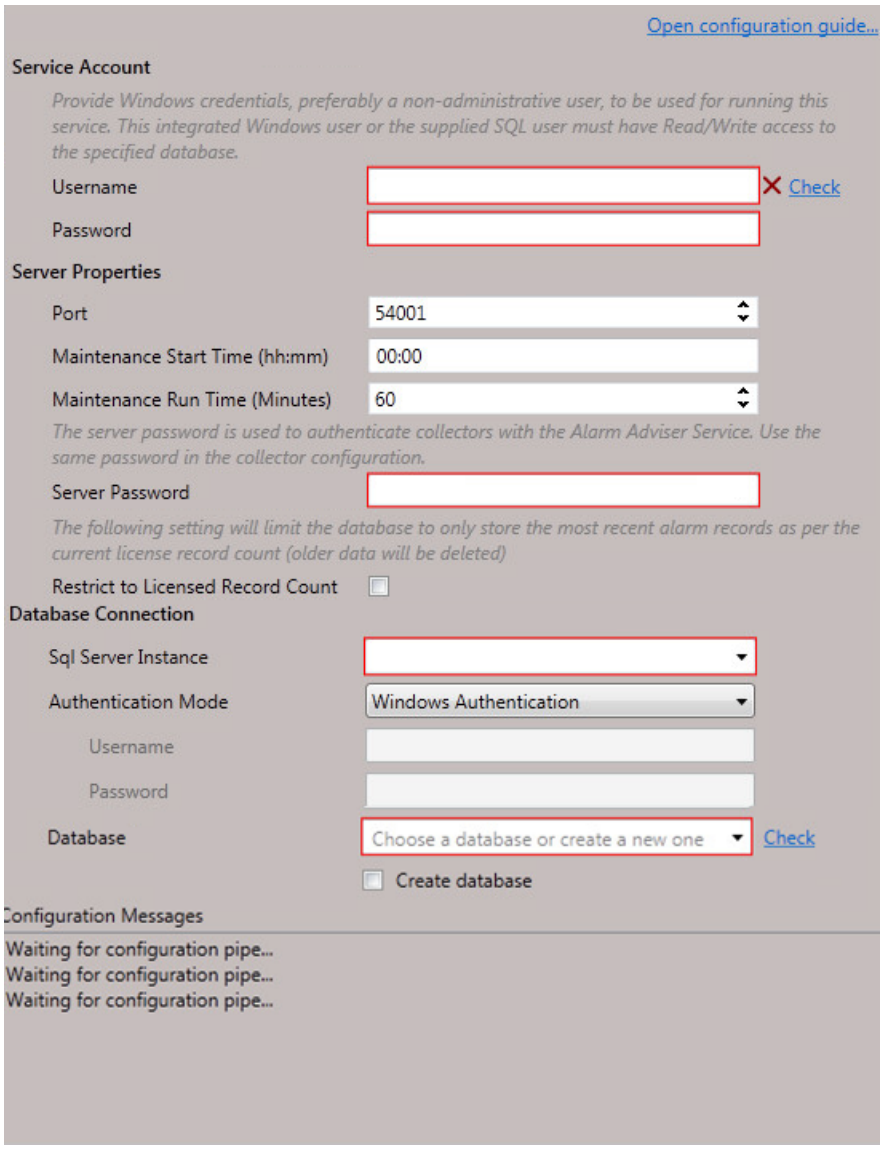
REF	SECTION	DESCRIPTION
2	Settings	Shows the configuration settings for the selected application component. Click the **Open configuration guide** link if you need information about the settings. This link opens the Alarm Adviser Installation and Configuration guide.
3	Configuration Messages	Shows the status of the configuration including any error messages resulting from invalid settings.
4	**Configure** button	Click to save and apply the settings for the selected component.
5	**All Messages** button	Click to view a list of all Configurator status and error messages. Messages are displayed in a tabular format in the Configurator Message Lists window with the following information: • Product Name • Feature Name • User Name • Time • Action • Short Message You can export this list to a file for reference by clicking the **Export to file** button. Click **Close** to exit the Configurator Message Lists window.
6	**Refresh** button	Click to refresh the screen to display updated settings.
7	**Close** button	Click to exit the Configurator.

Configuring the Alarm Adviser Service

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To configure the Alarm Adviser Service:

1. Open the Configurator utility.
2. In the left pane, click **Alarm Adviser > Adviser Service**. The configuration options appear in the right pane.



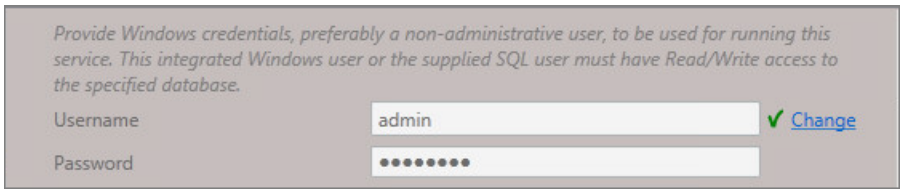
3. In the **Service Account** area, type the user account information to log on as and run the service. Do the following:

    a. In the **Username** box, type the name of the Windows local or domain account. If using a domain account, type it in the format *<domain name>\<user name>*.
    By default, the installer will install this service with a ***LocalSystem*** account, but the service will not run until it is configured with a Windows local or domain account in the Configurator. Your administrator can set up the Service Account with minimum privileges to run the Alarm Adviser Service.

Note: You may use the same account or a different account for running the Web Server, the Alarm Adviser Service and the Collector.

b. In the **Password** box, type the password for the Windows or domain user specified in the **Username** box.

c. Click **Check** to verify the user account and password. If the credentials are not valid, a red cross appears next to the **Username** box. If valid, the Configurator will display a green tick mark next to the box.



If you wish to use a different account, click **Change**, provide the account details, and click **Check**. Click **Configure**. The service is started.

Note: Service account credentials that are specified first for an Alarm Adviser service will be used for the other services unless changed manually. If you configure Adviser Service first and specify credentials for the service account, the same credentials will be applied as default for Web Server and Collector unless you change it manually while configuring those services.

4. In the **Server Properties** area, specify the details of the server on which the Adviser Service is installed. Do the following:

   a. In the **Port** box, type the TCP port number on which the Service will be hosted. Confirm that the port specified is not in use by another application. The default is 54001.
   It is possible to change the port at a later stage if required, but you need to check that all Collectors point to this port.

   b. In the **Maintenance Start Time (hh:mm)** box, type the time at which the Service should start running maintenance tasks each day. One of the main tasks is to delete records that are out of the data set range.
   Set the **Maintenance Start Time** to a time during the day when there is minimal user activity on the Alarm Adviser application.

c. In the **Maintenance Run Time (Minutes)** box, specify the duration for which maintenance will run. If a task is still in progress at the end of this duration, it will be resumed when the next maintenance operation starts. The default is 60 minutes.

d. In the **Server Password** box, type a password to be used to establish communication between the Collector and the Service. The specified keys need to be the same between the Alarm Adviser services.

e. Select the **Restrict to Licensed Record Count** check box to limit the number of records stored in Alarm Adviser Database to the most recent alarm records according to your Alarm Adviser license. When the record count is reached, older records will be deleted and new records will be written to the database.
Leaving this check box unselected will result in the web client showing a persistent message that the record count has been exceeded. Upgrade to a different license if you need to analyze a higher number of records in Alarm Adviser.

5. In the **Database Connection** area, specify database details for the Alarm Adviser Database. Do the following:

a. In the **SQL Server Instance** box, type or select the name of the SQL Server computer and instance on which the Alarm Adviser Database is located.

b. Select the **Authentication Mode**. Select **SQL Authentication** to use SQL Server credentials to connect to the Alarm Adviser database, or **Windows Authentication** to use the Service Account credentials to connect to the Alarm Adviser database. The Configurator automatically uses the credentials provided in the Service Account area.
If you selected SQL Server Authentication, type the SQL Server account **Username** and **Password** to be used to connect to the Alarm Adviser database.

Note: If Windows Integrated authentication mode is selected, both the **Username** and **Password** options will be disabled. The **Username** box will display the name of the Windows user account that will be used when the service is running.

c. In the **Database** box, type or select the name of the Alarm Adviser database. The drop down list shows all non-system databases on the selected SQL Server instance. Click **Configure**.
Alternatively, you can select a pre-created database, and then click **Configure**. The Configurator creates the database schema for the selected database.
If you want to create a new database on the selected SQL server instance, type a name for

the database, select **Create database** and then click **Configure**. The Configurator creates the new database using the credentials specified.
If additional privileges are required for creating the database, a dialog box appears. Enter the required credentials to proceed.

Note: Specific access may be required for the Alarm Adviser Database if it is installed remotely. Contact your database administrator to obtain access.

d. Click **Check** to confirm that you are able to connect to the Alarm Adviser database. While attempting to connect to the database, status messages are displayed in the Configuration Messages section. On successful connection, the message "Success" appears in the Configuration Messages section. If the credentials are not valid, a red cross appears next to the **Database** box. If valid, the Configurator shows a green tick mark next to that box.

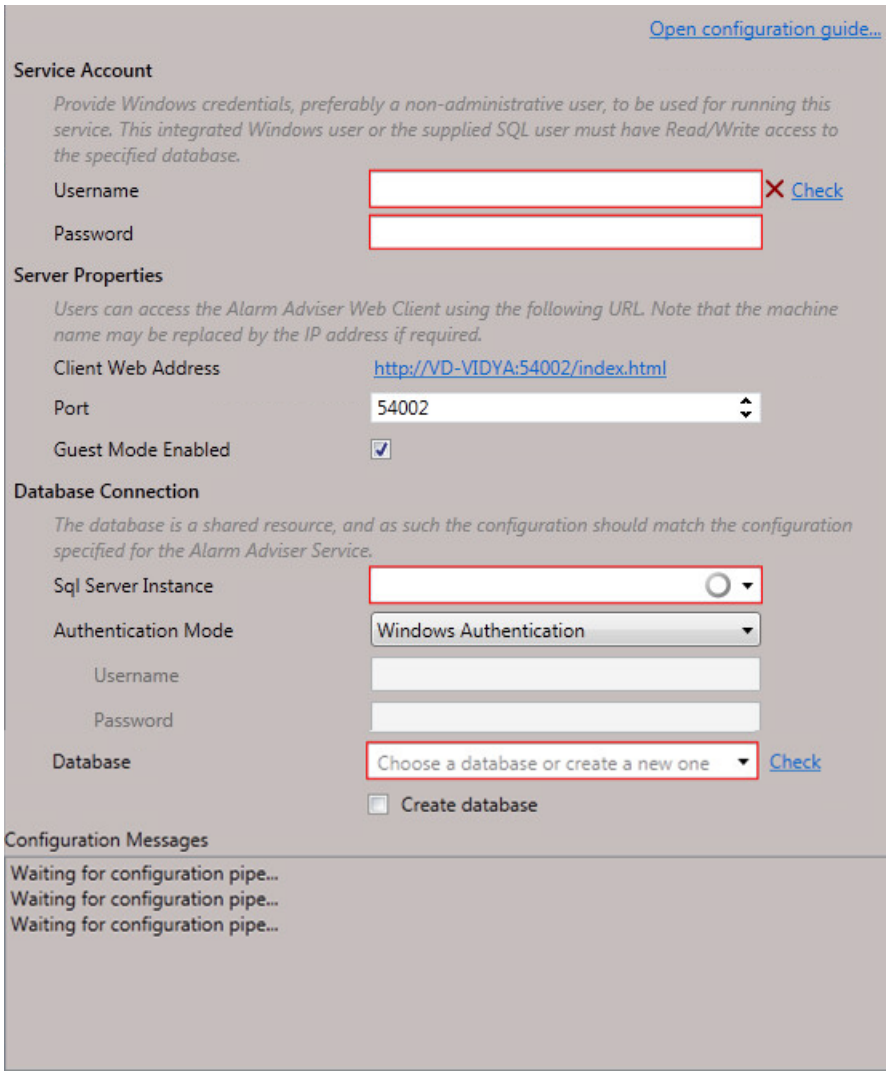
6. Click **Configure**. On successful configuration, the icon next to the service name in the left pane changes from  to  .

Configuring the Alarm Adviser Web Server

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To configure the Alarm Adviser Web Server:

1. Open the Configurator utility.
2. In the left pane, click **Alarm Adviser > Web Server**. The configuration options appear in the right pane.



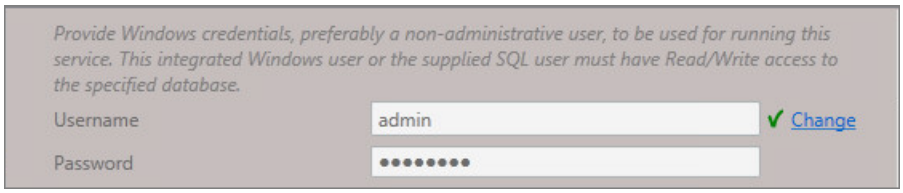
3. In the **Service Account** area, type the user account information to log on as and run the service. Do the following:

    a. In the **Username** box, type the name of the Windows local or domain account. If using a domain account, type it in the format *<domain name>\<user name>*.
    By default, the installer will install this service with a ***LocalSystem*** account, but the service will not run until it is configured with a Windows local or domain account in the Configurator. Your administrator can set up the Service Account with minimum privileges to run the Alarm Adviser Web Server service.

Note: You may use the same account or a different account for running the Web Server, Alarm Adviser Service and the Collector.

b. In the **Password** box, type the password for the Windows or domain user specified in the **Username** box.

c. Click **Check** to verify the user account and password. If the credentials are not valid, a red cross appears next to the **Username** box. If valid, the Configurator will display a green tick mark next to the box.



If you wish to use a different account, click **Change**, provide the account details, and click **Check**. Click **Configure**. The service is started.

Note: Service account credentials that are specified first for an Alarm Adviser service will be used for the other services unless changed manually. If you configure Adviser Service first and specify credentials for the service account, the same credentials will be applied as default for Web Server and Collector unless you change it manually while configuring those services.

4. In the **Server Properties** area, specify the details of the server on which the Web Server service is installed. Do the following:

a. The **Client Web Address** shows the URL for accessing the Alarm Adviser web client. The URL is made up of the computer name and port number specified in the **Port** box below.

b. In the **Port** box, type the TCP port number on which to host the Alarm Adviser application. Confirm that the port specified is not in use by another application. The default is 54002. This port number is used in the URL for connecting to the Web Server from a web browser. It is possible to change the port at a later stage if required, but you need to check that web clients point to the correct port in order to access Alarm Adviser.

c. The **Guest Mode Enabled** check box allows any user on the network to access and view information from Alarm Adviser without needing to log in. As a default, this check box is selected. In mode, users will have access to limited functionality. For more details, refer to

the Alarm Adviser Online Help.
If the **Guest Mode Enabled** check box is cleared, users need to have a valid account configured by an administrator to access Alarm Adviser.

5. In the **Database Connection** area, specify database details for the Alarm Adviser Database. Do the following:

    a. In the **SQL Server Instance** box, type the name of the SQL Server computer and instance on which the Alarm Adviser database is located, or select from a list of SQL servers available on the network.

    b. Select the **Authentication Mode**. Select **SQL Authentication** to use SQL Server credentials to connect to the Alarm Adviser database, or **Windows Authentication** to use the Service Account credentials to connect to the Alarm Adviser database. The Configurator automatically uses the credentials provided in the Service Account area.
    If you selected SQL Server Authentication, type the SQL Server account **Username** and **Password** to be used to connect to the Alarm Adviser database.

Note: If Windows Integrated authentication mode is selected, both the **Username** and **Password** options will be disabled. The **Username** box will show the name of the Windows user account that will be used when the service is running.

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    c. In the **Database** box, type or select the name of the Alarm Adviser database. The drop down list shows all non-system databases on the selected SQL Server instance. Click **Configure**.
    Alternatively, you can select a pre-created database, and then click **Configure**. The Configurator creates the database schema for the selected database.
    If you want to create a new database on the selected SQL server instance, type a name for the database, select **Create database** and then click **Configure**. The Configurator creates the new database using the credentials specified.
    If additional privileges are required for creating the database, a dialog box appears. Enter the required credentials to proceed.

Note: Specific access may be required for the Alarm Adviser database if it is installed remotely. Contact your database administrator to obtain access.

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    d. Click **Check** to confirm that you are able to connect to the Alarm Adviser database. While attempting to connect to the database, status messages are displayed in the Configuration

Messages section. On successful connection, the message "Success" appears in the Configuration Messages section. If the credentials are not valid, a red cross appears next to the **Database** box. If valid, the Configurator will display a green tick mark next to that box.

6. Click **Configure**.
   The Configurator applies the settings as specified above, and checks whether the *aaAdministrators* and *aaUsers* groups exist within the Windows local groups. If the groups have not been created, the Configurator creates them, and adds the built-in Windows *Administrators* group to *aaAdministrators* automatically. Therefore, all users in the Windows *Administrators* group are granted administrator access to Alarm Adviser. Any other user requiring administrator access needs to be added to the *aaAdministrators* group manually. Existing Windows users who require regular access to Alarm Adviser need be added to the *aaUsers* group manually. New local users can be created and added to the Alarm Adviser groups. Users belonging to a network domain can also be added to these groups. On successful configuration, the icon next to the service name in the left pane changes from  to  .

Note: No additional configuration is required on the web client.

Localizing Alarm Adviser

The Alarm Adviser Web Client is displayed in English by default. However, you may localize it to display interface elements in a language of your choice. The Web Client Online Help can also be translated into the selected language. This section provides instructions to localize the Web Client and Online Help.

Prerequisites for Localization

Windows SDK with the Resource File Generator (Resgen) utility version 4.0.30319.18444 or higher. For more information, see https://msdn.microsoft.com/en-us/library/ccec7sz1(v=vs.110).aspx.

Resgen is a command-line utility that allows you to compile updated resource files.

Localizing the Web Client

You can localize the Web Client through the resource files installed with the Alarm Adviser Web Service. By default, these files are located in the **C:\Program Files (x86)\Schneider Electric\Alarm Adviser\Web Service\ResourceSet** folder. You can work on the resource files from this location. Alternatively, copy the required resource files to a different folder, edit them and then copy them back to the **ResourceSet** folder.

Note: You need to have the required permissions to update the resource files.

To localize the web client:

1. Open the command prompt.
2. Create a text file for your language translation from the English resource file. To do this, type the following command:

   resgen SE.AlarmAdviser.Core.Resources.en.resources <MyTranslation.txt>

   This command will generate the MyTranslation.txt in which you will edit strings for your selected language.
3. Open the MyTranslation.txt file in a text editor.
4. Replace the strings to the right of each "=" with the translated strings.
5. Compile the MyTranslation.txt file to an Alarm Adviser resource file with the following command:

   resgen MyTranslation.txt SE.AlarmAdviser.Core.Resources.<Language Code>.resources

   Where Language Code is the code/region-specific code for the language into which you want to translate the web client. For example, "fr" for French or "fr-ca" for French (Canada). See Appendix A for a list of supported language codes.
6. Reload the web client. All user interface elements such as menu names, labels, number and date formatting will be displayed according to your locale setting.

Localizing the Online Help

1. Locate the Alarm Adviser Online Help folder, typically **C:\Program Files (x86)\Schneider Electric\Alarm Adviser\Web Service\Help**.
2. Copy the **Help** folder and name it as the language code for the desired translation. For example, "fr" for French or "fr-ca" for French (Canada).
3. Translate all Online Help HTML files as required.
4. Check the translated help by opening the **Default.htm** file in a browser.
5. Copy the folder containing the translations into the **Help** folder. For example, C:\Program Files (x86)\Schneider Electric\Alarm Adviser\Web Service\Help\fr\

Note: Context sensitive help links in the web client that are not found within a region specific help translation (for example, "fr-ca") will display the help in the base language translation if it exists (for example, "fr"). Otherwise, the Online Help will be displayed in English.

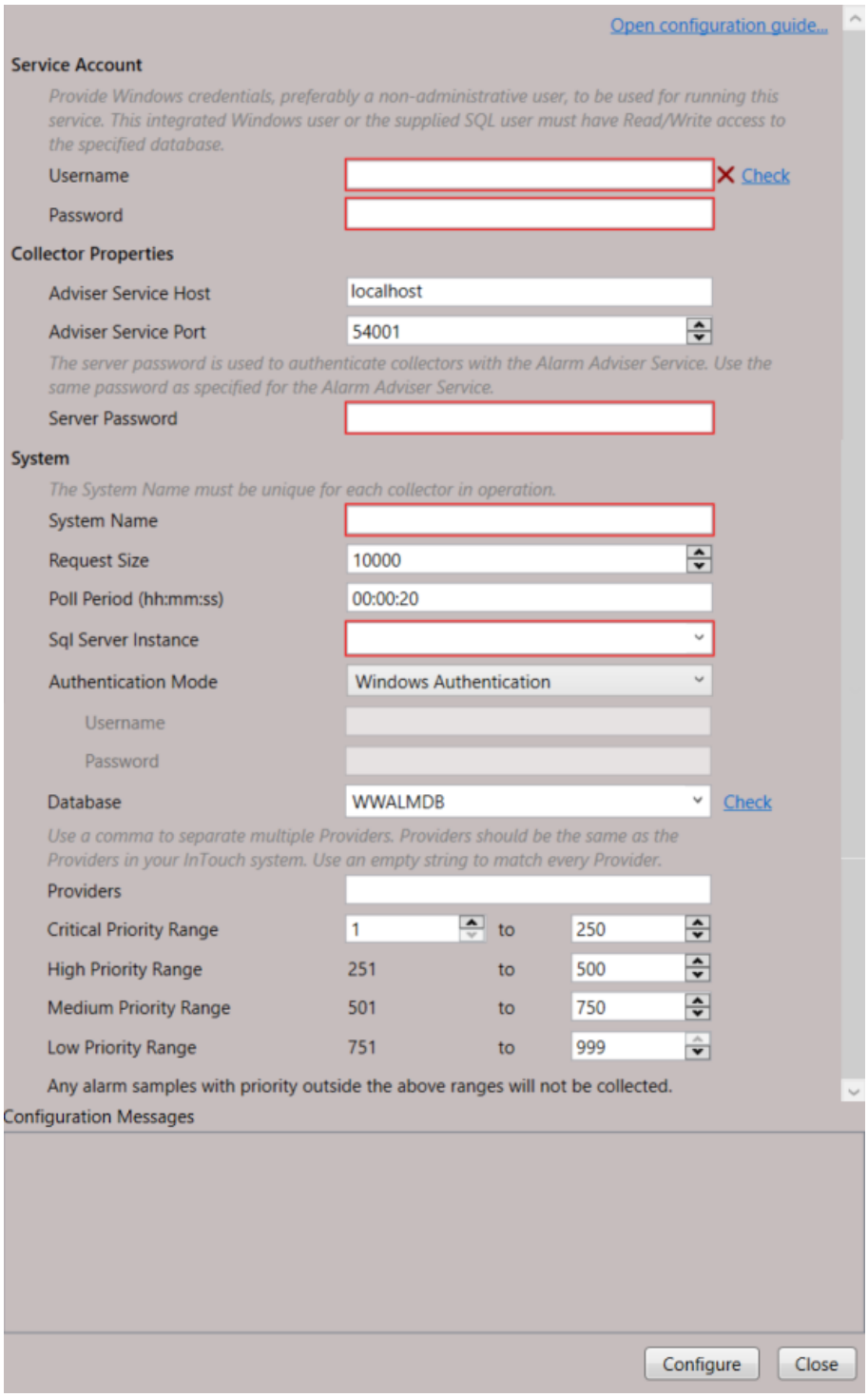
Note: Do **NOT** translate product and license names. For example, Alarm Adviser, Standard, Professional, Premier and Unlimited.

Configuring Wonderware Collector

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To configure the Wonderware Collector (for InTouch or System Platform systems):

1. Open the Configurator utility.
2. In the left pane, click **Alarm Adviser > Wonderware Collector**. The configuration options appear in the right pane.


Open configuration guide...
Service Account
Provide Windows credentials, preferably a non-administrative user, to be used for running this service. This integrated Windows user or the supplied SQL user must have Read/Write access to the specified database.
Username
Check
Password
Collector Properties
Adviser Service Host
localhost
Adviser Service Port
54001
The server password is used to authenticate collectors with the Alarm Adviser Service. Use the same password as specified for the Alarm Adviser Service.
Server Password
System
The System Name must be unique for each collector in operation.
System Name
Request Size
10000
Poll Period (hh:mm:ss)
00:00:20
Sql Server Instance
Authentication Mode
Windows Authentication
Username
Password
Database
WWALMDB
Check
Use a comma to separate multiple Providers. Providers should be the same as the Providers in your InTouch system. Use an empty string to match every Provider.
Providers
Critical Priority Range
1
to
250
High Priority Range
251
to
500
Medium Priority Range
501
to
750
Low Priority Range
751
to
999
Any alarm samples with priority outside the above ranges will not be collected.
Configuration Messages
Configure
Close

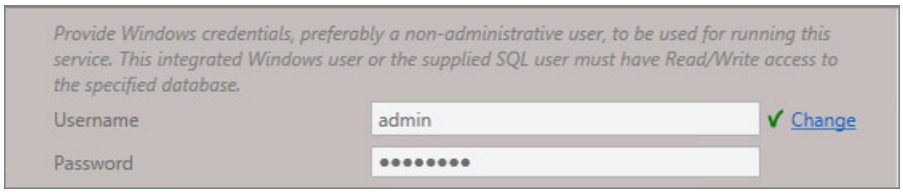
3. In the **Service Account** area, type the user account information to log on as and run the service. Do the following:

    a. In the **Username** box, type the name of the Windows local or domain account. If using a domain account, type it in the format <*domain name*>\<*user name*>.
    By default, the installer will install this service with a ***LocalSystem*** account, but the service will not run until it is configured with a Windows local or domain account in the Configurator. Your administrator can set up the Service Account with minimum privileges to run the Wonderware Collector service.

Note: You may use the same account or a different account for running the Web Server, Alarm Adviser Service and the Collector.

    b. In the **Password** box, type the password for the Windows or domain user specified in the **Username** box.

    c. Click **Check** to verify the user account and password. If the credentials are not valid, a red cross appears next to the **Username** box. If valid, the Configurator will display a green tick mark next to the box.



If you wish to use a different account, click **Change**, provide the account details, and click **Check**. Click **Configure**. The service is started.

Note: Service account credentials that are specified first for an Alarm Adviser service will be used for the other services unless changed manually. If you configure Adviser Service first and specify credentials for the service account, the same credentials will be applied as default for Web Server and Collector unless you change it manually while configuring those services.

4. In the **Collector Properties** area, do the following:

    a. In the **Adviser Service Host** box, type the name or IP Address of the server on which the Adviser Service is installed.

b. In the **Adviser Service Port** box, specify the TCP port to connect to on the server where the Adviser Service is installed.

c. In the **Server Password** box, type a password to be used to establish communication between the Collector and the Adviser Service. The specified keys need to be same between the Alarm Adviser services.

5. In the **System** area, do the following:

a. In the **System Name** box, type a valid system name that matches the System name specified for the system in the Alarm Adviser Web Client.

IMPORTANT: Launch the Alarm Adviser client, and configure the system name on the client *before* you complete this option, and restart the Alarm Adviser Web Server service. This system name returned by the Collector is validated against the system name in the client. If a matching system name is not found, the Collector will not be able to register with the Adviser Service. If you change the system **SQL Server Instance** or **Database**, confirm that a new system name is used and updated on the Alarm Adviser client. Retaining the same system name may result in data from two different systems being merged.

b. In the **Request Size** box, type the maximum number of records the Collector Service retrieves from the SCADA system database. The **Request Size** can be between 1000 and 50000 records. The default is 10000.
A very high number of records may result in high resource usage.

c. In the **Poll Period** box, type the interval at which the Collector Service requests records from the SCADA system database. The **Poll Period** can be between 100 milliseconds and 10 minutes. The default is 20 seconds.
Check that the poll period is configured to optimize system performance without an adverse impact on the SCADA system.

d. In the **SQL Server Instance** box, type the name of the SQL Server computer and instance on which the SCADA system database is located, or select from a list of SQL servers available on the network.

e. Select the **Authentication Mode**. Select **SQL Authentication** to use SQL Server credentials to connect to the Alarm Adviser database, or **Windows Authentication** to use the Service Account credentials to connect to the SCADA system database. The Configurator automatically uses the credentials provided in the Service Account area.

If you selected SQL Server Authentication, type the SQL Server account **Username** and **Password** to be used to connect to the SCADA system database.

Note: If Windows Integrated authentication mode is selected, both the **Username** and **Password** options will be disabled. The **Username** box will display the name of the Windows user account that will be used when the service is running.

f. In the **Database** box, type or select the name of the InTouch alarms database. The drop down list shows all non-system databases on the selected SQL Server instance.

Note: Specific access may be required for the Alarm Adviser database if it is installed remotely. Contact your database administrator to obtain access.

g. Click **Check** to confirm that you are able to connect to the InTouch alarms database. While attempting to connect to the database, status messages are displayed in the Configuration Messages section. On successful connection, the message "Success" appears in the Configuration Messages section. If the credentials are not valid, a red cross is appears next to the **Database** box. If valid, the Configurator shows a green tick mark next to the box.

h. Use the **Providers** box if you want the collector to collect alarm data from one or more specific providers. Type the name of the provider from which you want to collect data. To specify more than one provider, type each provider name separated by a comma. Leave the **Provider** box empty to collect alarm data from all providers defined in your system. For more information, refer to Alarm Adviser Licensing for InTouch.
For large amounts of data to be collected from multiple providers, you can configure multiple Collectors to collect data from the required providers. Each Collector needs to be installed on a separate computer.

i. Set the priority range for **Critical**, **High**, **Medium** and **Low** priority types.
Use these options to create a mapping between InTouch alarm priority and Alarm Adviser alarm severity. Define a numeric range for the following severities: Critical, High, Medium and Low. The numeric range represents the alarm priority as recorded and stored by the InTouch system, and maps to the specified Alarm Adviser severity. For example, if you set the Critical range as 1 to 250, all Wonderware alarms with values that lie in this range will be categorized as Critical in Alarm Adviser.
When you specify the upper limit for a severity, the Configurator automatically sets the lower limit for the following severity to the next consecutive number. The Configurator will not allow you to set overlapping values. The default range for alarm severity is:

- Critical – 1 to 250
- High – 251 to 500
- Medium – 501 to 750
- Low – 751 to 999

Note: Alarm severity values that lie outside the range will not be collected for analysis.
If you change the mapping, only new data collected will use the updated severity mapping.

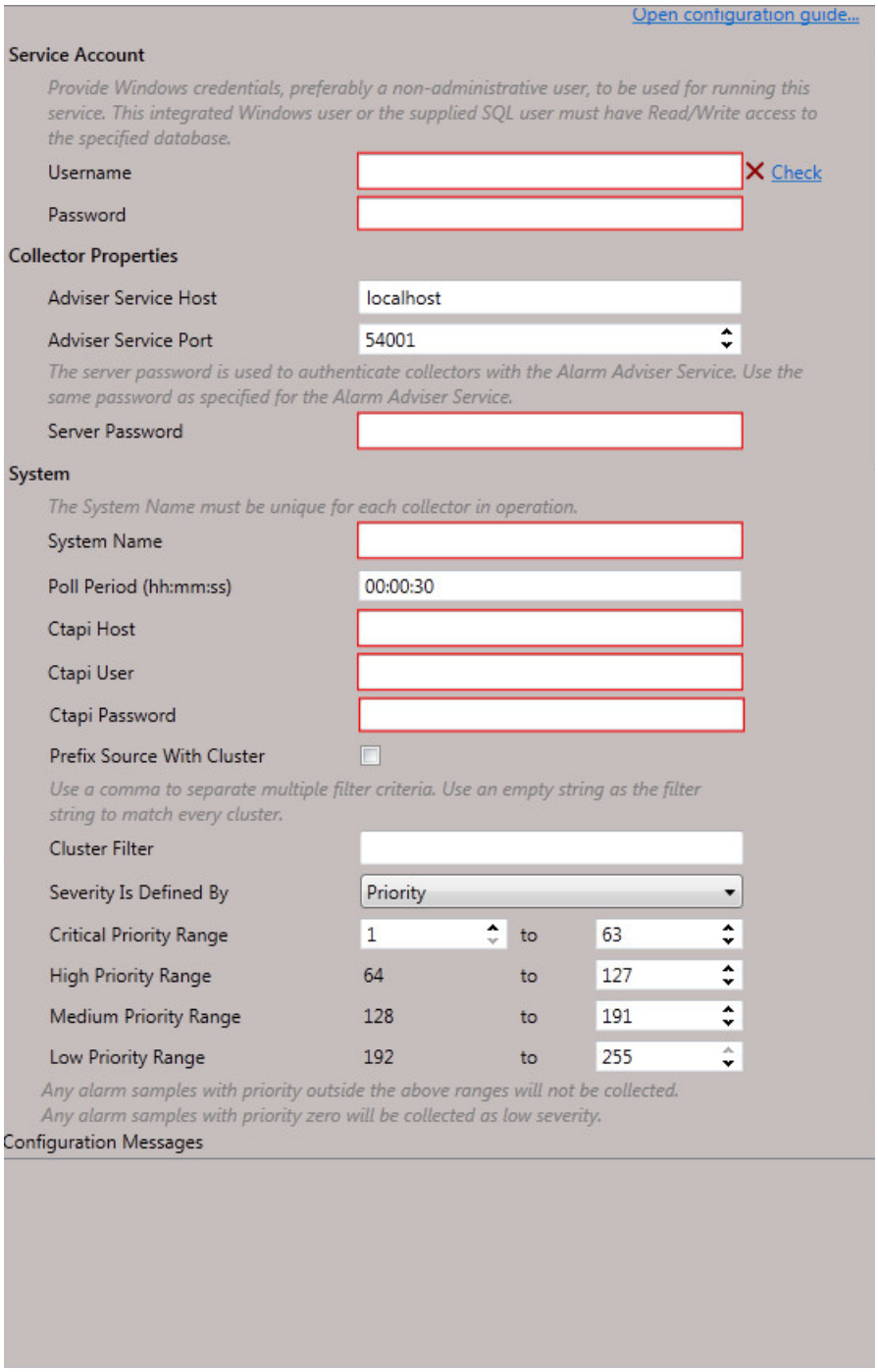
j. Click **Configure**. On successful configuration, the icon next to the service name in the Navigation page changes from  to  .

Configuring the Vijeo Citect Collector

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To configure the Vijeo Citect Collector:

1. Open the Configurator utility.
2. In the left pane, click **Alarm Adviser > Vijeo Citect Collector**. The configuration options appear in the right pane.



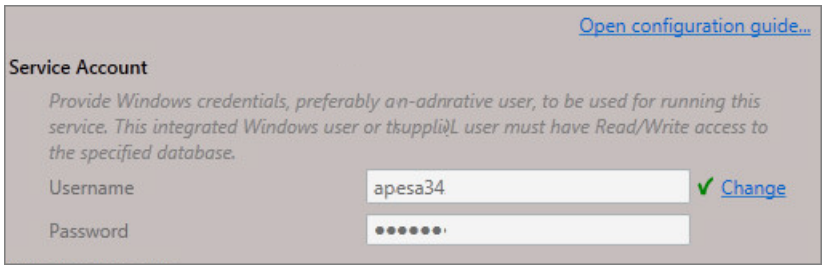
3. In the **Service Account** area, type the user account information to log on as and run the service. Do the following:

a. In the **Username** box, type the name of the Windows local or domain account. If using a domain account, type it in the format *<domain name>\<user name>*.
By default, the installer will install this service with a ***LocalSystem*** account, but the service will not run until it is configured with a Windows local or domain account in the Configurator. Your administrator can set up the Service Account with minimum privileges to run the Vijeo Citect Collector service.

Note: You may use the same account or a different account for running the Web Server, Alarm Adviser Service and the Collectors.

b. In the **Password** box, type the password for the Windows or domain user specified in the **Username** box.

c. Click **Check** to verify the user account and password. If the credentials are not valid, a red cross appears next to the **Username** box. If valid, the Configurator will display a green tick mark next to the box.



If you wish to use a different account, click **Change**, provide the account details, and click **Check**. Click **Configure**. The service is started.

4. In the **Collector Properties** area, do the following:

a. In the **Adviser Service Host** box, type the name or IP Address of the server on which the Adviser Service is installed.

b. In the **Adviser Service Port** box, specify the TCP port to connect to on the server where the Adviser Service is installed.

c. In the **Server Password** box, type a password to be used to establish communication between the Collector and the Adviser Service. The specified keys need to be same between the Alarm Adviser services.

5. In the **System** area, do the following:

    a. In the **System Name** box, type a valid system name that matches the System name specified for the system in the Alarm Adviser Web Client.

IMPORTANT: Launch the Alarm Adviser client, and configure the system name on the client *before* you complete this option, and restart the Alarm Adviser Web Server service. This system name returned by the Collector is validated against the system name in the client. If a matching system name is not found, the Collector will not be able to register with the Adviser Service. If you change the CtAPI host, confirm that a new system name is used and updated on the client. Retaining the same system name may result in data from two different systems being merged.

    b. In the **Poll Period** box, type the interval at which the Collector Service requests records from the SCADA system database. The poll period can be between 100 milliseconds and 1 hour. The default is 30 seconds.
    Check that the poll period is configured to optimize system performance without an adverse impact on the SCADA system.

    c. In the **CtAPI Host** box, type the name or IP address of the computer on which Vijeo Citect is installed.

    d. In the **CtAPI User** box, type the Vijeo Citect user name to be used for accessing the Vijeo Citect system. This user needs to have access to all Vijeo Citect alarms. The alarms to which the specified user does not have access will not be collected for analysis.

    e. In the **CtAPI Password** box, type the password for the CtAPI user specified above.

Note: CtAPI remote connection needs to be enabled in Vijeo Citect for the Collector to establish connection.

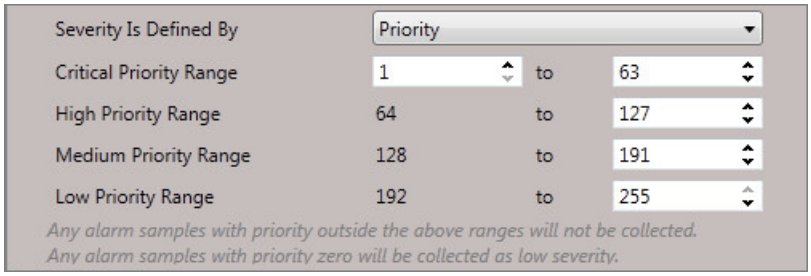
    f. Select the **Prefix source with cluster** check box to add the cluster name to the alarm source name, such as Cluster1.Alarm1. Changing this setting will apply changes to alarm data that has already been collected. For example, an alarm collected as Alarm1 will change to Cluster1.Alarm1 if the setting is enabled after data has been collected.

    g. Vijeo Citect can be configured to store alarm data in multiple clusters. Use the **Cluster Filter** box if you want the Vijeo Citect Collector to collect alarm data from one or more specific clusters. Type the name of the cluster from which you want to collect data. To specify more than one cluster, type each cluster name separated by a comma. Leave the **Cluster Filter** box empty to collect alarm data from all clusters defined in your Vijeo Citect

system.
For large amounts of data to be collected from multiple clusters, you can configure multiple Collectors to collect data from the required clusters. Each Collector needs to be installed on a separate computer.

h. From the **Severity is defined by** list, select the basis on which alarm severity should be mapped for display in the Alarm Adviser web client. You can choose **Priority** or **Category**.

- If you have chosen to define severity based on priority, define a numeric range for the following severities: Critical, High, Medium and Low. The numeric range represents the alarm priority as recorded and stored by the Vijeo Citect system, and maps to the specified Alarm Adviser severity. For example, if you set the Critical range as 1 to 63, all Vijeo Citect alarms with values that lie in this range will be categorized as Critical in Alarm Adviser.



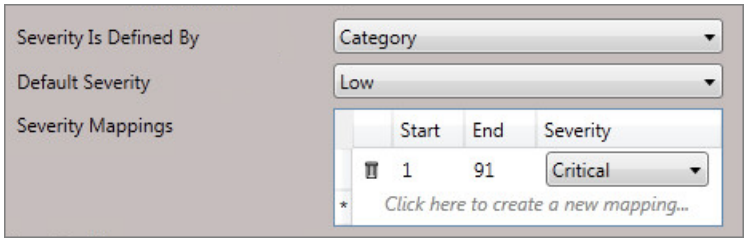
When you specify the upper limit for a severity, the Configurator automatically sets the lower limit for the following severity to the next consecutive number. The Configurator will not allow you to set overlapping values.

The default range for alarm severity is:

- Critical – 1 to 63
- High – 64 to 127
- Medium – 128 to 191
- Low – 192 to 255

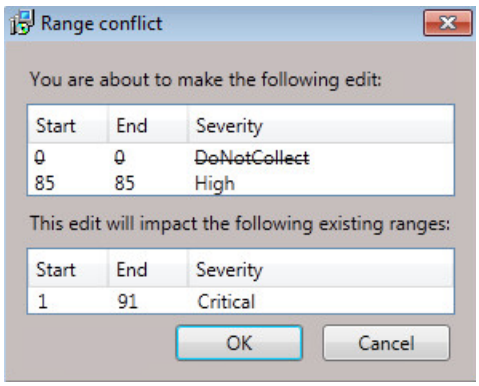
- If you have chosen to base Alarm Adviser severity on **Category**, you can set a default severity that will be applied to categories for which a severity has not been assigned. To do this, select a severity value from the **Default Severity** drop down list.
  Then, add the required severity values for selected alarm category ranges.

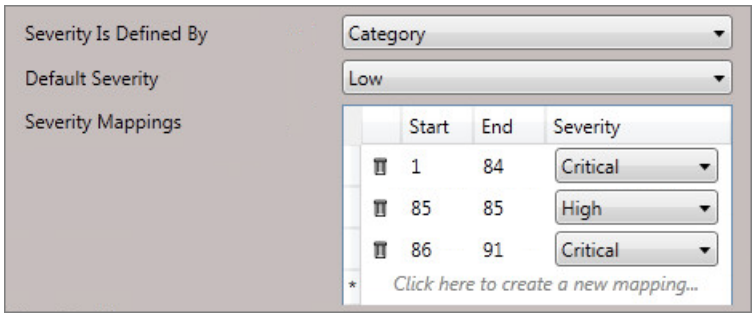
To do this, you need to set the **Start** and **End** values, and then associate a severity to this range. For example, in the figure below, all alarms in the range 1 to 91 will be assigned the severity "Critical" in Alarm Adviser. You may add as many severity values as you wish. Each range and associated severity is displayed in a tabular format.



It is possible to define a single value as the range by assigning the same value to the **Start** and **End** values, and then assigning a severity to it. For example, you can assign the severity "High" to the value 85 as shown below. This, however, splits the existing range of 1 to 91.

Click **Configure**. The Configurator will prompt you to confirm the change by displaying the Range conflict dialog, which shows the change and its impact on the existing settings. Click **OK** to confirm the change. The Configurator will assign the severity "Critical" to the values 1 to 84 as well as to the values 86 to 91.





Note: To exclude a range of category values from being displayed in Alarm Adviser, select the values, and set the severity value to "Do not collect". The Vijeo Citect Collector will not collect the severity values marked "Do not collect" for analysis.

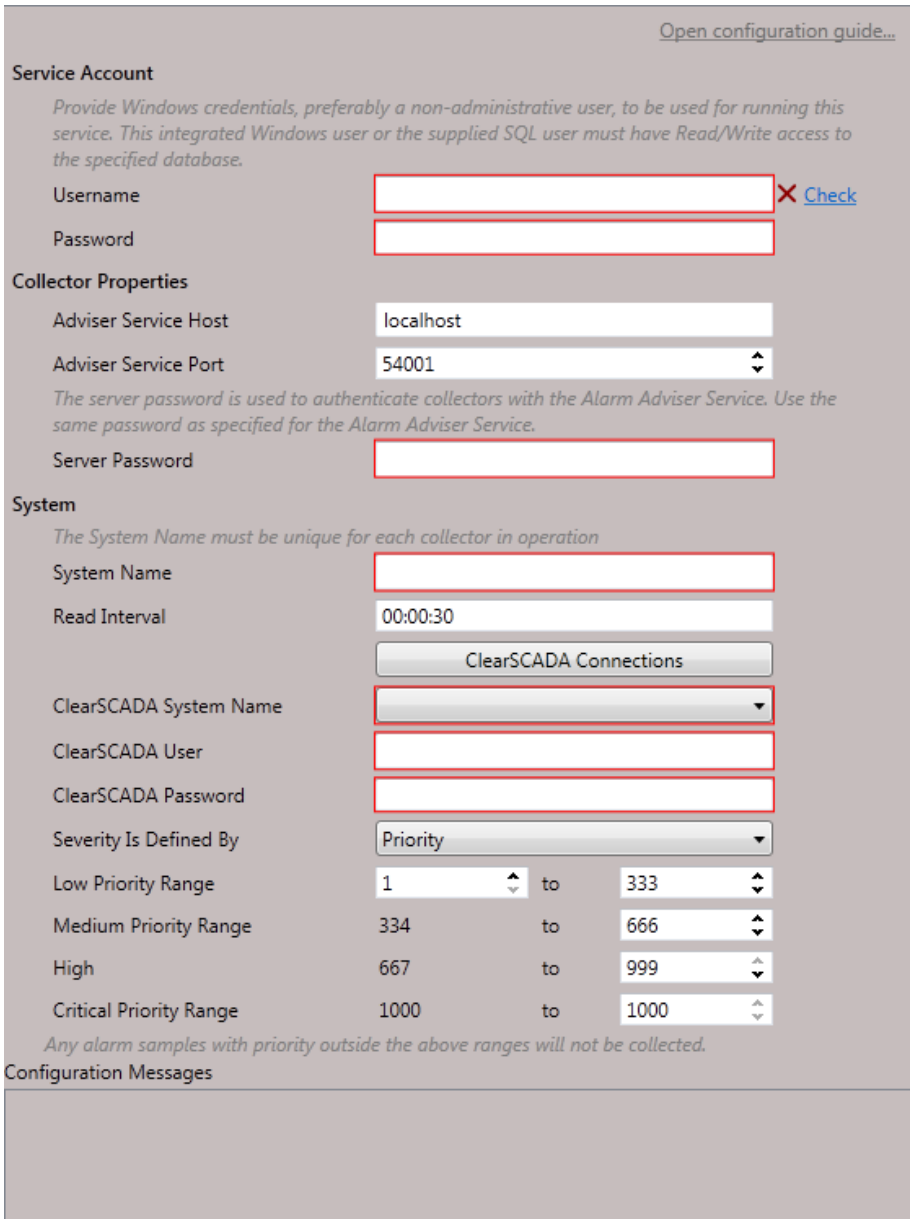
6. Click **Configure**. On successful configuration, the icon next to the service name in the Navigation pane changes from  to  .

Configuring the ClearSCADA Collector

Note: Once you have configured Alarm Adviser and the collectors, additional configuration for adding systems is required. This is done through the Web Client. The instructions for adding systems is covered in the Alarm Adviser online help.

To configure the ClearSCADA Collector:

1. Open the Configurator utility.
2. In the left pane, click **Alarm Adviser > ClearSCADA Collector**. The configuration options appear in the right pane.



3. In the **Service Account** area, type the user account information to log on as and run the service. Do the following:

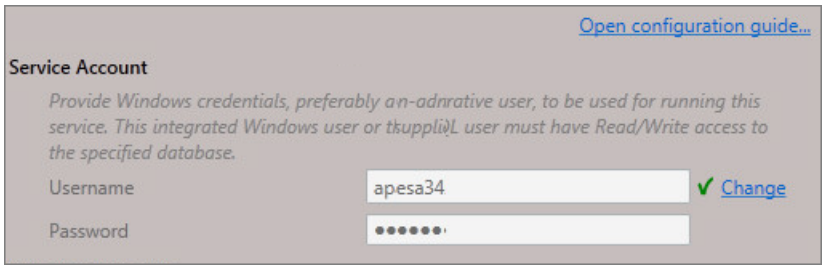
    a. In the **Username** box, type the name of the Windows local or domain account. If using a domain account, type it in the format <*domain name*>\<*user name*>.
    By default, the installer will install this service with a ***LocalSystem*** account, but the service will not run until it is configured with a Windows local or domain account in the Configurator.

Your administrator can set up the Service Account with minimum privileges to run the ClearSCADA Collector service.

Note: You may use the same account or a different account for running the Web Server, Alarm Adviser Service and the Collectors. In each case, you need to ensure that the account that is being used has sufficient privileges.

b. In the **Password** box, type the password for the Windows or domain user specified in the **Username** box.

c. Click **Check** to verify the user account and password. If the credentials are not valid, a red cross appears next to the **Username** box. If valid, the Configurator will display a green tick mark next to the box.



If you wish to use a different account, click **Change**, provide the account details, and click **Check**. Click **Configure**. The service is started.

4. In the **Collector Properties** area, do the following:

   a. In the **Adviser Service Host** box, type the name or IP Address of the server on which the Alarm Adviser Service is installed.

   b. In the **Adviser Service Port** box, specify the TCP port to connect to on the server where the Alarm Adviser Service is installed.

   c. In the **Server Password** box, type a password to be used to establish communication between the Collector and the Alarm Adviser Service. The specified passwords need to be same between the Alarm Adviser services.

5. In the **System** area, do the following:

   a. In the **System Name** box, type a valid system name that matches the System name specified for the Alarm Adviser system in the Alarm Adviser Web Client.

IMPORTANT: Launch the Alarm Adviser client, and configure the system name on the client *before* you complete this option, and restart the Alarm Adviser Web Server service. The **System Name** returned by the Collector is validated against the system name in the Alarm Adviser server. If a matching system name is not found, the Collector will not be able to register with the Alarm Adviser Service. Likewise, if you change the ClearSCADA system, confirm that a new **System Name** is used and updated on the client. Retaining the same system name may result in data from two different ClearSCADA systems being merged.

b. In the **Read Interval** box, type the interval at which the Collector Service requests records from the SCADA system database. The poll period can be between 100 milliseconds and 1 hour. The default is 30 seconds.
Check that the poll period is configured to optimize system performance without an adverse impact on the SCADA system.

c. Select the **ClearSCADA Connections** button to display the ClearSCADA Client window and configure the connections to the ClearSCADA system (*see* Add a Connection to the ClearSCADA System).

d. From the **ClearSCADA System Name** list, select the ClearSCADA database from which you want to collect alarm data. The entries in the list are defined using the ClearSCADA Client window (see above).
If you want to collect data from more than one ClearSCADA database, you can configure multiple Collectors to collect data from the required databases. Each Collector has to be installed on a different computer.

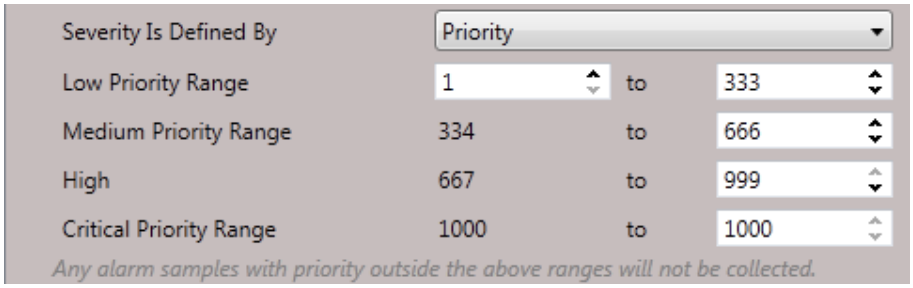
e. In the **ClearSCADA User** box, type the ClearSCADA user name to be used for accessing the ClearSCADA system. This user needs to have access to all ClearSCADA alarms that need to be reported to Alarm Adviser. The alarms that the specified user does not have access to will not be collected for analysis.

f. In the **ClearSCADA Password** box, type the password for the ClearSCADA user specified above.

g. From the **Severity is defined by** list, select the basis on which alarm severity should be mapped for display in the Alarm Adviser web client. You can choose **Priority** or **Category**.

- If you have chosen to define severity based on priority, define a numeric range for the following severities: Critical, High, Medium and Low. The numeric range represents the alarm priority as recorded and stored by the

ClearSCADA system, and maps to the specified Alarm Adviser severity. For example, if you set the Low range as 1 to 63, all ClearSCADA alarms with values that lie in this range will be categorized as Low in Alarm Adviser.

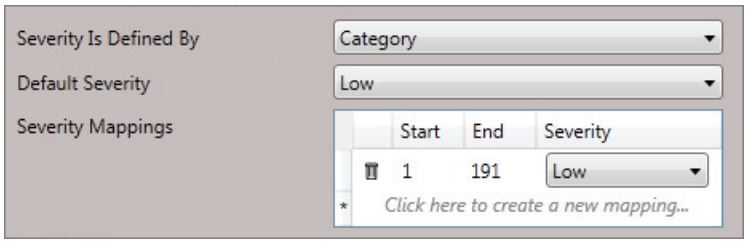


When you specify the upper limit for a severity, the Configurator automatically sets the lower limit for the following severity to the next consecutive number. The Configurator will not allow you to set overlapping values.

The default range for alarm severity is:

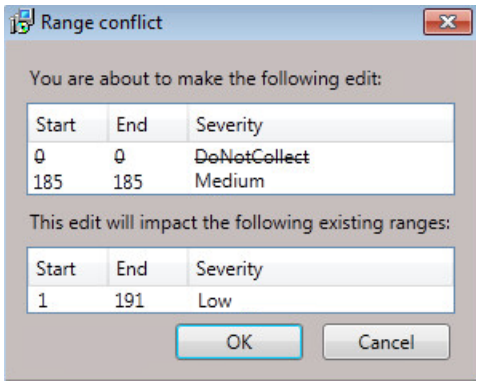
- Low – 1 to 333
- Medium – 334 to 666
- High – 667 to 999
- Critical – 1000 to 1000

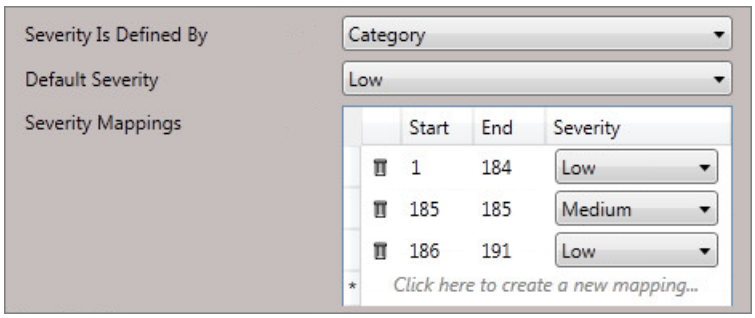
- If you have chosen to base Alarm Adviser severity on **Category**, you can set a default severity that will be applied to categories for which a severity has not been assigned. To do this, select a severity value from the **Default Severity** drop down list.
  Then, add the required severity values for selected alarm category ranges. To do this, you need to set the **Start** and **End** values, and then associate a severity to this range. For example, in the figure below, all alarms in the range 1 to 191 will be assigned the severity "Low" in Alarm Adviser. You may add as many severity values as you wish. Each range and associated severity is displayed in a tabular format.



It is possible to define a single value as the range by assigning the same value to the **Start** and **End** values, and then assigning a severity to it. For example, you can assign the severity "Medium" to the value 185 as shown below. This, however, splits the existing range of 1 to 191.

Click **Configure**. The Configurator will prompt you to confirm the change by displaying the Range conflict dialog, which shows the change and its impact on the existing settings. Click **OK** to confirm the change. The Configurator will assign the severity "Low" to the values 1 to 184 as well as to the values 186 to 191.





Note: To exclude a range of category values from being displayed in Alarm Adviser, select the values, and set the severity value to "Do not collect". The ClearSCADA Collector will not collect the severity values marked "Do not collect" for analysis.

6. Click **Configure**. On successful configuration, the icon next to the service name in the Navigation pane changes from to .

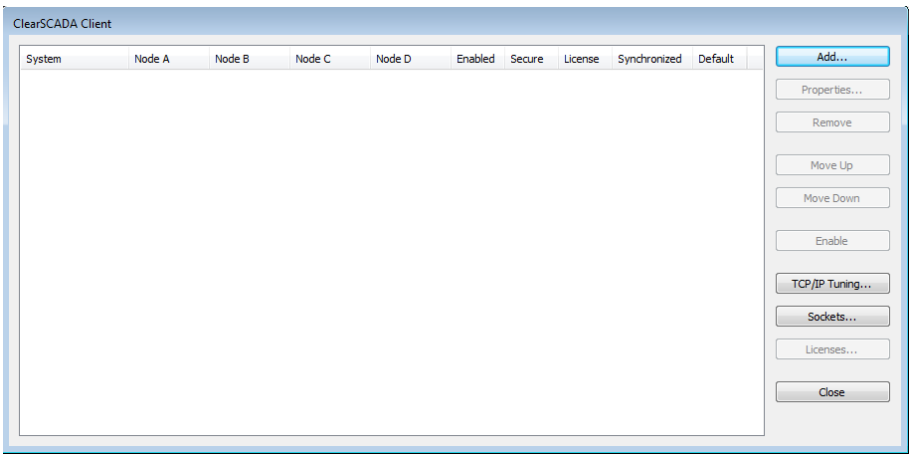
Add a Connection to the ClearSCADA System

In order for Alarm Adviser to collect alarm data from ClearSCADA, a connection has to exist between the ClearSCADA Collector and the ClearSCADA system. You use the ClearSCADA Client applet to view and configure the connections you require.

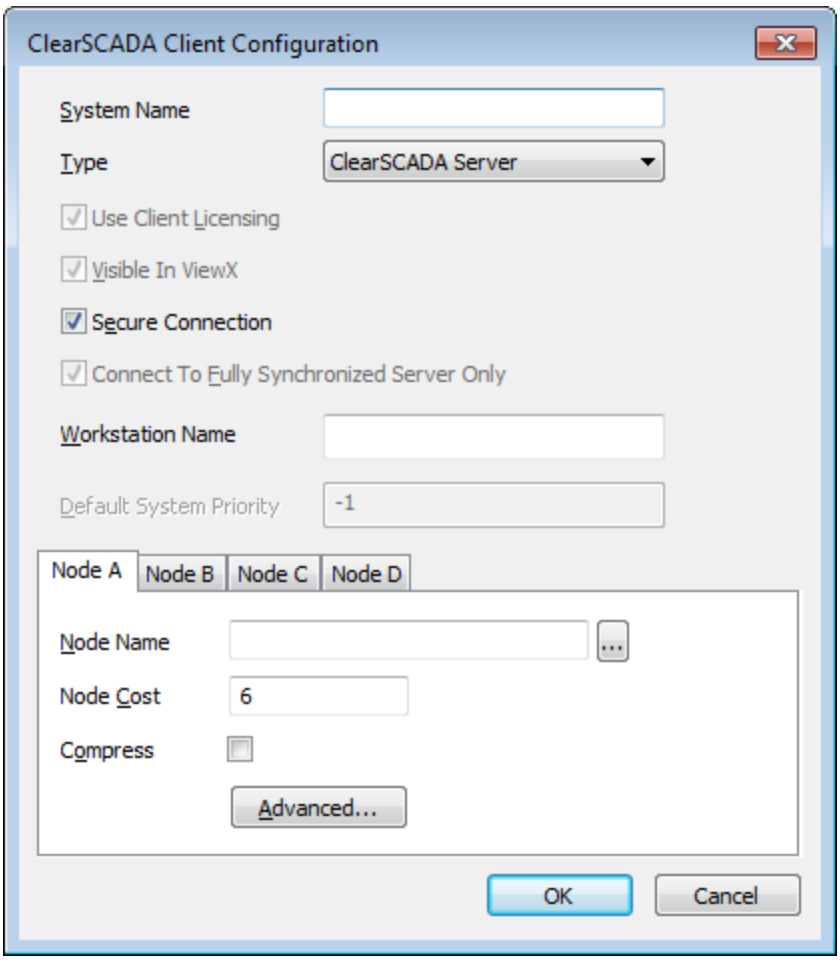
Note: The first time that you attempt to access the ClearSCADA Client applet, a diagnostic message will be displayed, advising that configuration is incomplete. Ensure that you complete the mandatory configuration once you have set up the required connection(s).

To add and configure a connection to the ClearSCADA system:

1. Access the ClearSCADA Client applet. To do this, select the **ClearSCADA Connections** box in the Configurator for the ClearSCADA Collector (*see* Configuring the ClearSCADA Collector).



2. Select the **Add** button on the applet.
   The Client Configuration window is displayed.



Note: Some of the properties in the window are 'grayed out' and unavailable for use. These properties do not apply to ClearSCADA Collector connections.

3. In the **System Name** box, enter the name that you want to use to identify the ClearSCADA system from which Alarm Adviser is to collect data.

   The name that you specify here will appear in the drop-down list that is available from the **ClearSCADA System Name** box in the ClearSCADA Collector's Configurator window (*see* Configuring the ClearSCADA Collector).

4. The **Type** box specifies the type of system to which the ClearSCADA Collector is to connect. The box only has one option: **ClearSCADA Server**.
5. To help protect the system from unauthorized use, we recommend that the ClearSCADA Collector uses a secure connection to ClearSCADA. To do this, ensure that the **Secure Connection** check box is selected. (With new ClearSCADA installations that are running ClearSCADA 2015 R1 onwards, the ClearSCADA server only accepts secure connections from client machines by default.)

   Although we do not recommend it, you can use an insecure connection to communicate with the ClearSCADA server. To do this, you clear the **Secure Connection** check box.

⚠ WARNING
SECURITY RISK
To avoid compromising your system, the **Secure Connection** check box should be selected. This will help to prevent unauthorized users from gaining access to your system.
Failure to follow these instructions can result in death, serious injury, or equipment damage.

6. We recommend that you leave the **Workstation Name** box empty unless you are asked to do otherwise.

   If you are required to enter a **Workstation Name** to identify the ClearSCADA Collector in ClearSCADA, ensure that the name is unique. The name must differ to that used for any other clients in ClearSCADA or Alarm Adviser.

7. Configure the **Node** settings for the ClearSCADA Collector (*see* Configure the Node Settings). You use these settings to specify the individual ClearSCADA server(s) to which the Collector can connect in a single ClearSCADA system. A ClearSCADA Collector can connect to any one of up to four ClearSCADA servers, with the **Node Cost** property being used to determine the order of preference for the connections.
8. Click **OK**. The client-system connection is added to the ClearSCADA Client applet.

Note: If you want to collect data from more than one ClearSCADA database, you can configure further Collectors to collect data from the other databases. Each Collector has to be installed on a different computer.

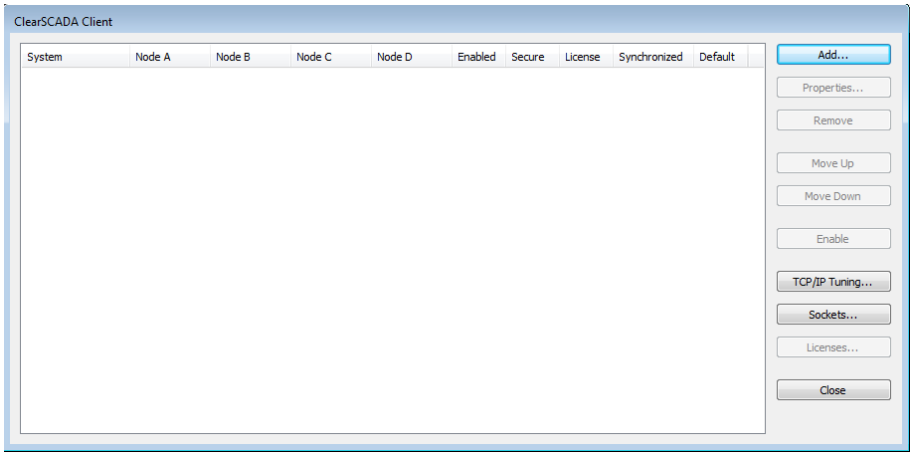
Note: The ClearSCADA Collector will only connect to a fully synchronized ClearSCADA server. This is to help ensure that the Collector only reads data from a server that contains the latest current and historic data. On a multi-server ClearSCADA system, Standby and Permanent Standby servers synchronize with the Main server to ensure that they contain the latest data. The process of synchronization can take some time, particularly when a new server is added to the system. For more information about synchronization, see the ClearSCADA online help.

The rest of the topics in this section provide information about:

- The Client Applet Buttons
- Node Cost
- How to Configure the Connection Port and Poll Timeouts and Intervals.

Client Applet Buttons

The ClearSCADA Client applet that you use to configure the connections between the ClearSCADA Collector and the ClearSCADA system contains these buttons:



BUTTON	DESCRIPTION
Add	Use to add a connection between the ClearSCADA Collector and the ClearSCADA system (*see* Add a Connection to the ClearSCADA System).
Properties	Use to edit the properties of the selected entry in the list. You can also edit an entry by double-clicking the entry in the list. The Client Configuration window is displayed. You can use the window to view or edit the connection's settings. For more information about the Client Configuration window, *see* Add a Connection to the ClearSCADA System.
Remove	Use to remove the connection to a ClearSCADA system. Click on the obsolete entry in the list and then click this button to remove that entry. The ClearSCADA Collector will no longer be able to collect data using a connection that has been removed.
Move Up	Use to change the order in which the entries appear in the ClearSCADA Client window, by moving the selected entry further up the list. (The order in which the entries appear in the window is irrelevant to the ClearSCADA Collector; the buttons are provided in case you want to rearrange the entries to suit your own preferences.)
Move Down	Use to change the order in which the entries appear in the ClearSCADA Client window, by moving the selected entry further down the list.
Disable/Enable	Use to disable or enable the selected connection entry.

TCP/IP Tuning	Use to configure the port settings if asked to do so by Schneider Electric *(see* Configure the TCP/IP Tuning Settings).
Sockets	Use to Change the Range of Ports for the Collector's Incoming Data.
Close	Use to close the ClearSCADA client window.

The other buttons on the ClearSCADA Client window are unused.

For information about displaying the ClearSCADA Client applet, *see* Add a Connection to the ClearSCADA System.

Configure the TCP/IP Tuning Settings

This topic explains how to use the TCP/IP Tuning setting to determine how TCP/IP manages communications when a connection between the ClearSCADA Collector and ClearSCADA system has been lost.

You can use the **TCP/IP Tuning** button on the Client Configuration window to access the Transmission Control Protocol/Internet Protocol settings. These settings determine how TCP manages the transmission of data over the Collector - system connection.

By using the TCP/IP Tuning settings on the Client Configuration window, you can define:

- How TCP/IP manages the communications when the Collector - system connection has been lost
- The number of times TCP/IP will attempt to resend failed transmissions over a healthy, established Collector - system connection.

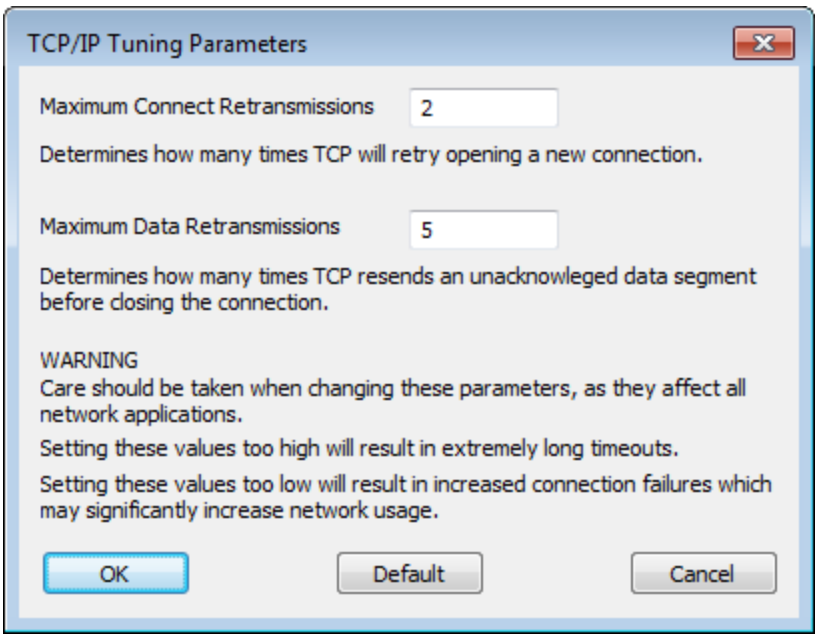
Attention: You should only change the TCP/IP settings if you have been advised to do so by Schneider Electric staff. Changing the TCP/IP settings will affect every application on the Collector machine, not just the ClearSCADA Collector.

If the ClearSCADA Collector is experiencing long timeouts or is unable to connect, you may be advised to change the TCP/IP settings by a member of Schneider Electric staff. If you are advised to change the settings, you should:

1. Display the Client Configuration Window (*see* Add a Connection to the ClearSCADA System).

2. Select the entry for the connection for which you want to define the TCP/IP Tuning settings.

3. Click the **TCP/IP Tuning** button on the Client Configuration window.
   The TCP/IP Tuning window is displayed.



4. In the **Maximum Connect Transmissions** box, enter the maximum number of times that TCP will attempt to make a connection between the Collector and the server. If a connection is not made within the defined limit, TCP regards the connection as failed. TCP will make no further connection attempts.

5. In the **Maximum Data Retransmissions** box, enter the maximum number of times that TCP will attempt to re-transmit data that has not received an acknowledgment (in normal circumstances, the Collector will send data to the server, and the server will respond with an acknowledgment before sending any further data). If TCP has not received an

acknowledgment within the defined limit, it will make no further attempts to send the data segment and will close the connection.

6. Click **OK** to confirm the changes, or click **Default** to restore the default settings. Alternatively, click **Cancel** to abort any changes you have made.

Change the Range of Ports for the Collector's Incoming Data

This topic explains how to use the Sockets window to change the range of ports for the ClearSCADA Collector's incoming data.

You can use the ClearSCADA Client applet to change the range of ports that can be used for the Collector's incoming data (one of the ports in the range will be used to receive data via the server advise connection). By default, the range of ports is 5500 to 5509 inclusive which allows a maximum of ten client applications per Collector machine. You may need to change this range if the ports 5500 to 5509 are already allocated to other applications or you need to increase or reduce the number of client applications permitted on the Collector machine.

To check which ports are available on the Collector machine:

You can use Microsoft Windows to access information about the ports being used by the applications on the machine on which the ClearSCADA Collector is installed. This is useful if you suspect that multiple applications are attempting to use the same ports on the machine.

When you have identified the ports that are being used by the active (running) programs on the machine, you can define a more appropriate range of ports in the machine's Sockets settings. This will stop ClearSCADA from attempting to use ports that are used by other programs.

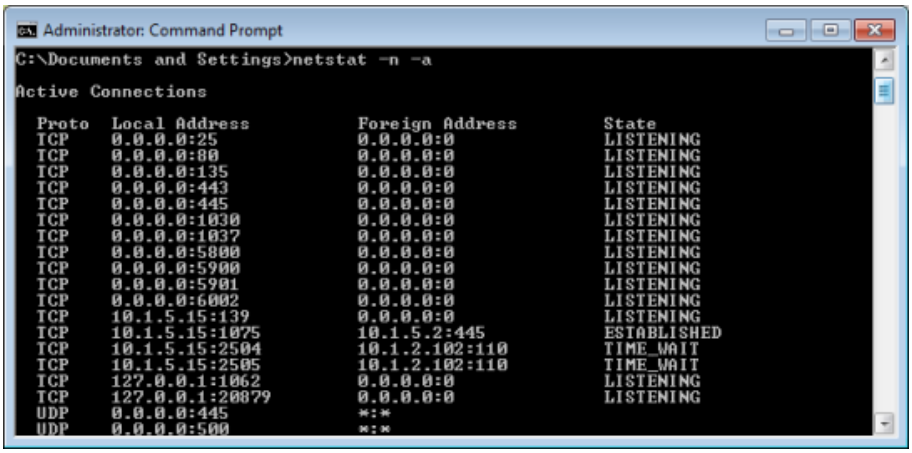
To check which ports are being used by the programs that are currently running on the machine on which the ClearSCADA Collector is installed:

1. Disable any active ClearSCADA Collector connections to the ClearSCADA system. To do this, you use the **Disable** button on the ClearSCADA Client applet (*see* Client Applet Buttons).
2. On the Collector machine, run the command prompt in Windows.

3. Enter the following at the prompt:

   netstat -n -a

4. Press the **Enter** key.
   The command prompt displays a list of the ports that are currently in use.

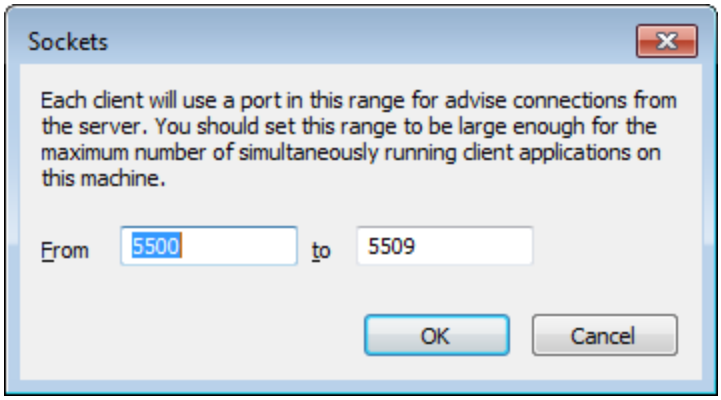


5. Examine the entries in the list. The entries that have a **Proto** of TCP and **State** of LISTENING are the local ports that are in use. You can see the actual port numbers by viewing the entries in the **Local Address** column: the last part of the address is the port number, for example, 0.0.0.0:445 means port 445 is in use.
6. Make a note of the ports that are being used so that when you change the sockets settings, you can choose a range that is available.

To change the range of ports that can be used for the ClearSCADA Collector's incoming data:

1. On the Collector machine, display the ClearSCADA Client applet (*see* Add a Connection to the ClearSCADA System).
2. Select the entry for the connection for which you want to alter the port settings.

3. Click the **Sockets** button to display the Sockets window.



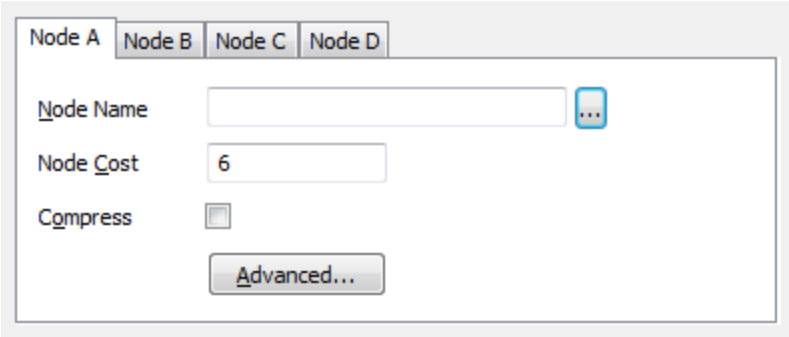
4. Enter the new port range (lowest port number in the left-hand field, highest port number in the right-hand field).
5. Click **OK** to confirm your selection and close the Sockets window.

Remember that if you are using a firewall between the ClearSCADA Collector and ClearSCADA server, you will need to configure the firewall to allow the ClearSCADA server to send data to the ClearSCADA Collector via the ports in the defined range.

Configure the Node Settings

A ClearSCADA Collector can use a single connection to connect to one of up to four nodes (servers) in a single ClearSCADA system. The number of servers to which a Collector might have to connect varies according to the architecture of the system, for example, a Hot-Standby system has two servers (Main and Standby) whereas a Triple Standby system with a Permanent Standby has four (the Main, Standby 1, Standby 2, and the Permanent Standby).

You use the **Node** tabs on the Client Configuration window to specify the server(s) to which the ClearSCADA Collector is to connect (*see* Add a Connection to the ClearSCADA System).



Use the **Node** tabs as follows:

SYSTEM ARCHITECTURE	DESCRIPTION
Lone Server	Use the **Node A** tab to define the connection between the Collector and the server.
Hot-Standby Pair	Use the **Node A** tab to define the connection between the Collector and the first server, and the **Node B** tab to define the connection between the Collector and the second server.
Hot-Standby Pair with one Permanent Standby Server	Use the **Node A** tab to define the connection between the Collector and the first server, and the **Node B** tab to define the connection between the Collector and the second server. Use the **Node C** tab to define the connection between the Collector and the permanent standby server.
Hot-Standby Pair with two Permanent Standby Servers	Use the **Node A** tab to define the connection between the Collector and the first server, and the **Node B** tab to define the connection between the Collector and the second server. Use the **Node C** tab to define the connection between the Collector and the first permanent standby server, and the **Node D** tab to define the connection between the Collector and the second permanent standby server.

System Architecture	Description
Hot-Standby Pair (Dual Ethernet)	Use the **Node A** and **Node B** tabs to define the connections between the Collector and one server (each Node tab defines the settings for a network card), and use the **Node C** and **Node D** tabs to define the connections between the Collector and the alternate server. For Dual Ethernet arrangements, you have to define the Node Name as an IP address so that the Collector machine can differentiate between the network cards.
Triple Standby	Use the **Node A** tab to define the connection between the Collector and the first server. Use the **Node B** and **Node C** tabs to define the connections between the Collector and the second and third servers.
Triple Standby with a Permanent Standby Server	Use the **Node A** tab to define the connection between the Collector and the first server. Use the **Node B** and **Node C** tabs to define the connections between the Collector and the second and third servers. For the Permanent Standby server, use the **Node D** tab.

With each **Node** tab that is required, specify the following:

1. Define the **Node Name**. This is either the name of the server, or the IP address of the server to which the ClearSCADA Collector is to connect. For dual Ethernet arrangements you have to enter the IP addresses for the nodes (servers).

   You can either type in the IP address or server name, or use the browse button next to the field to display a Reference browse window. Select the computer that comprises the server from the window.

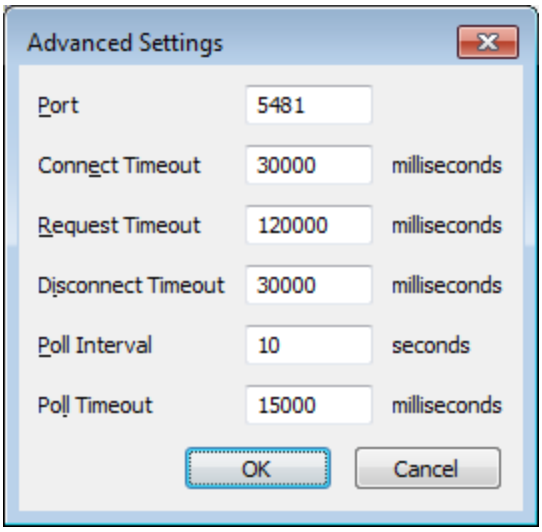
2. Specify the **Node Cost**. This helps to determine the order of preference for the server connection—the connection with the lowest cost is the preferred connection. For more information, *see* Node Cost.

3. Use the **Compress** check box to define the data compression requirements. Select the check box to enable compression; clear the check box to disable compression.

   The advantage of having compression enabled is that it reduces the amount of data that has to be sent, and so the data can be transferred more quickly. However, compression makes greater use of the processor and so there is an overhead on the server and Collector machine. For this reason, you typically only enable compression on systems that are connected via slow networks (low bandwidth, typically less than 10Mb per second).

   When using the Compress feature, consider which servers in the ClearSCADA system architecture are needed to compress and decompress the data (each server in ClearSCADA has a Compress feature).

4. Select the **Advanced** button to display the Advanced Settings window. Use the window to Configure the Connection Port and Poll Timeouts and Intervals.



5. Repeat the above steps for any other nodes that you need to configure.

Ensure that the rest of the fields on the Client Configuration window are suitably populated. For information about other properties on the Client Configuration window, *see* Add a Connection to the ClearSCADA System.

Node Cost

Each connection between a ClearSCADA Collector and a ClearSCADA system is allocated a cost number that is used by the ClearSCADA Collector to determine the order of preference for the connections—the connection with the lowest cost is the preferred connection. Usually, the cost indicates the efficiency of the connection, with the more efficient connections having the lowest cost numbers.

When a ClearSCADA Collector connects to a ClearSCADA server, it automatically connects to the server that has the lowest **Node Cost** value and is available. This sets the Collector to use the most suitable route to the ClearSCADA system. If there are two connections that have the same **Node Cost**, the ClearSCADA Collector will connect to the server that is Main.

To define the **Node Cost** for a Collector - system connection, you need to have an appreciation of the type and efficiency of each connection:

- **If the connections are via a Local Area Network (LAN)**, you should set the **Node Cost** of each connection to the same amount. This means that the Collector only connects to the Main server.
- **If there are Wide Area Network (WAN) connections**, you should set the **Node Cost** to indicate the amount of bandwidth that is available for the connection. The connections that provide the lowest amount of bandwidth are the least efficient and so should have the highest **Node Cost** values.

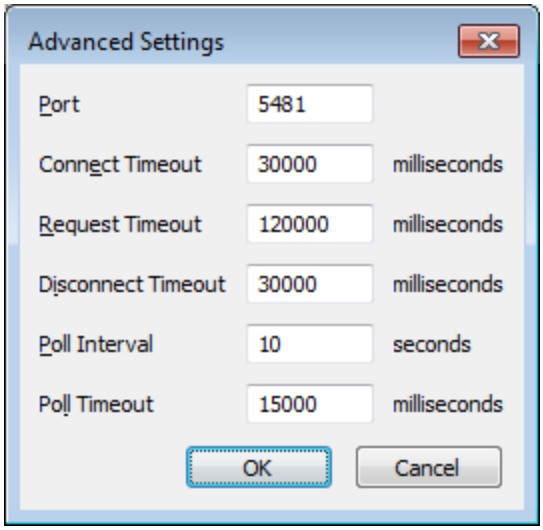
You can use any numbers to define the Node Costs, but the lowest number in your chosen range must be allocated to the connection that provides the most suitable route to the ClearSCADA server. You can allocate subsequent numbers to the next most efficient connection (the least efficient connection will have the highest number in the range). For more information on how the **Node Cost** settings are used, see the ClearSCADA online help.

When configuring the connection between a ClearSCADA Collector and a ClearSCADA system, you define the **Node Cost** on the tab for the relevant node (ClearSCADA server) on the ClearSCADA Client Configuration window (*see* Configure the Node Settings).

For examples that demonstrate how **Node Cost** can help determine the ClearSCADA server to which a ClearSCADA Collector connects, see the ClearSCADA online help.

Configure the Connection Port and Poll Timeouts and Intervals

This topic explains how to use the Advanced Settings window to define the Port Number, Timeouts, and Poll Interval for each node on a connection between a ClearSCADA Collector and a ClearSCADA server.



You access the Advanced Settings window by selecting the **Advanced** button on the Client Configuration window (*see* Configure the Node Settings). You can use the window to change the port number for the connection as well as its timeout and interval settings. While it is unlikely that you will need to adjust the default timeout and interval settings, you may need to make adjustments on systems that are especially small or large. For example, with small ClearSCADA systems you may choose to reduce the timeouts to improve response times and on some larger systems you may need to increase the default time settings.

The time settings you need to enter will vary according to the amount of network traffic and the capabilities of your network. As a general rule, you should set the timeouts to be slightly more than the worst-case scenario response time for your network.

To change the Port Number, Timeouts, or Poll Interval settings for the connection between the ClearSCADA Collector and the ClearSCADA system:

1. Display the Advanced Settings window. To do this, select the **Advanced** button on the Client Configuration window (*see* Configure the Node Settings).
2. Alter the settings as required:
    - **Port:** Enter the number of the port that the ClearSCADA Collector uses for outgoing communications to the ClearSCADA server.
    - **Connection Timeout:** The amount of time, in milliseconds, in which the Collector can attempt to make a connection. If the Collector cannot establish a connection to the server within this time, the connection is deemed to have failed.
    - **Request Timeout:** The amount of time, in milliseconds, in which the Collector can request data from a ClearSCADA server. If the server has not responded by the end of this time, the request is deemed to have failed.
    - **Disconnect Timeout:** The amount of time, in milliseconds, in which the Collector can attempt to terminate a connection to the server. If a disconnection has not occurred by the end of this time, the disconnection is deemed to have failed. The Disconnect Timeout is used when the network connection can be detected, but the Collector cannot disconnect from the system.
    - **Poll Interval:** The amount of time, in seconds, that can elapse between each update request sent from the Collector to the ClearSCADA server. The latest data is requested in every poll.
    - **Poll Timeout:** The amount of time, in milliseconds, that the Collector will wait for a response to a poll request. If no response is received within the defined time, the timeout is logged in the Wonderware event viewer.
3. Click **OK** to confirm the changes. The Advanced Settings window is closed and you are returned to the Client Configuration window (*see* Add a Connection to the ClearSCADA System).

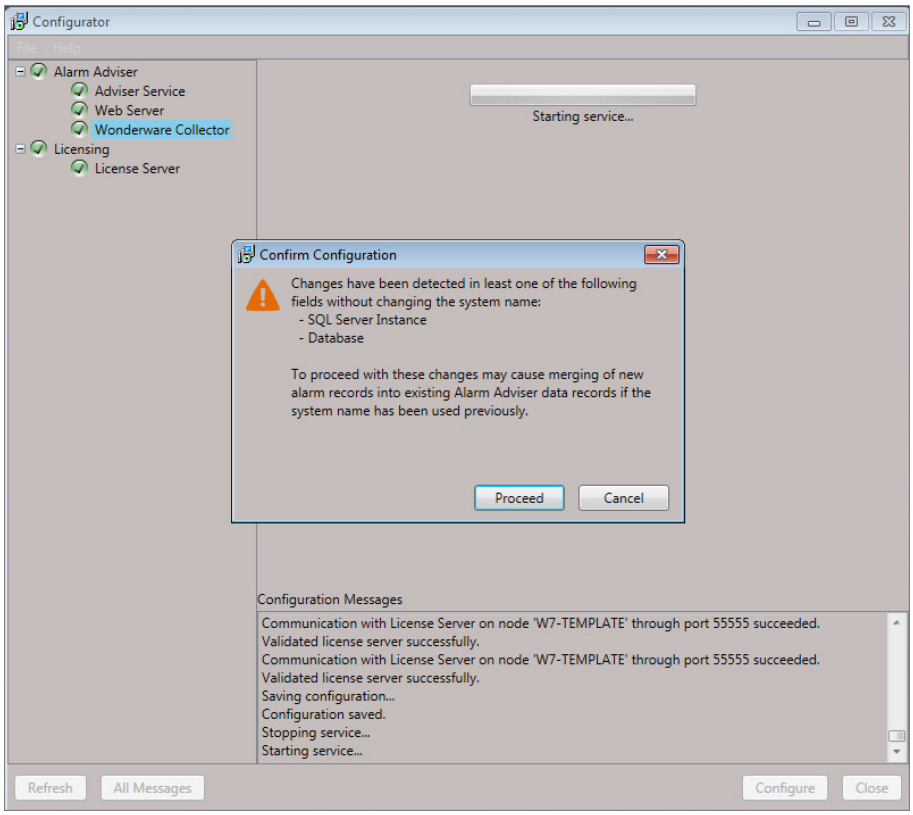
Changes to Collector Configuration

Changes to the Collector configuration re-starts data collection from the backfill period (see Moving Web Server, Service or Collector for more information). For example:

- For the Wonderware Collector: Changing the SQL Server instance, the database (changes such as moving the database to a different SQL Server instance, or pointing to a new database or a copy of the existing one)
- For the Vijeo Citect Collector: Changing the CtAPI Host
- For the ClearSCADA Collector: Changing the ClearSCADA System Name with which the Collector is associated (when using the Alarm Summary means of data collection).

When configuring a Collector, a unique system name that you specify is associated with the SQL Server instance and database from which data is collected. If you retain this system name while modifying the Collector configuration, Alarm Adviser will merge records from the SCADA system with the records in the Alarm Adviser Database. If you create a new system, the Alarm Adviser Database will be updated with the new system name, the SQL Server instance and database associated with it, and data from the SCADA system for the backfill period.

Since these changes affect the alarm data that users will see on the client, Alarm Adviser shows the following message.

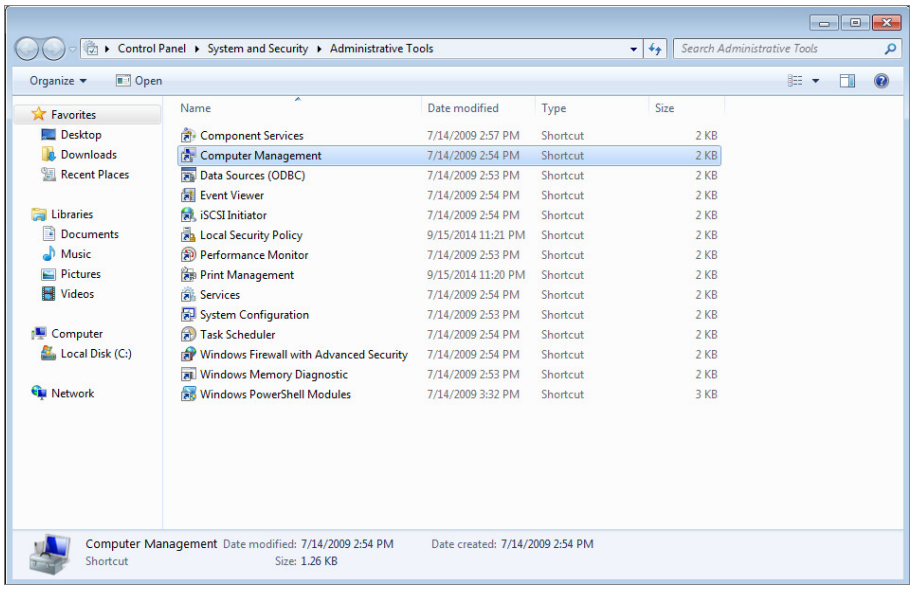


To continue with the modifications, click **Proceed**, and then click **Configure**. Otherwise, click **Cancel**, modify the system name, and then click **Configure**.

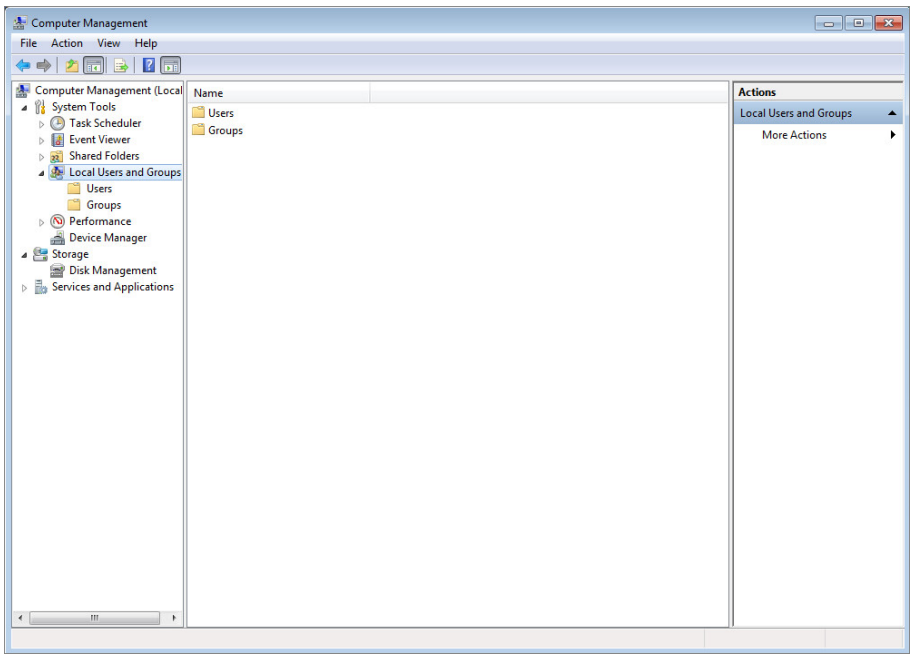
Creating Alarm Adviser Users

To create Alarm Adviser users:

1. From the Windows **Start** menu, select **Control Panel > System and Security > Administrative Tools**.

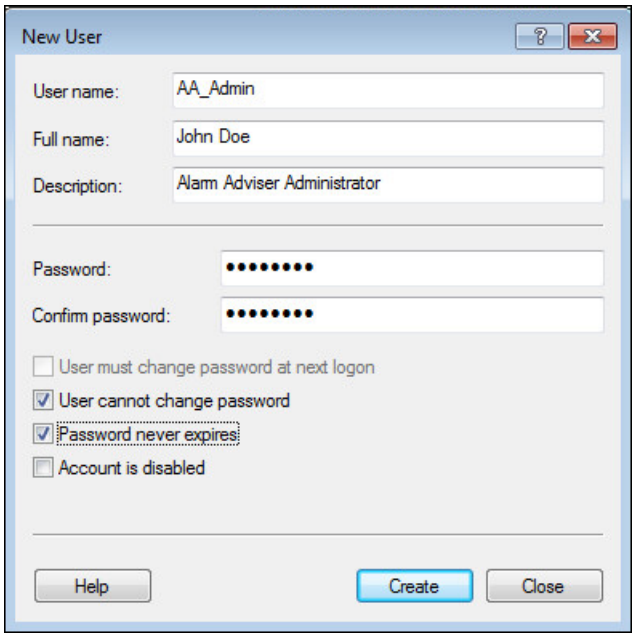


2. From the list of Administrative Tools, double-click on **Computer Management**. The Computer Management window appears. Navigate to the Local Users and Groups node in the left pane.



3. Expand the **Users** folder by clicking on it. All existing users on the computer are listed.

4. From the **Action** menu, select **New User**. The New User dialog box appears.

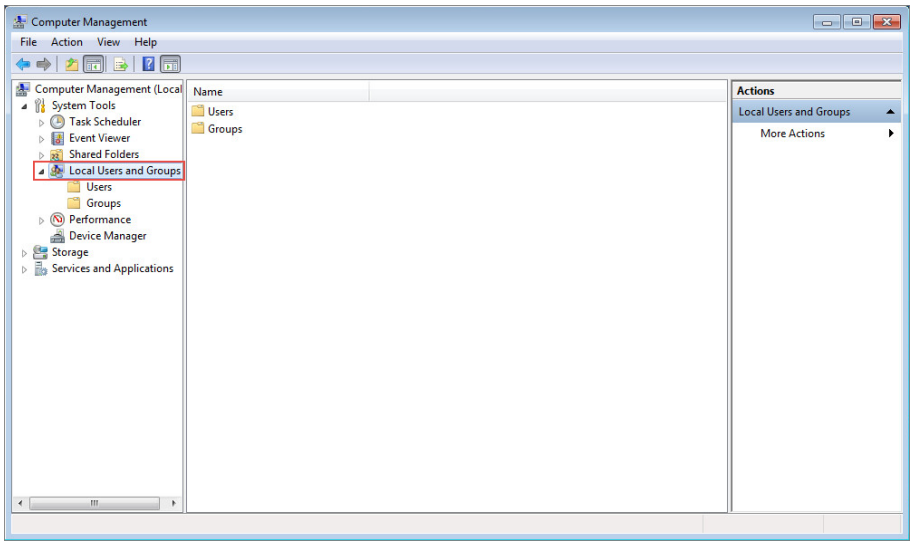


5. Specify the following:
    a. **User name**: Type the name of the Alarm Adviser user you want to create.
    b. **Description**: Type a relevant description for the name as shown above.
    c. **Password**: Type a password for this user. The password can be a combination of letters and numbers, and up to 8 characters in length.
    d. **Confirm Password**: Re-type the password specified in the **Password** box.
    e. Select the required password settings.
6. Click **Create**.

Adding Users to an Alarm Adviser Group

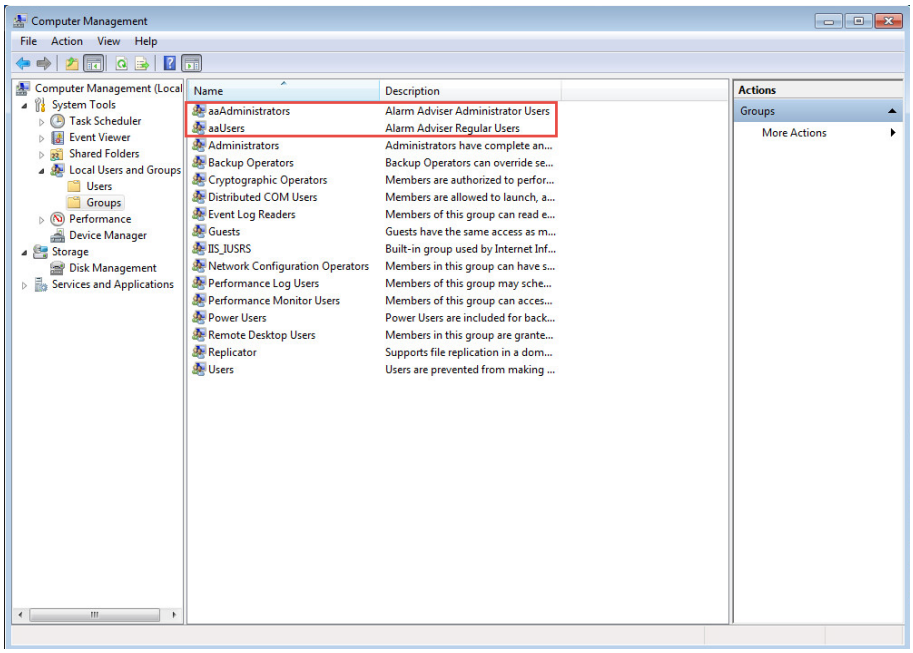
To add the users to an Alarm Adviser group:

1. From the Windows Start menu, select **Control Panel > System and Security > Administrative Tools > Computer Management**.

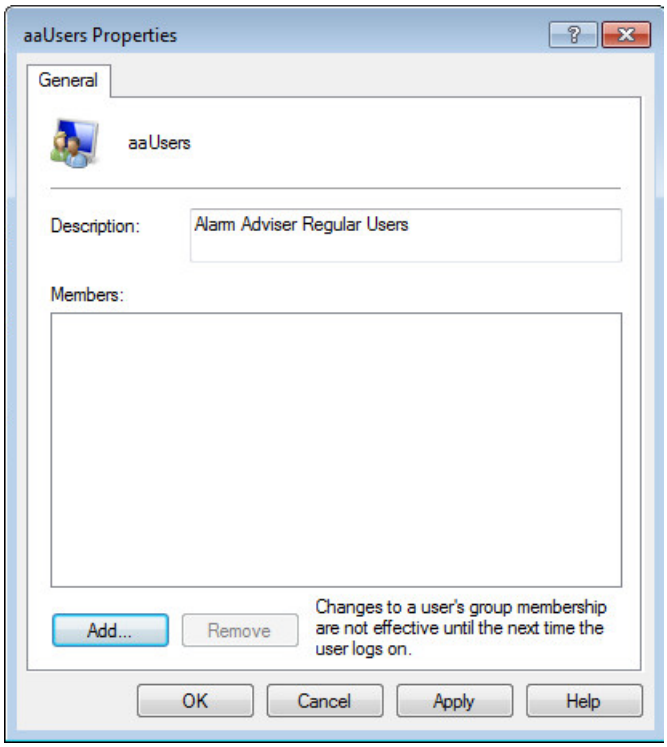


2. Click the arrow next to the **System Tools > Local Users and Groups** node in the left pane.

3. Click the arrow next to the **Groups** folder. All existing groups on that computer are listed in the right.



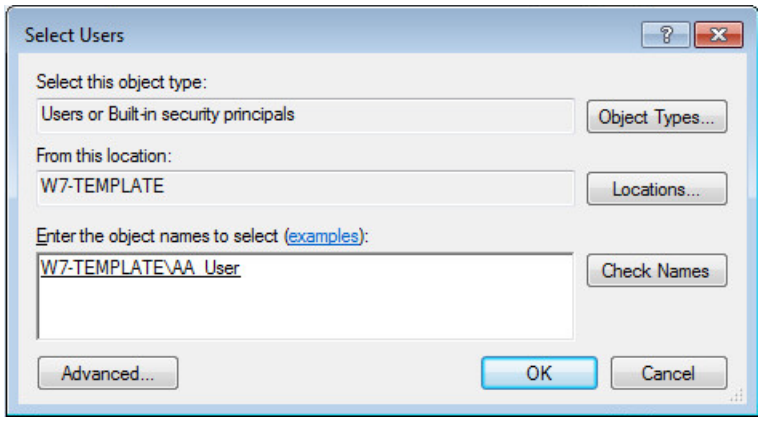
4. Double-click on the Alarm Adviser group to which you want to add a user. The selected group's Properties dialog box appears.



5. Click **Add**. The Select Users dialog box appears.

6. In the **Enter the object names to select** box, type the complete name of the user you wish to add to the selected group.

7. Click **Check Names**. If the name is a valid user, the name is shown as underlined text in the format **<computer name>\username**.



8. Click **OK**. The user is added to the group and the Properties dialog box appears with the user name in the **Members** box.

9. Click **OK**. Communicate the sign in information to the Alarm Adviser user for whom you have created the account. They should be able to log on to Alarm Adviser with this account.

Chapter 7: Alarm Adviser Licensing

Alarm Adviser offers the following types of licenses based on a combination of the number of SCADA systems that Alarm Adviser can analyze and the maximum number of records available in the database:

License Type	No. of SCADA Systems Analyzed	No of Records Processed
Demo (free)*	1	1 million
Alarm Adviser Lite for InTouch **	1	1 million
Standard	1	1 million
Professional	5	10 million
Premier	10	Unlimited
Unlimited	Unlimited	Unlimited

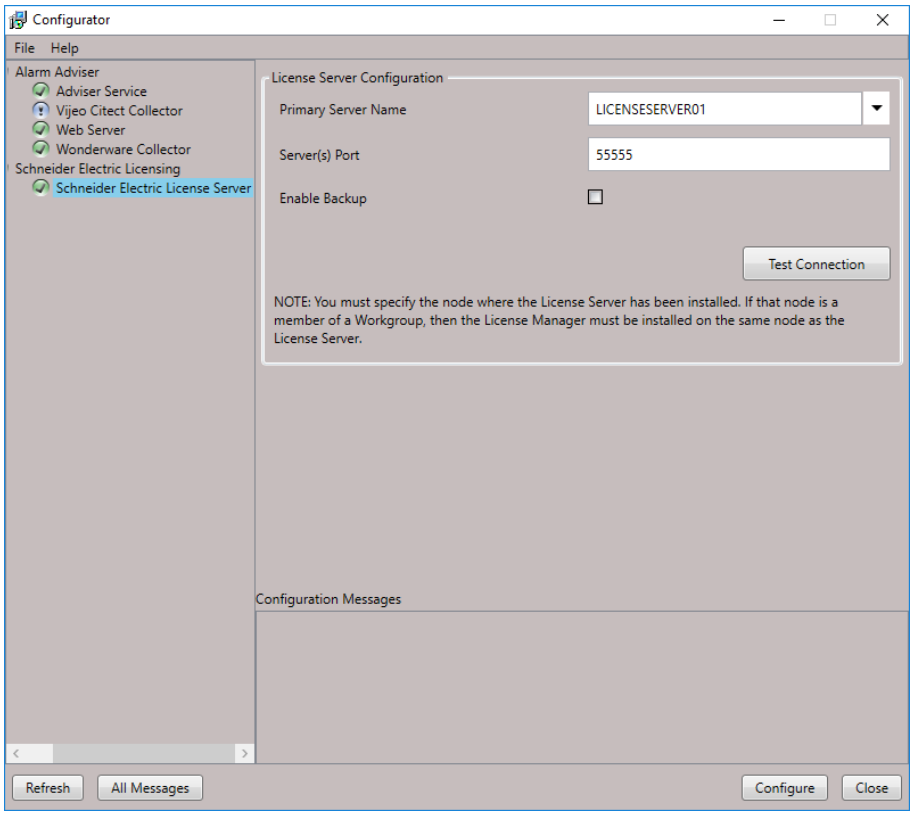
* The alarm name is not displayed on graphs and charts if you are running Alarm Adviser in Demo mode; it is replaced by a series of hashes (#).

** This license is applicable for InTouch systems only and is limited to a maximum of five providers. For more information, see Alarm Adviser Licensing for InTouch.

License information for Alarm Adviser resides on the License Server, which needs to be installed along with the License Manager. For more information, refer to the Schneider Electric Licensing Guide.

Once installed, the License Server configuration options are automatically completed for you in the Configurator. Click **Configure** to establish communication with the Alarm Adviser Service.

IMPORTANT: As per the EULA (Section 2.2.2), the Software data collector (s) may only be used on the site(s) or by the business entity for which the Software was originally purchased and configured for. You must not re-configure the Software data collector (s) between systems, sites or other business entities.



After the License Server is configured successfully, you need to activate your Alarm Adviser license before you can start using the application. This can be done through the License Manager, which is a web-based application that allows you to manage your Alarm Adviser license.

For information on activating and deactivating your Alarm Adviser license, refer to the section "Activating Licenses" in the Schneider Electric Licensing Guide.

Alarm Adviser Licensing for InTouch

A new license type, Alarm Adviser Lite for InTouch, has been introduced in version 2017 of Alarm Adviser. This license is applicable to InTouch systems. It is recommended that you go through the following additional information about this license type:

- This license type is applicable only if you are running an InTouch system.
- With this license type, you can specify a list of InTouch providers when configuring the Wonderware collector.
- The maximum number of providers that you can specify is limited to five.
- If the InTouch Alarm Database has more than five providers, while configuring the collector you can specify a comma-separated list of the providers to include for collection. If your list of providers exceeds five, data will be collected from the first five providers only.
- If there is a large amount of data to be collected, data from all specified providers may not be collected if the data limit (number of records) is exceeded.
- If you change the list of providers specified, you will need to delete the existing system definition and add a new system definition for your InTouch system. For information about creating and deleting systems, refer to the Alarm Adviser Online Help topic "Configuring Systems".

Note: The System Management Console (SMC) logs a warning message if you exceed the number of allowed providers. For more information about using the SMC, refer to Running System Diagnostics.

Chapter 8: Maintaining an Alarm Adviser Installation

Maintenance tasks you will need to perform include:

- Maintaining the database periodically.
- Backing up the database.
- Re-installing components to reflect changes in your Alarm Adviser deployment architecture.
- Modifying an installation.
- Repairing an installation.
- Uninstalling Alarm Adviser.

Maintaining the Alarm Adviser Database

As part of the Alarm Adviser Service configuration, you can set up a time and duration for maintenance tasks to run every day. At the specified time, the following tasks are carried out:

- Old records that are out of the data set range are deleted from the database automatically.
- Any system that an administrator might have deleted using the web client is purged along with any corresponding data.
- The Alarm Adviser Database is re-indexed in order to optimize performance.

Note: Data collection is temporarily stopped while the maintenance tasks are running. It resumes as soon as the tasks are completed.

Backing Up the Alarm Adviser Database

After you have installed Alarm Adviser, it is recommended that you establish a regular backup routine. For information about backing up and restoring the database, refer to the backup information for SQL Server. Also see, http://msdn.microsoft.com/en-us/library/ms187048(v=sql.110).aspx.

Re-deploying Installed Components

Changes to the infrastructure may necessitate re-deploying Alarm Adviser components on a different computer. It is recommended that you take a backup of the Alarm Adviser Database before you make any changes to your installation. For information about backing up and restoring the database, refer to backup information for SQL Server. Also see, http://msdn.microsoft.com/en-us/library/ms187048(v=sql.110).aspx.

Moving Web Server, Service or Collector

To move the Alarm Adviser Web Server or Service to a different computer:

1. Stop all Alarm Adviser services.
2. Backup the Alarm Adviser Database.
3. Uninstall the component to be redeployed.
4. Install and configure the component on the new computer.

Note: Installing the Collector on a different computer will re-start data collection from the backfill period of six months, or from the time of the earliest available data (if shorter). This may result in additional processing load on the computer. Any overlapping data is merged.
If the Collector is re-installed on a computer where it has previously been installed, the Collector's cache file will be deleted, and the Collector will re-start data collection from the backfill period of six months, or from the time of the earliest available data (if shorter).

Moving the Alarm Adviser Database

To move the Alarm Adviser Database to a different SQL Server instance:

1. Stop all Alarm Adviser services.
2. Backup the Alarm Adviser Database.
3. Restore the database on the new computer.
4. Delete the old database, or detach it from the SQL Server instance, or set its status to Offline.
5. Re-configure the Adviser Service and Web Server to point to the new SQL Server instance and database.

Note: All running Collectors will automatically connect to the Adviser Service on the new computer after the Service is re-configured, and subsequently data collection will resume.

6. Verify that connection to the database is successful.

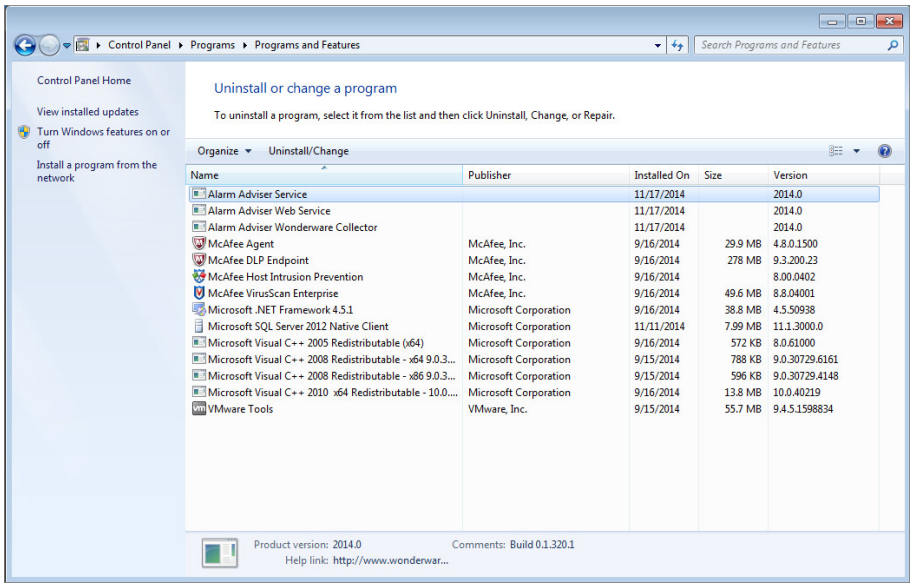
Modifying an Installation

You can change the Alarm Adviser components installed on your computer. You can add new components or remove the existing ones.

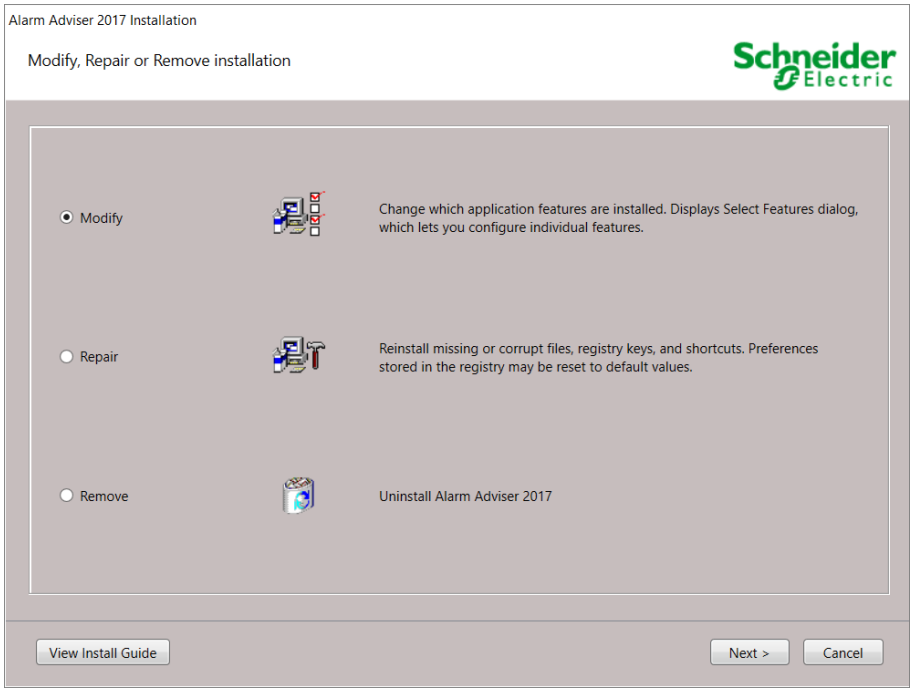
Note: You need to have the installation CD inserted in the CD-ROM drive before you can modify a component, or have access to other media where the installation files are located.

To modify an installation:

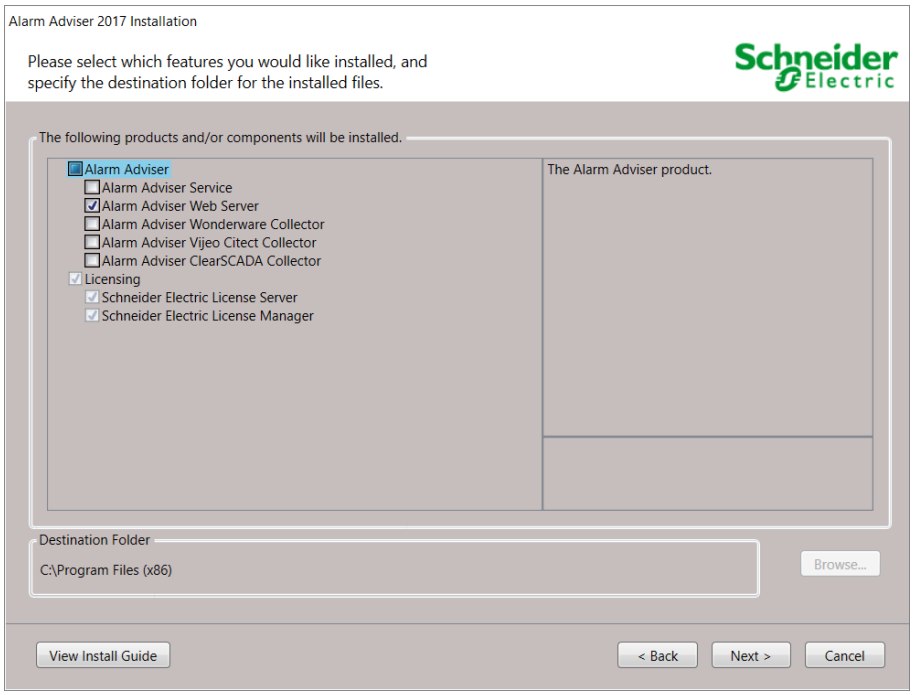
1. From the Windows **Start** menu, select **Control Panel > Programs > Programs and Features > Uninstall a Program**. All programs installed on your computer are listed.
2. Select Alarm Adviser.



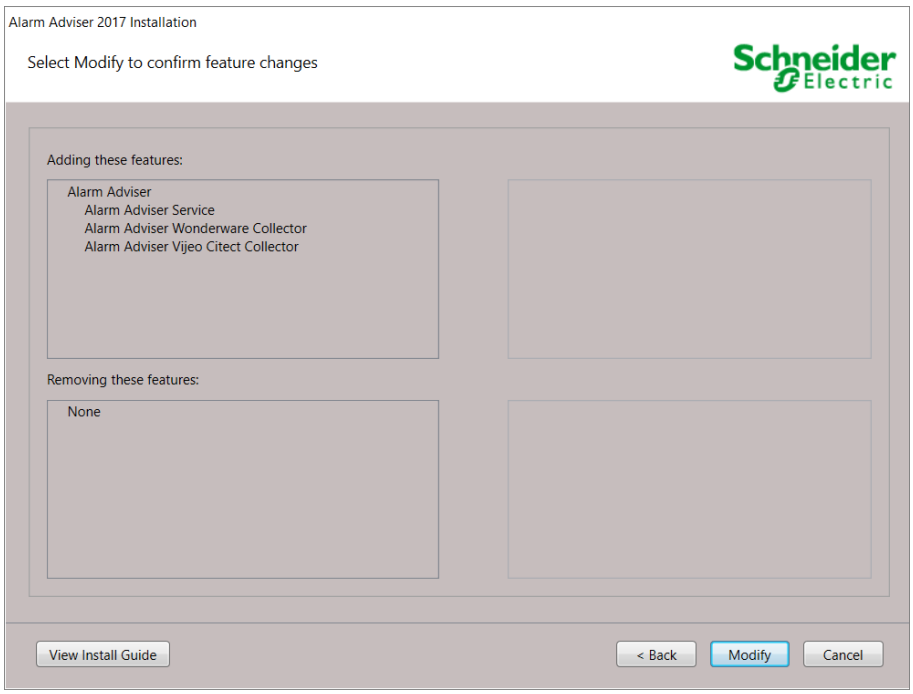
3. Click **Uninstall/Change**. The Modify, Repair or Remove Installation dialog box appears.



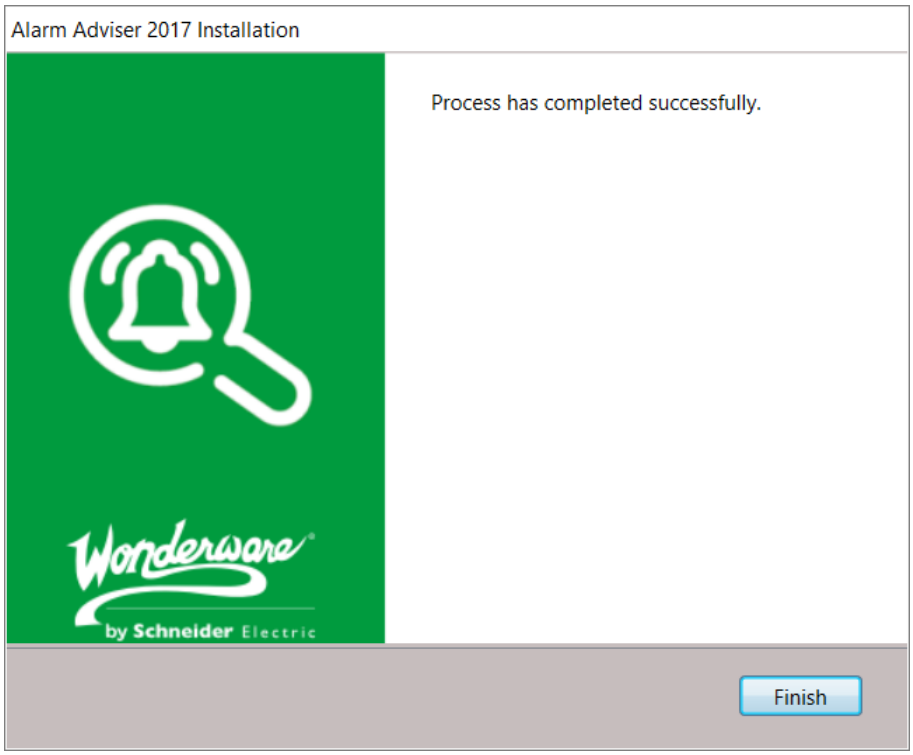
4. Click **Modify**, and then click **Next**. A list of installed Alarm Adviser components appears, with the components installed on the computer selected.



5. Change the selection if required, and then click **Next**. A screen prompting you to confirm the modify operation appears.



6. Click **Modify**. The selected components are added or removed, and a message confirming successful update appears.



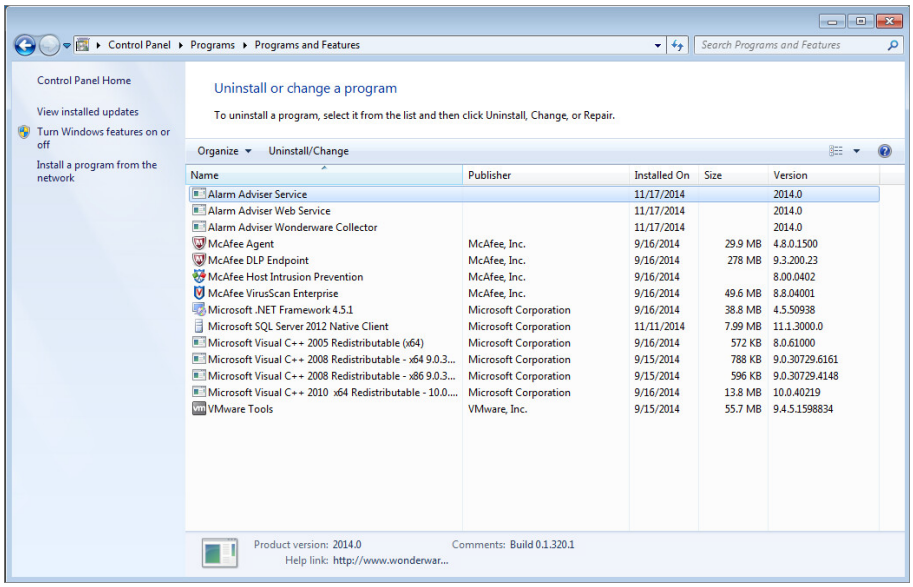
Repairing an Installation

You can use the installation program to repair corrupt files of installed Alarm Adviser components.

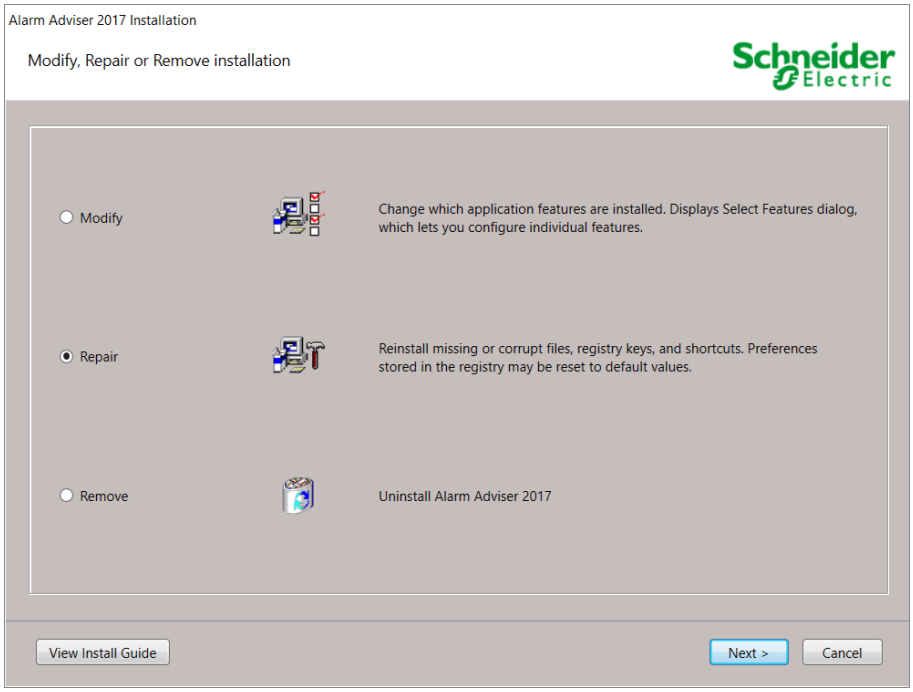
Note: You need to have the installation CD inserted in the CD-ROM drive before you can modify a component, or have access to other media where the installation files are located.

To repair an installation:

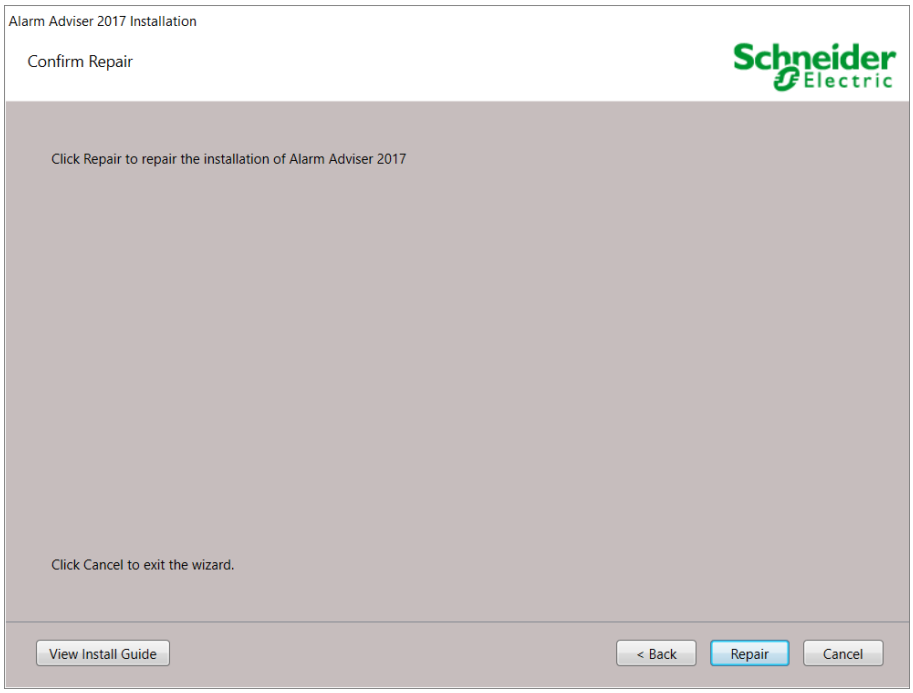
1. From the Windows **Start** menu, select **Control Panel > Programs > Programs and Features > Uninstall a Program**. All programs installed on your computer are listed.
2. Select the required Alarm Adviser component.



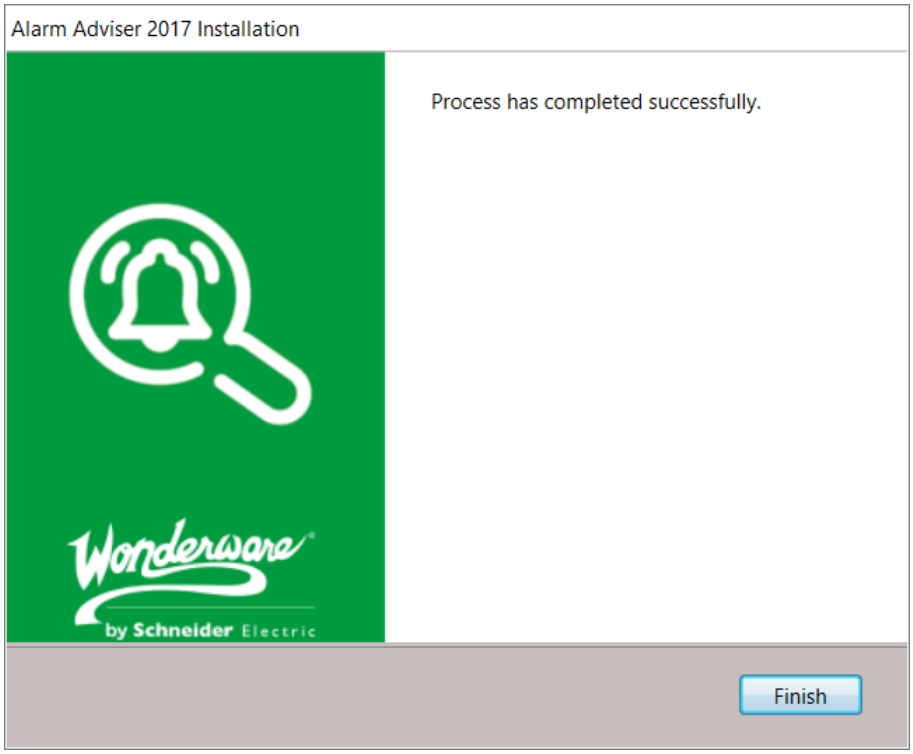
3. Click **Uninstall/Change**. The Modify, Repair or Remove Installation dialog box appears.



4. Click **Next**. A screen prompting you to confirm the repair operation appears.



5. Click **Repair**. All installed Alarm Adviser components are repaired, and a message confirming successful repair appears.



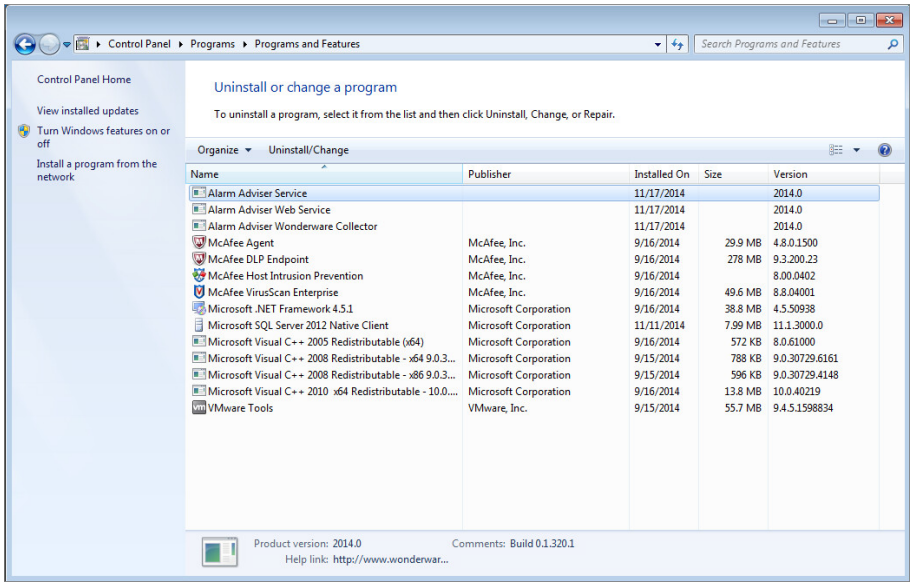
Uninstalling Alarm Adviser

You may wish to remove installed Alarm Adviser components. This section provides instructions for uninstalling Alarm Adviser.

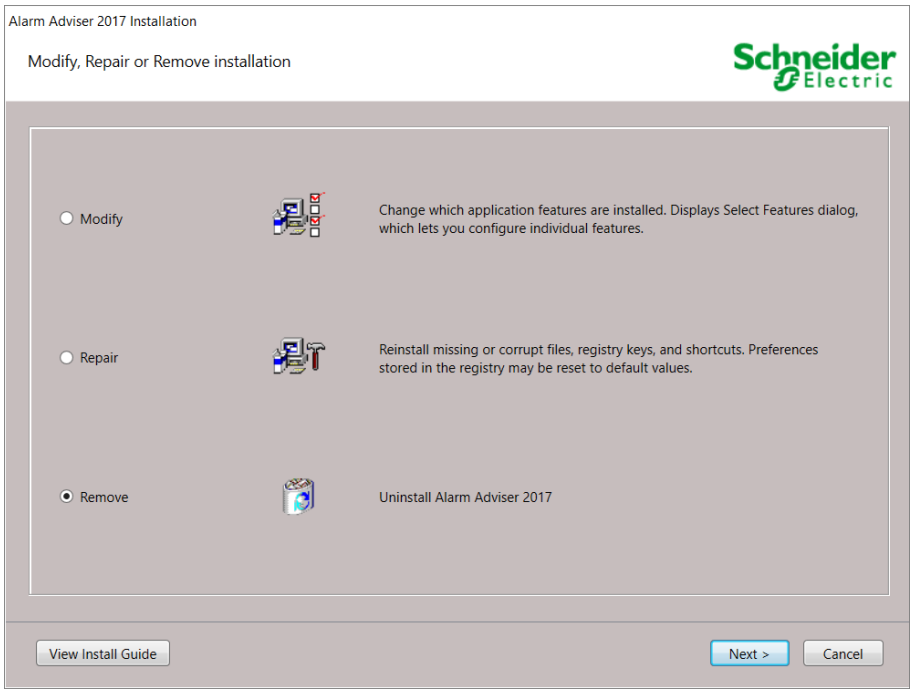
Note: Licensing components cannot be uninstalled using the Alarm Adviser installer.

To uninstall Alarm Adviser:

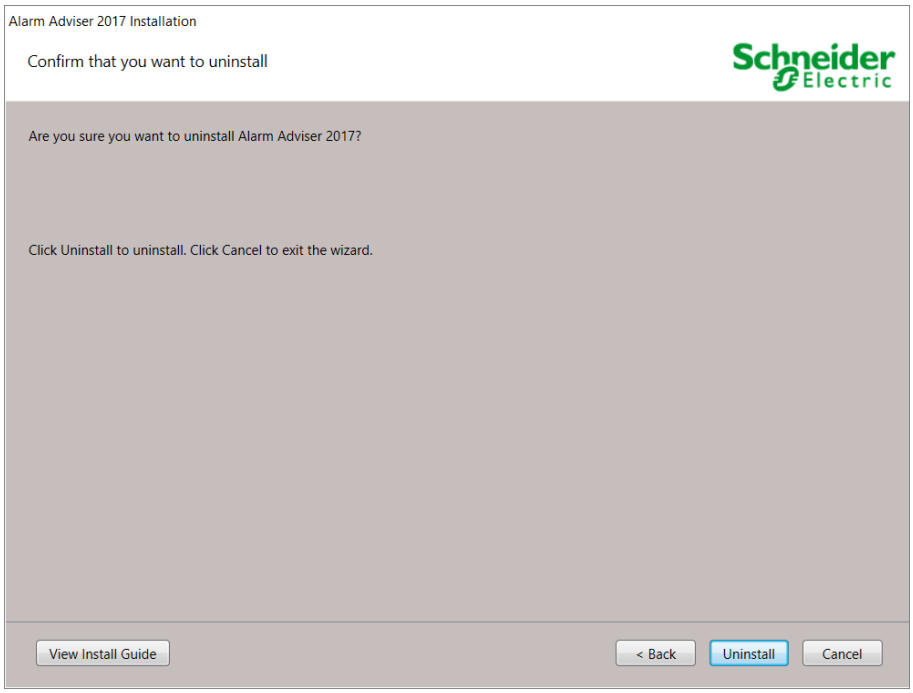
1. From the Windows **Start** menu, select **Control Panel > Programs > Programs and Features > Uninstall a Program**. All programs installed on your computer are listed.
2. Locate the Alarm Adviser component you wish to uninstall. Click to select the component.



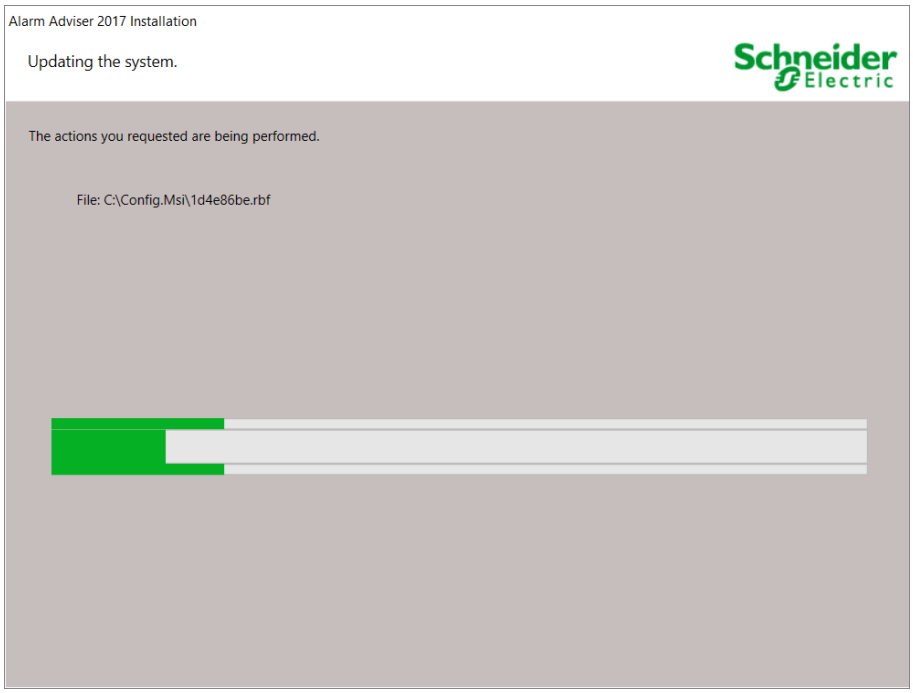
3. Click **Uninstall/Change**. The Modify, Repair or Remove Installation dialog box appears.



4. Click **Remove**, and then click **Next**. A screen confirming the uninstall appears.



5. Click **Uninstall**. The selected component is uninstalled.



6. Repeat steps 2 through 5 to uninstall the other components, if required.

The Uninstall wizard removes the selected Alarm Adviser components, deletes the installation folder and all files within the folder, and removes the Alarm Adviser component options from the Configurator and SMC.

Note: The Configurator and SMC will remain on the computer if there are other applications on the computer that require these applications.

Note: Alarm Adviser components can also be uninstalled by running **setup.exe** from the Alarm Adviser product CD.

Chapter 9: Running System Diagnostics

Web Page Error Notifications

If viewing the web pages causes the browser to operate in an unexpected way, an error message will display with information describing the cause. The information can be used to troubleshoot the situation and resolve it.

Alarm Adviser Component Logging

Alarm Adviser components (Web Server, Adviser Service and Collector) run in the background and are therefore unable to display information about any unexpected occurrences that may be encountered during operation. However, Alarm Adviser component events are logged using the System Management Console (SMC) through which you can troubleshoot such occurrences. *For example, if users are not able to connect to Alarm Adviser Web Server through the client, you can refer to the event log for the component using SMC*.

The following scenarios can cause one or more Alarm Adviser components to behave unexpectedly:

- Not being able to connect to the web client resulting from an issue with the Web Server
- The web client not displaying any data potentially indicating an issue with the Adviser Service
- Updated data not being retrieved from your SCADA system as a result of an issue with the Collector

Confirm that your system is configured correctly, and check that the related service is running before you proceed.

Note: SMC is initially configured to log Error, Warning and Info (Information) messages only. The event type to be logged can be configured in SMC. Events will only be logged when they are selected for logging. Event types are referred to as log flags in SMC.

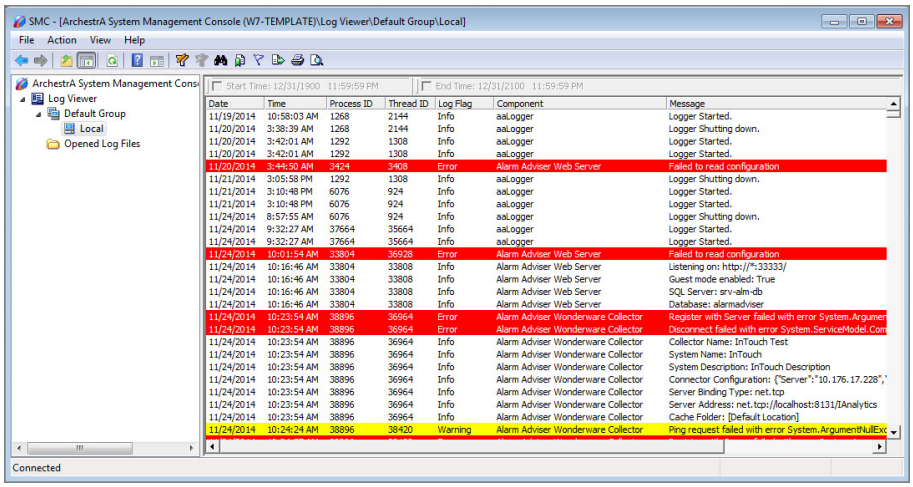
For each component, it is recommended that the following types of events be configured for logging:

Event Type (Log Flag)	Description
Error	Unexpected conditions resulting from misconfiguration or inadequate user privileges. The entire exception is logged to assist with troubleshooting.
Warning	Expected conditions that most likely cause issues with the system. For example, the Wonderware Collector is unable to register with Alarm Adviserr Service.
Info	General one off information messages. For example, values configured on startup or Collector registration with Alarm Adviser Service.
Trace	Additional information logged to assist with troubleshooting. For example, status of collection of alarm records.
Start-Stop	Log of each time an Alarm Adviser service is started or stopped.
SQL	Log of SQL statements executed on the Alarm Adviser Database.

Viewing Alarm Adviser Events

To view Alarm Adviser events:

1. Start SMC by clicking **Start > Programs > Wonderware > System Management Console**.
2. From the navigation menu in the left, select **ArchestrA System Manager > Log Viewer > Default Group > Local**.



Alarm Adviser events appear under the following names in the **Component** column:

- Alarm Adviser Web Server
- Alarm Adviser Service
- Alarm Adviser Wonderware Collector
- Alarm Adviser Vijeo Citect Collector
- Alarm Adviser ClearSCADA Collector

Note: The columns displayed in SMC can be configured. To do this, click on the column name, and then click **Show** to display the column or **Hide** to hide the column. Alternatively, from the **View** menu, select **Choose Columns** to configure the columns to be displayed. For more information on using SMC, refer to the SMC help or documentation.

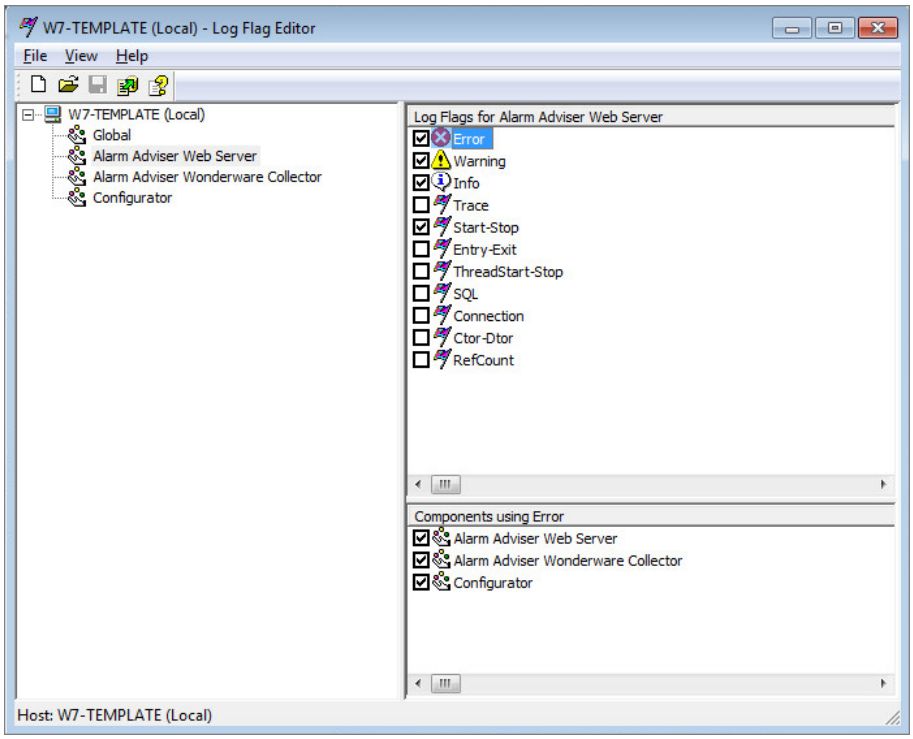
Note: It is recommended that the event log be cleared at regular intervals in order to free-up disk space. To do this, select **ArchestrA System Manager > Log Viewer > Default Group > Local** and right-click on it. A context menu appears. From the menu, select **Messages > Purge**. The Purge dialog box appears. Select the **Purge All log messages** option and click **OK**. You can take a backup of the messages before deleting them, if required.

Configuring Events for Display

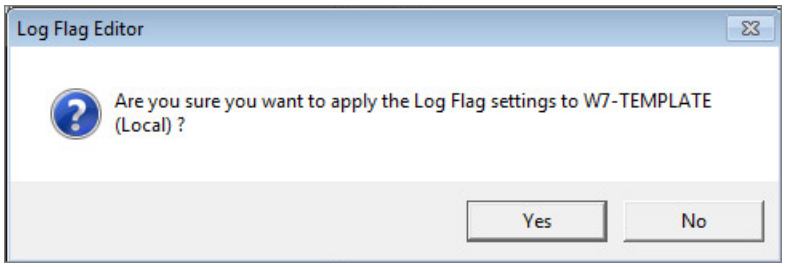
Events displayed for each component may be different depending on your configuration within SMC. Event types are referred to as Log Flags in SMC.

To configure the events you want logged:

1. Start SMC by clicking **Start > Programs > Wonderware > System Management Console**.
2. From the **Action** menu, select **Log Flags**. The Log Flag Editor appears.
3. From the left pane of the editor, select the Alarm Adviser component for which you want to configure log flags. The right pane shows available and selected log flags for this component.



4. Select the required log flags, or clear the selection for log flags that are not required.
5. From the **File** menu, select **Apply** or click . You will be prompted to confirm the changes.



6. Click **Yes** to apply the changes.

7. Repeat steps 2 through 6 to configure log flags for other Alarm Adviser components.

Unhandled Exceptions

Unexpected occurrences that are not written to the SMC are classified as "unhandled". When an unhandled event occurs, Alarm Adviser will display an "Unhandled Exception" message, and provide details of the file to which an error message has been logged.

Typically, the file is named with the date and time of the occurrence with the extension "mdmp", and is located in the Windows users/<user name>/AppData/Local/Alarm Adviser directory. For example, 201501113024010.mdmp. To help resolve the situation, create a ZIP file of the log file and send it to your customer support representative.

Appendix A: Supported Language Codes

A language code needs to be supplied to the Resgen command-line utility for localizing the Alarm Adviser web client to a specific language. Given below is a list of supported language codes.

LANGUAGE	CODE
Afrikaans	af
Moroccan Arabic	ar-ma
Arabic Saudi Arabia	ar-sa
Arabic	ar
Azerbaijani	az
Belarusian	be
Bulgarian	bg
Bengali	bn
Tibetan	bo
Breton	br
Bosnian	bs
Catalan	ca
Czech	cs
Chuvash	cv
Welsh	cy
Danish	da
Austrian German	de-at
German	de
Modern Greek	el
Australian English	en-au

LANGUAGE	CODE
Canadian English	en-ca
Great Britain English	en-gb
Esperanto	eo
Spanish	es
Estonian	et
Euskara	eu
Persian	fa
Finnish	fi
Faroese	fo
Canadian French	fr-ca
French	fr
Galician	gl
Hebrew	he
Hindi	hi
Hrvatski	hr
Hungarian	hu
Armenian	hy-am
Bahasa Indonesia	id
Icelandic	is
Italian	it
Japanese	ja
Georgian	ka
Khmer	km
Korean	ko
Luxembourgish	lb
Lithuanian	lt
Latvian	lv

LANGUAGE	CODE
Macedonian	mk
Malayalam	ml
Marathi	mr
Bahasa Malysia	ms-MY
Burmese	my
Norwegian BokmÃ¥l	nb
Nepal/Nepalese	ne
Dutch	nl
Norwegian Nynorsk	nn
Polish	pl
Brazilian Portugese	pt-br
Portuguese	pt
Romanian	ro
Russian	ru
Slovak	sk
Slovenian	sl
Albanian	sq
Serbian-cyrillic	sr-cyrl
Serbian-latin	sr
Swedish	sv
Tamil	ta
Thai	th
Tagalog/Filipino	tl-ph
Turkish	tr
Morocco Central Atlas TamaziÉ£t in Latin	tzm-lat
Morocco Central Atlas TamaziÉ£t	tzm
Ukrainian	uk

LANGUAGE	CODE
Uzbek	uz
Vietnamese	vi
Chinese	zh-cn
Traditional Chinese	zh-tw

Index

A

about

this guide 9

Alarm Adviser

architecture 10

collector

configuring ClearSCADA Collector 74

configuring Vijeo Citect Collector 67

configuring Wonderware Collector 62

components 11

configuring 9, 49

deploying 15

install 31

installing 9, 31

licensing 103

localizing 60

maintaining 107

Service

configuring 52

system requirements 19

uninstall 115

uninstalling 115

Alarm Adviser Database 13

Alarm Adviser Service 13

Alarm Adviser users

creating 97

architecture 10

Alarm Adviser 10

introduction 10

C

collector 12

components 11
moving 108-109
re-deploying 108
configurator 49
configuring Alarm Adviser Service 52
configuring ClearSCADA Collector 74
configuring Vijeo Citect Collector 67
configuring Web Server 56
configuring Wonderware Collector 62
creating
Alarm Adviser users 97

D

database backup
maintenance 108
database clean-up
maintenance 107
deploying 15
Alarm Adviser 15
deployment
scenarios 15
deployment scenarios 15

I

installation 21, 31
modifying 109
prerequsites 21
repairing 112
installed components
moving 108-109
installing 31
Alarm Adviser 9
ClearSCADA Collector 42
Vijeo Citect Collector 39
Wonderware Collector 36
introduction
architecture 10

L

licensing 103

localizing 60

M

maintaining

Alarm Adviser 107

maintenance 107

database backup 108

database clean-up 107

modifying 109

installation 109

moving 109

installed components 108-109

P

prerequisites 21

R

re-deploying 108

components 108

related documents 9

repairing 112

installation 112

S

system diagnostics 119

running 119

system requirements 19

T

this guide

about 9

U

uninstalling 115

users 97